Corero Management Server User Guide

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Welcome to the Corero Management Server

Configure the CMS

The CMS manages appliances and policy configurations that protect your servers and other critical resources inside your organization's firewall from DDoS attacks and cyber threats, and also protects and optimizes the performance of critical infrastructure such as servers and firewalls. This overview describes how the CMS does this.

- Basic CMS setup and configuration checklist. This is a list of steps most users will need to perform to define a minimal working configuration.
- Add a Corero appliance for the CMS to manage.
- Save your configuration.
- Optional and Advanced Configuration. These topics cover configuration tasks that are outside of a minimal working configuration.

Analyze Traffic

- Use the CMS dashboard to see overall status for the site. The dashboard is displayed by default when you log in to the CMS.
- View blocked and detected events. Rules can log traffic (detect it), allow it to pass through, or block it completely. These events are from the traffic your appliances are detecting and blocking.
- View security events. Generally, traffic generates a security event every time it triggers a rule (this can vary according to the details of your configuration). This shows you all of the security events that have been generated.

Restrict Access

These are temporary measures that enable you to block malicious traffic immediately, while you define one or more policies to implement a more sophisticated long-term solution.

- Block Specific IP Headers. Block traffic outright, based on its packet header type.
- Block IP Addresses. Block an IP address completely by adding it to a list of addresses that is consulted before any other part of the corresponding packet is examined. A packet from a blocked address is discarded immediately, and no additional processing is performed on any portion of its contents.
- Smart-Rule. Configure threshold values and rate limits for protecting servers from various types of flood traffic.
- Flex-Rule. Define custom packet-blocking filters that enable you to respond quickly to new threats you encounter that are not yet addressed by formal rules in Corero's SmartWall software, or, otherwise, to threats for which you want a very specific or nuanced response that is not provided to your liking by the built-in software.

Define Policies For Long Term Protection

The appliance uses profiles to determine the acceptability of traffic seeking entry to your servers, and responds to that traffic based on server protection instances that you define.

- Configure client profiles. These are sets of traffic characteristics that the appliance uses to distinguish between acceptable and unacceptable traffic coming in to the network.
- Configure server protection instances. These are assemblies of profiles and responses that are associated with particular servers to guard them from DDoS attacks and cyber threats.
- Block traffic based on its country of origin. If you know that you don't have any legitimate users in a particular country, you can just block all traffic from that country's IP address ranges.

Contact Corero Network Security

- Contact Corero Customer Support
- Version and Legal Information For This CMS
- Go to www.corero.com, the Corero Network Security homepage
- Corero Product Overview: The First Line of Defense

Corero Product Overview: The First Line of Defense

Corero Network Security provides the solution to a common problem that often impairs the performance and effectiveness of many organizations' IT infrastructures.

Recognizing a Common IT Infrastructure Problem

Today, an organization’s network can be overwhelmed by DDoS attacks and cyber threats coming in from the Internet. This can comprise a number of hazards, from undesired users and services on your network to distributed denial of service (DDoS) attacks and advanced evasions that target your servers. Unfortunately, today’s security infrastructure is not sufficient to stop these encroachments on your network. Specifically:

- The state tables of current generation firewalls get saturated with large numbers of connections from unwanted traffic.
- Next generation firewalls don’t have adequate DDoS protection, and therefore cannot discriminate good traffic from bad traffic.
- IT infrastructure and servers sitting downstream from the firewall end up processing unnecessary traffic and therefore are less responsive to good traffic.

This not only exposes you to threats, but costs service degradation, site downtime, brand damage, and, most importantly, lost business.

The Solution: Corero's First Line of Defense

The solution to this problem, Corero's First Line of Defense®, stops DDoS attacks, protects infrastructure, and eliminates downtime from its position *in front* of your firewall. The First Line of Defense stops all types of DDoS attacks, targeted server attacks, advanced evasions and a host of other attack vectors including mitigation of DDoS attacks and cyber threats before they even enter your network. As a result, it makes your IT infrastructure more efficient and effective, improving performance of applications and servers, and ultimately eliminating downtime. More importantly, the assumption you had in the beginning, that your customer traffic flows through unabated, now actually holds true. You have satisfied customers and foiled attackers.

Corero's First Line of Defense mitigates DDoS attacks and cyber threats with a systematic approach comprising five steps:

1. Allow expected traffic while restricting known attackers. Examples of known attackers are IP addresses known to be malicious and geographies that you don't do business with. Corero's ReputationWatch® filters them out ahead of time so that no traffic from these sources comes in.
2. Thwart attackers based on their behavior. Examples include an attacker who throws large volumes of traffic at your website, or who opens an excessive number of connections, tying up your server resources. Corero's dynamic threat assessment and DDoS rejection capabilities mitigate these hazards by evaluating their behavior and limiting their rate, or blocking them altogether.
3. Enforce correct traffic behavior by checking whether traffic conforms to RFC specifications or de facto standards. For example, normal behavior does not send payload traffic in tiny fragments. Corero's protocol validation modules (PVMs) stop such protocol abuse and advanced evasion techniques (AETs).
4. Analyze traffic integrity by looking at whether it contains any known intrusions. Corero uses signatures to inspect traffic and block any buffer overflows, code injections, and exploits before they compromise your servers. With Protection Packs, key components of Corero's threat update service, Corero keeps its customers up to date with the latest security protection by delivering malware signatures and vulnerability protection on a weekly basis.
5. Provide you with insight into what is happening on your network. This includes the ability to collate data from SYSLOG events, SNMP gets & traps, and proprietary APIs into a meaningful stream of information. With that, you can reassess the threat landscape to see if your security policies are effective or need modifications. Corero's SecureWatch®, a maintenance and monitoring service, provides infrastructure and expertise to keep Corero deployments up to date and assure sustained availability.

Corero supports this approach with a product that acts as the First Line of Defense for your network: the SmartWall Network Threat Defense Appliance (referred to simply as an "appliance"). This device is managed using the Corero Management Server, an application designed expressly for configuring, managing, and monitoring the appliances that protect your network from harm.

CMS Overview

The Corero Management Server (CMS) is a comprehensive and centralized tool for managing all aspects of network protection for your servers and other computing resources that reside inside your organization's firewall. The CMS achieves this through its control of appliance configuration for each

site. The CMS enables you to define the server and client profiles, shun labels, header blocking conditions, and server protection policies that protect your organization's critical infrastructure. In addition, the CMS provides detailed real-time visibility into the traffic entering your network at each step, as it is first observed, assessed for threats, treated for threats, and subsequently allowed to enter protected areas of your network.

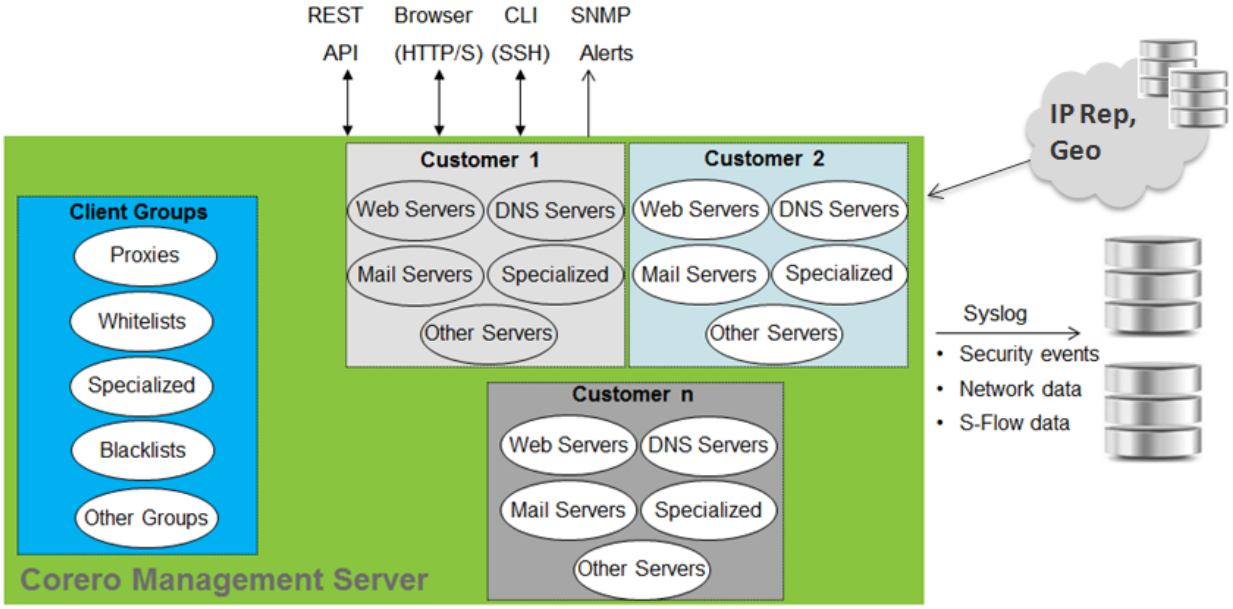
The CMS's organization reflects the structure of the underlying data models that make up the appliance configuration, enabling you to access and configure all aspects of appliance behavior. The CMS may be accessed primarily through its web UI (all major web browsers, including Internet Explorer, Firefox, Chrome, and Safari are supported), but a fully functional command line interface (CLI) and REST interface are present, as well. All user interfaces provide access and control to the same configuration elements, but the web interface provides the most dense display of configuration controls and richest visualization tools for statistics, events, and reports.

The CMS is run on a Linux virtual machine, delivered in a Corero Managed Service Appliance.

The CMS's support for configuring and managing appliances includes:

- a single, unified management view of the site, including appliances and policies
- a high-level dashboard that shows real-time site status and provides at-a-glance summaries of security events and various attack statistics
- ability to view aggregated statistics and aggregated data reports
- ability to view, manage, and modify appliance configurations
- ability to view, manage, and define security policies
- ability to respond immediately to threats by shunning IP addresses and blocking IP header types
- ability to acquire and distribute operating software and Defense Packs that keep the appliance's threat awareness up to date

This illustration shows a simple representation of the CMS and its relationship to appliances and various user interfaces:



CMS Web UI Tool Bar

The tool bar across the top of the CMS web UI provides convenient access to all the screens and controls in the web UI, making it easy to configure custom security for your servers in a simple series of clicks across just a few screens for any configuration task. Most CMS functionality supports server protection, enabling you to define and manage profiles and other definitions for protecting your critical network infrastructure.

- **Defense Monitoring** – Display high-level summaries of what is happening in the site; these include Security Event Viewer and Blocked and Detected Attacks. In addition, administer packet recording and forensics queries using the Packet Capture screen.
- **Defense Configuration** – Configure protection for your servers. This includes IP Address Shunning, IP Packet Header Blocking, Client Security Profiles, Server Security Profiles, and Server Protection.
- **Site Configuration** – Administer Corero appliances and manage their software.
- **CMS** – Administer the CMS itself, including users, customers, syslog and SNMP access, and redundancy (high availability).
- **CLI** – Access the command line interface as an alternative means of configuring the appliance.
- **Help** – Access this help set for descriptions of CMS screens and controls, and instructions for configuring various behaviors. The Help menu provides several entry points to the help set, including the ability to search for a particular string.
- **Save** – Retain changes you've made to the configuration and distribute them to the devices you're managing. Refer to Save Your Configuration for a detailed description.
- Status icons – Beneath the main menu bar is a series of icons that summarize the site's status. A green icon indicates that there are no problems; while an orange icon indicates that your attention is required in that area. Each icon is accompanied by a number that indicates the number of objects requiring your attention. Floating the cursor over the icon displays a balloon containing a brief descriptive summary of the associated status.
  - **Uncleared Alarms** – Click this icon to display the Alarm Center screen.
  - **Configuration** – Click this icon to display the Device Status screen or to immediately deploy the latest saved CMS configuration to any out of sync appliances.
  - **Connectivity** – Click this icon to display the Device Status screen or the Cluster Configuration screen.
  - **Notifications** – Click this icon to display or clear notifications.

Essential Server Protection Practices

Your servers may encounter unwanted and malicious network traffic that includes:

- *Undesired services* – Traffic from protocols or ports that you know ahead of time have no legitimate reason to access your servers.
- *DDoS attacks and competitive abuse* – Attempts to make your servers unavailable to desired or legitimate traffic by overwhelming them with spurious requests. These can take a variety of forms.

- *Protocol violations and advanced evasion techniques (AETs)* – These may include traffic types that are expected or otherwise legitimate, but that include packets that do not comply with the standard defined by the appropriate RFC. Such packets often indicate an attack, as they have been malformed intentionally to circumvent and exploit the protocol's correct behavior.
- *Targeted server attacks* – Intrusions that are crafted specifically to exploit some characteristics of your servers in order to gain access and cause harm to your infrastructure.

Protecting your servers from such threats relies on these essential practices:

- *Allow only expected traffic* – Block known malicious IP addresses and geographic locations; allow desired ports and services. The CMS supports this by enabling you to restrict access to specific clients and groups of clients. This is supported by features such as the real time reputation engine, geographic IP address blocking, on-demand shunning, IP address blocking, and IP header blocking.
- *Evaluate the amount of traffic* – Limit excessive requests and connections. The CMS supports this by enabling you to configure rate limits, such as packet rate limiting.
- *Enforce the correctness of traffic* – Stop protocol abuse and RFC violations, and prevent AETs. The CMS supports this with its stateful protocol validation capability.
- *Analyze the integrity of traffic* – Inspect traffic and block intrusion, buffer overflows, code injections, and exploits. The CMS supports this by preventing intrusions; it identifies known attack signatures, vulnerability signatures, overflows, injections .and brute force attacks.

In addition, the CMS provides rich visibility into threats as they are recognized, assessed, and mitigated, and furnishes a variety of tools for displaying the relevant data.

Configuring Custom Server Protection

The CMS and appliance provide a lot of protection for your servers automatically with rules, rule sets, service protocols, profiles, and server protection that is built in to the product, supporting the essential server protection practices that are critical to keeping your network safe. To achieve the maximum possible protection for your servers, however, you will want to customize the configuration so that it is optimized for the particular traffic that your organization's network encounters. Clicking **Defense Configuration** in the web UI tool bar provides access to the following commands for configuring server protection explicitly:

- **Server Protection** – Configure how the appliance responds to traffic bound for your servers.
- **Server Profiles** – Specify traffic characteristics to associate .with servers in your network.
- **Client Groups** – Specify sets of clients that should have their traffic handled the same by your appliances.
- **Client Profiles** – Specify traffic characteristics to associate with clients outside your network.
- **Service Protocols** – A service protocol specifies how traffic corresponding to a particular application is handled by the appliance.
- **Rule Sets** – A rule set is a named collection of rules that is configured to identify and address traffic for a particular purpose.
- **IP Packet Header Blocking** – Block access to your network for types of traffic that you never expect to encounter from legitimate users.

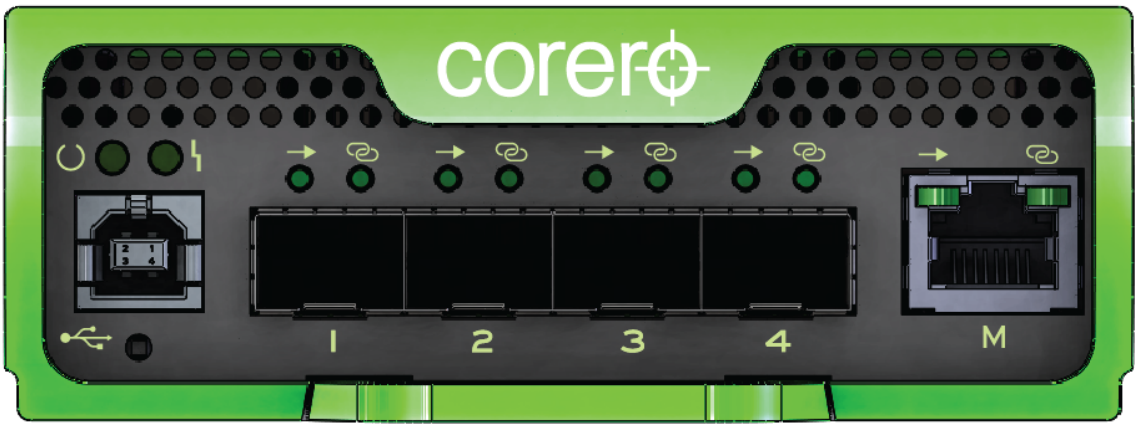
- **IP Address Shunning** – Shunning is a temporary measure that refuses entry to all traffic from a specific IP address for a finite amount of time. Shunning is useful when you encounter problematic traffic from a particular IP address; you can shun the traffic from that address for a period of time until you decide on a policy for handling that traffic over the long term.
- **IP Address Blocking** – Blocking an IP address completely adds that IP address to a list of addresses that is consulted before any other part of the corresponding packet is examined. A packet from a blocked address is discarded immediately, and no additional processing is performed on any portion of its contents.
- **Flex Rule based Blocking** – The Flex-Rule utility provides a means of defining custom packet blocking filters. These custom filters enable you to respond quickly to new threats you encounter that are not yet addressed by formal rules in Corero's SmartWall software, or, otherwise, to threats for which you want a very specific or nuanced response that is not provided to your liking by the built-in software.

Overview of the Corero SmartWall Threat Defense System

The Corero SmartWall Network Threat Defense Appliance is a network security appliance that protects network infrastructure against distributed denial of service (DDoS) attacks. It is the foundation of Corero's First Line of Defense solution.

The SmartWall Network Threat Defense Appliance is available as a purpose-built hardware unit dedicated to running Corero's SmartWall Network Threat Defense Appliance software.

The SmartWall Network Threat Defense Appliance can stand alone as an individual appliance, or it can be combined with one or more other SmartWall Network Threat Defense Appliances in a high-availability configuration, or in a networked series of SmartWall Network Threat Defense Appliances, each of which inspects and handles different aspects of the traffic entering your network.



The SmartWall Network Threat Defense Appliance provides the following protection:

- Sophisticated protection against DDoS attacks
- Broad support for intrusion prevention

Attack Detection Methodology

Practice the following methodology for identifying and mitigating attacks on your network:

1. **Research**. Peruse dashboard data and performance statistics for your network regularly, and develop an understanding of what constitutes normal or typical behavior for your network. Noticing aberrant behavior is the first step in recognizing an attack.
2. **Protect**. Define security policies that guard your network against unwanted or dangerous traffic. This will stop many attacks before they have any chance to impair your network's performance.
3. **Detect**. Use dashboard data and performance statistics for your network to recognize aberrant traffic that may indicate an attack. Unusual increases in certain types of traffic or unusual reductions in network performance may well indicate an attack.
4. **Triage**. Examine unusual dashboard data and performance statistics to assess the attack. How does the attack manifest? What is its apparent target? What is its apparent source?
5. **Mitigate**. Take immediate action to reduce the effectiveness of the attack. This may mean shunning one or more client IP addresses, disabling access for a particular service, or changing a limit on the number of connections or number of packets that are allowed for a service. Even if you cannot stop the attack completely without also inhibiting desired traffic, you can reduce the attack's ability to impair your network's proper function.
6. **Rectify**. Adjust your security policies to prevent the attack from impairing your network performance again.

Using the REST API

The Corero Management Server supports the use of a REST API over HTTP for performing configuration and monitoring tasks. You can use a variety of REST tools to exercise the API; the descriptions in this help set center around the UNIX/Linux command line tool, cURL, and the Postman REST client for the Google Chrome web browser.

Supported HTTP Methods

The CMS REST server supports these HTTP methods:

- GET – Retrieve a representation of a known resource. The XML representation has the list or container name as the XML document root.
- PATCH – Edit a known resource.
- POST – POST to the parent resource to create a new resource. The XML document root element specifies which resource is created. The new resource URI is returned in the "Location" HTTP header.
- DELETE – Remove a known resource.

You can use either XML or JSON format, both for data IN (PATCH or POST) and data OUT (GET).

GET Request query parameters

These parameters are supported for use with GET requests to provide more specific control over which resources should be returned from the request.

Use "?" to separate the parameter from the resource path. For example, `curl -k -v -u admin:admin https://`

```
[cmsipaddress]:443/api/running/defense/profiles/service-
protocols/service-protocol-group?deep --header "Accept:
application/vnd.yang.data+json"
```

- deep – Retrieve a resource and all its subresources.
- shallow – Retrieve a resource and none of its subresources.
- offset – Retrieve a subset of resources.
- limit – Retrieve a subset of resources.
- select – Specify exactly which nodes and subresources to retrieve from a resource.

HTTP Return Codes

The REST server supports the following HTTP return codes:

- 200 – *OK*. The request was successfully completed, and a response body is returned containing a representation of the resource.
- 201 – *Created*. A resource was created, and the new resource URI is returned in the "Location" header.
- 202 – *Accepted*. The request has been accepted, but has not been executed, and may not be executed at all.
- 204 – *No Content*. The request was completed successfully, but no response is returned.
- 400 – *Bad Request*. The request could not be processed because it contains missing or invalid information (such as a validation error on an input field, a missing required value, and so on).
- 404 – *Not Found*. The requested resource does not exist.
- 406 – *Not Acceptable*. The resource identified by this request is not capable of generating the requested representation, specified in the "Accept" header or in the "format" query parameter.
- 409 – *Conflict*. This code is used if a request tries to create a resource that already exists.

Media Type

The media type of the resources exposed through the REST API is "application/vnd.yang.datastore".

Secure and Insecure Connections

Connections to the CMS web UI are made over SSL and port 443. When accessing the CMS via REST, you need either to specify a certificate that will verify your identity to the CMS, or to use the *-k/--insecure* option to execute the cURL command using a default certificate. For brevity, the examples shown throughout this help set use the *-k/--insecure* option.

Learning the CMS Data Model

Resources are exposed to the REST API through the CMS data model, and it is necessary to be familiar with the data model in order to exercise the API effectively and use it to execute commands for retrieving data or configuring appliances. Prior to assembling a script or application that will execute numerous or complex instructions via the REST API, it is helpful to become familiar with the data model and the API at a smaller scale, by executing individual requests using cURL or a browser client.

One way of learning the CMS data model is by using the CMS CLI. to execute commands. While the CMS web UI sometimes abstracts the data model in order to collect multiple related resources in a single screen, CLI commands always match the corresponding nodes in the data model in a one-to-one manner. Another helpful approach to learning the CMS data model is to prepare a configuration using the CMS (using either the web UI or the CLI), then using HTTP GET over the REST API to retrieve all or a portion of that configuration. This gives you a clear, reliable template that you can edit and use subsequently to modify the configuration via HTTP POST or PATCH.

The data model is structured in hierarchical levels such that it is easy to access multiple resources that reside at a single level, but it is difficult to work with resources across different levels.

Resources

The available resources reside under /api, as either "running" or "operational", and are accessible via XML or JSON. The media type of the resources is "application/vnd.yang.datastore".

operational

The read-only "operational" resource represents the device's state data. Use this resource with the GET method to view the existing data model and configuration.

running

The "running" resource represents the running configuration datastore, and is present on all devices. Use this resource with the POST, PATCH, and DELETE methods to create or change things.

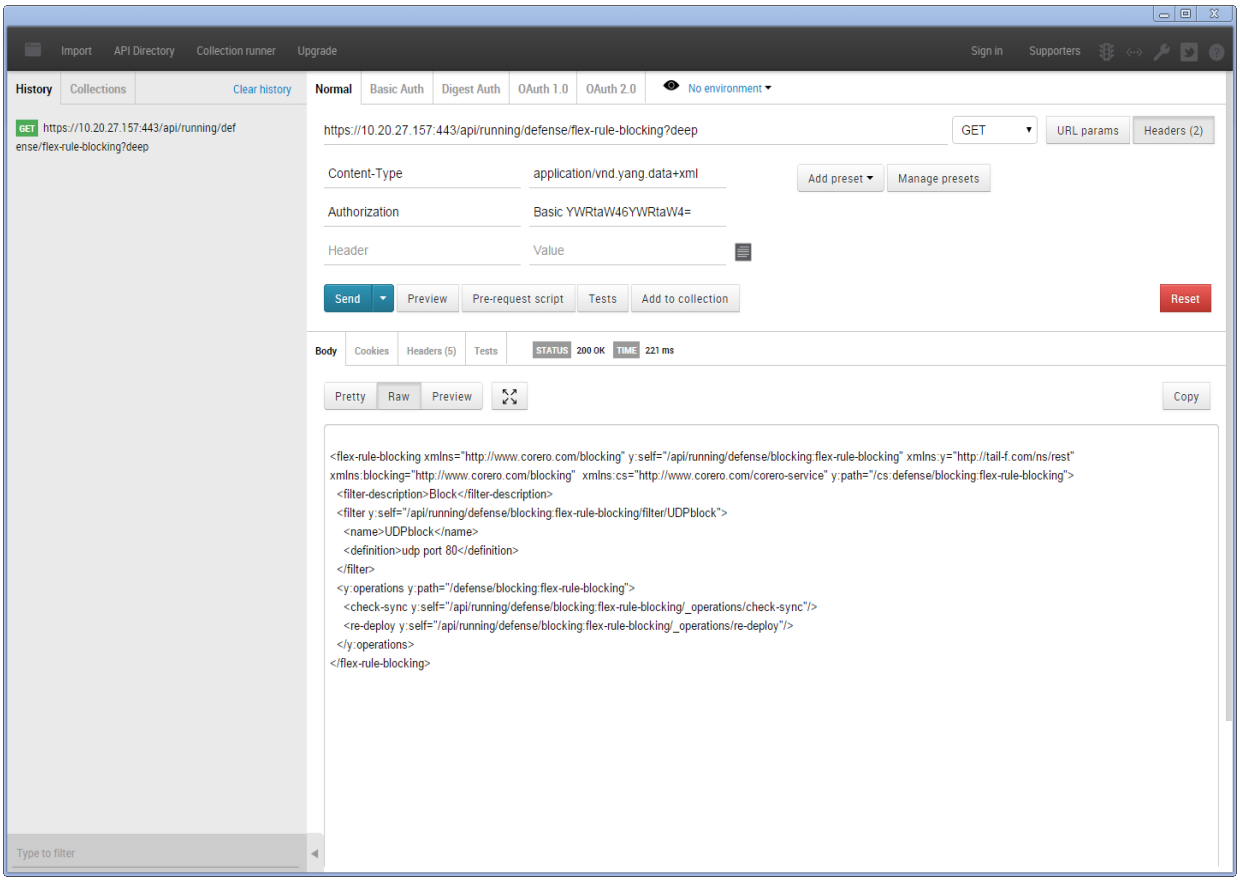
Simple Tools: cURL and REST Clients

The UNIX/Linux command line tool, cURL, provides a simple means of composing and transmitting REST requests, though it lacks some of the amenities provided by browser-based REST clients (such as Postman) that can make it easier to compose requests.

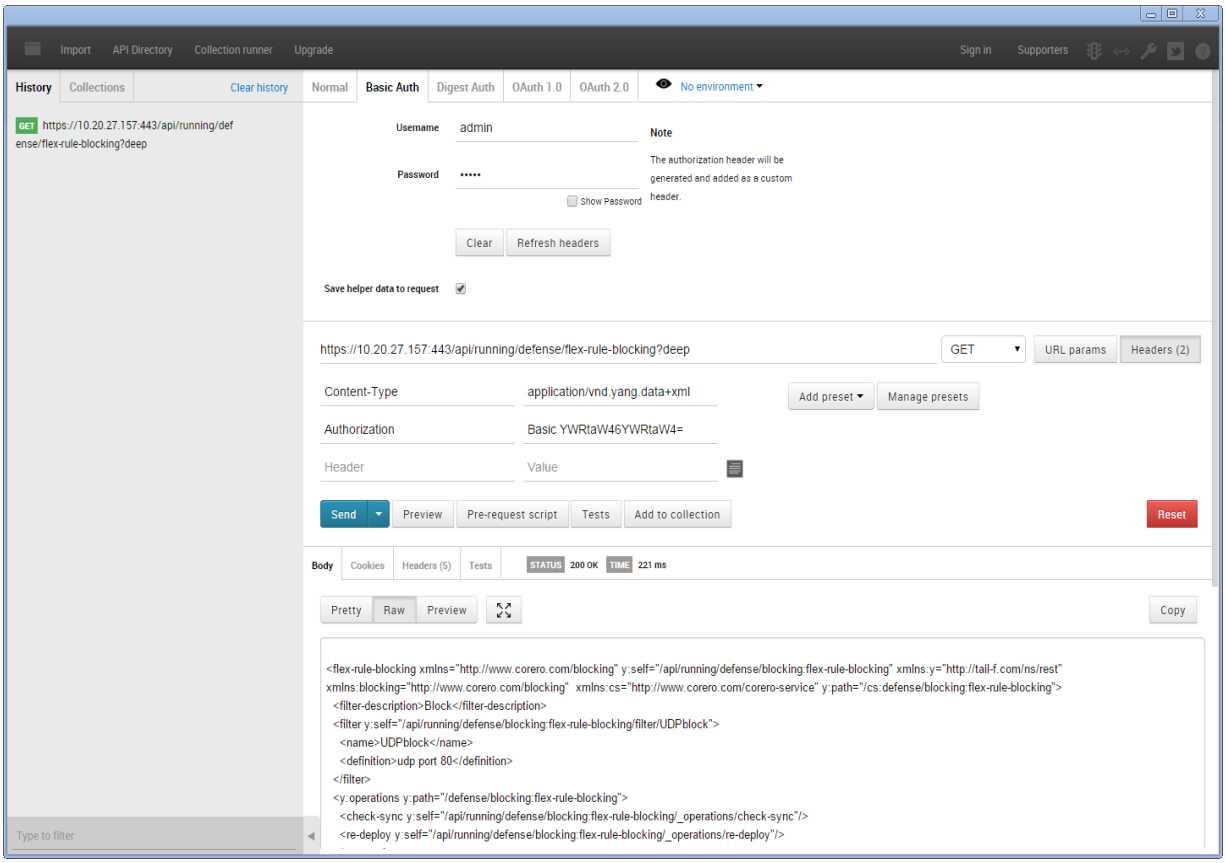
The examples in this section use the Postman extension for Google Chrome, but any REST client plug-in for any major web browser probably will provide the functionality you need. You can download the Postman REST client from Google's Chrome Web Store.

Using a REST Client

An example Postman screen:



A Postman screen with the Basic Authentication header shown:



One good way of becoming familiar with the data model is to use a REST client to perform a GET of some or all of an existing CMS configuration.

This example shows the use of the GET method to obtain the default AAA configuration from the CMS:

- Type `https://`**`cmsipaddress`**`:443/api/running/aaa?deep` in the URL field
- Choose **GET** as the method
- Specify the content type: `application/vnd.yang.data+xml`
- Click **SEND** to initiate the query.

The result will look approximately like this:

```
<aaa xmlns="http://tail-f.com/ns/aaa/1.1" xmlns:y="http://tail-
f.com/ns/rest" xmlns:aaa="http://tail-f.com/ns/aaa/1.1">
<authentication>
<users>
<user>
<name>admin</name>
<uid>9251</uid>
<gid>9251</gid>
<password>$1$zjFs9SD/$qt7GOBe9y87zo5f3psJAu/</password>
<ssh_keydir>/var/corero/homes/admin/.ssh</ssh_keydir>
<homedir>/var/corero/homes/admin</homedir>
</user>
<user>
```

```
<name>support</name>
<uid>9250</uid>
<gid>9250</gid>
<password>$1$dH2fOcDQ$wBECONVrKEN25GlSxpVIX/</password>
<ssh_keydir>/var/corero/homes/support/.ssh</ssh_keydir>
<homedir>/var/corero/homes/support</homedir>
</user>
</users>
</authentication>
<external-authentication xmlns="http://www.corero.com/extauth">
</external-authentication>
</aaa>
```

Use the results of the GET request in the previous example as a template for changing the password of the admin user:

```
<aaa xmlns="http://tail-f.com/ns/aaa/1.1" xmlns:y="http://tail-
f.com/ns/rest" xmlns:aaa="http://tail-f.com/ns/aaa/1.1">
<authentication>
<users>
<user>
<name>admin</name>
<uid>9251</uid>
<gid>9251</gid>
<password>adminpassword</password>
<ssh_keydir>/var/corero/homes/admin/.ssh</ssh_keydir>
<homedir>/var/corero/homes/admin</homedir>
</user>
</users>
</authentication>
</aaa>
```

This example uses the results of the GET request in the first example as a template for adding a new user named "supv":

```
<aaa xmlns="http://tail-f.com/ns/aaa/1.1" xmlns:y="http://tail-
f.com/ns/rest" xmlns:aaa="http://tail-f.com/ns/aaa/1.1">
<authentication>
<users>
<user>
<name>supv</name>
<uid>875</uid>
<gid>80</gid>
<password>supvpassword</password>
<ssh_keydir>/var/corero/homes/supv/.ssh</ssh_keydir>
<homedir>/var/corero/homes/supv</homedir>
</user>
</users>
</authentication>
</aaa>
```

Considerations For Using a REST Client

Some considerations to bear in mind when using a REST client such as Postman:

- Authenticate to your CMS first using the browser associated with your REST client. Also, make certain to enter your credentials in the **Basic Auth** tab.
- Type the URI and choose the HTTP method from the pulldown menu.
- Use the **raw** tab for viewing and composing content.
- Some REST clients may display more markup content than others. When using GET to retrieve resources to use as a template, you may find it best to edit out all markup that does not pertain explicitly to the resource you want to modify.

REST Examples

Some simple examples follow. These are intended to illustrate the execution of some different basic tasks using the REST API, and are not intended to demonstrate a comprehensive configuration.

The REST API examples provided in this help set's procedural topics use cURL for two reasons: 1.) it's a convenient way of including the REST command along with the corresponding web UI and CLI commands, and, 2.) the cURL syntax indicates each resource's position in the CMS data model more explicitly than the corresponding syntax entered into a REST client session. This, in turn, is part of why you may find it more comfortable and effective to use such a REST client, and Corero encourages you to do so, unless you are already familiar with cURL and prefer it.

In general, when using the REST API:

- For the REST API, **Save**/`commit` occurs automatically upon execution of each request.
- As with the web UI and the CLI, you can type an IP address range as a spread of IP addresses (200.1.1.1 - 200.1.1.255), with a subnet mask (200.1.1.0 255.255.255.0), or as an offset (200.1.1.0/24).

There are several things to note in these examples, specifically:

- For simplicity, each example shows the built-in admin user account being used along with the default admin account password of "admin", shown in cleartext. Make certain to take appropriate security precautions when using passwords for your own accounts.
- The examples uses the "running" resource, except for the one that demonstrates the display of defense statistics. That example uses the "operational" resource.
- Note the differences in the headers that are used for GET and POST/PATCH operations, specifying the format of each operation.

Perform a GET of all resources at a single level

This example shows how all resources at a single level are obtained at once; note that this request does not obtain subresources:

```
curl -k -v -u admin:admin -X GET https://
[cmsipaddress]:443/api/running/defense --header "Accept:
application/vnd.yang.data+json"
```

Perform a recursive GET of all resources

This example shows how all resources and their corresponding subresources are obtained at once. Note how the "?deep" parameter is appended to the GET request to cause it to be recursive:

```
curl -k -v -u admin:admin -X GET https://
[cmsipaddress]:443/api/running/site?deep --header "Accept:
application/vnd.yang.data+json"
```

Display a List of Connected Devices

This example shows the use of the GET method to display a list of devices connected to the CMS.

```
curl -k -v -u admin:smartwall -X GET
https://cmsipaddress:443/api/operational/site/topology-settings/device-
list --header "Accept: application/vnd.yang.data+json"
```

Enable/disable a rule

This example shows the PATCH method used to change the configuration of an existing rule. (It shows rule CNS-000006 being disabled.)

```
curl -k -v -u admin:admin -X PATCH
https://cmsipaddress]:443/api/running/defense/profiles/rule-
groups/packet-rules/rule/ -d '{"rule": {"rule-name": "cns-000006",
"admin-state": "disabled"} }' --header "Content-
Type:application/vnd.yang.data+json"
```

Define a new Flex Rule

This example shows how to define a new filter using Flex Rule. Note the use of the PATCH method, rather than POST.

```
curl -k -v -u admin:admin -X PATCH https://
[cmsipaddress]:443/api/running/defense/flex-rule-blocking/filter/ -d '
{"filter": {"name": "TCPblock", "definition": "tcp port 80", "admin-
state": "disabled"} }' --header "Content-
Type:application/vnd.yang.data+json"
```

Delete an existing Flex Rule

This example shows how to delete an existing Flex Rule:

```
curl -k -v -u admin:admin -X DELETE https://
[cmsipaddress]:443/api/running/defense/flex-rule-
blocking/filter/TCPblock/ --header "Content-
Type:application/vnd.yang.data+json"
```

Configure Smart Rule parameters

This example demonstrates Smart Rule configuration, using the PATCH method:

```
curl -k -v -u admin:admin -X PATCH https://
[cmsipaddress]:443/api/running/defense/early-access-capabilities/smart-
rule-parameters/ -d '{"smart-rule-parameters": {"udp-server-bit-rate-
limit": "5G", "udp-server-bit-threshold": "8G"} }' --header "Content-
Type:application/vnd.yang.data+json"
```

Display defense statistics

This example shows the use of the GET method and the operational resource to display defense statistics:

```
curl -k -v -u admin:admin -X GET https://
[cmsipaddress]:443/api/operational/defense/statistics?deep --header
"Accept: application/vnd.yang.data+json"
```

Display a specific statistic

This example shows the use of the GET method and the operational resource to display a single specific statistic, that for in-use suspicious addresses:

```
curl -k -v -u admin:admin -X GET https://
[cmsipaddress]:443/api/operational/defense/statistics/ip-address-
statistics/in-use-suspicious-addresses --header "Accept:
application/vnd.yang.data+json"
```

Configuring CMS Basic Behavior

Topics in this section explain how to configure frequently-used general functions on the appliance and CMS.

Refer to the topics in Analyzing Traffic and Blocking Threats for explanations of more specialized tasks.

Basic Configuration Checklist

Perform these procedures to define a typical basic configuration for your site. It may be that you don't need to configure every one of these features for your particular site, but this list comprises the basic, typical, recommended set of configuration tasks.

1. Add an Appliance to the configuration.
2. Obtain and install new software.
3. Configure user accounts and authentication.
4. Configure client profiles.
5. Configure server protection instances.
6. Block IP addresses by geography.

Make sure to save your configuration frequently throughout this process. At a minimum, you should save after completing each of these steps.

Adding a Corero Appliance to Manage

A Corero appliance has to be added to the site in order for the CMS to be able to manage it. Corero appliances include all members of the SmartWall product family, including SmartWall Network Threat Defense Appliance, SmartWall Network Threat Defense Appliance, SmartWall Threat Defense System, and SmartWall Threat Defense System.

Note that adding an appliance does not cause any existing configuration to be pushed to it. The configuration must be pushed to the new appliance using **Save** in the web UI or `commit` in the CLI. This is described in the instructions for saving your configuration.

Modify these settings to add a Corero appliance to a site:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Specify the new Corero appliance to add to the site, and specify the cluster and protection group to which the appliance belongs, as well as its position (physical location) relative to the other	Web UI: **Site Configuration** > **Cluster Configuration** > **Add** to display the Add a New Cluster dialog.	The web UI combines these tasks in the Add a New Cluster dialog. Specify the appliance and define the cluster and protection group to which it belongs, along with the appliance's position in the

Action	Where to perform it	What to do
appliances in the protection group.		equipment rack. • **Cluster Name** – Type a unique descriptive name for the cluster you are adding. • **The Cluster Has A** – Indicate whether your appliances operate individually or in a redundant (HA) configuration. • **Each Protection Group Looks Like** – Choose a representation of the set of appliances you are adding to the site. • **The Device Settings Are** – Type the IP address and a descriptive name for each appliance you are adding. For a Forensics appliance, click **Forensics Settings** to display a set of pop-up fields for specifying the IP address and subnet mask for port 3 and port 4. The protection group name is added automatically, and is based on the cluster name that you specified.
	CLI: `set devices device` **`descriptivename`** `address` **`ipaddress`** `module-type` **`moduletype`** `set site topology-settings cluster` **`clustername`**	The CLI requires you to use two separate commands to do this; one to add the appliance, and another to specify its cluster,

Action	Where to perform it	What to do
	`protection-group` **`groupname`** `module` **`devicename`**	protection group, and position in the equipment rack. Add the appliance using the `set devices` command, and specify the cluster, protection group, and position using the `set site topology-settings` command.
	REST API: `curl -k -v -u admin:smartwall -X POST https://` **`cmsipaddress`** `/api/running/devices -d '{"device": {"name": "`**`descriptivename`**`", "address": "`**`ipaddress`**`"} }' --header "Content-Type:application/vnd.yang.data+json"` `curl -k -v -u admin:smartwall -X PATCH https://`**`cmsipaddress`**`/api/running/cs:site/topology:topology-settings/cluster -d '{ "cluster": [ {"name": "`**`clustername`**`", "protection-group": [ {"name": "`**`protectiongroupname`**`", "module": [ { "name": "`**`devicename`**`", "position": 1 } ] } ] } ] }' --header "Content-Type:application/vnd.yang.data+json"`	As with the CLI, you need to execute two separate cURL commands to add the appliance and specify its cluster, protection group, and position in the equipment rack.
Note that adding an appliance does not cause any existing configuration to be pushed to it. The configuration must be pushed to the new appliance using **Save** in the web UI or `commit` in the CLI. This is described in the instructions for saving your configuration.		

Return to the Basic Configuration Checklist if you're configuring your site for the first time.

Save Your Configuration

Save your configuration frequently to ensure that the changes you make in the CMS are pushed to the appliance and are retained persistently.

Clicking **Save** in the web UI tool bar saves all configuration changes that have been made since the last save. By default, saving pushes the configuration changes to the Corero appliances managed by the CMS. The `commit` command in the CLI is equivalent to the **Save** button in the web UI.

Note: When using the REST API for configuration, **Save**/`commit` occurs automatically upon execution of each cURL command.

Return to the Basic Configuration Checklist if you're configuring your site for the first time.

Creating an IP Access Control List

Blocking/always allowing an IP address adds that IP address to a list of addresses that is checked before any other part of the corresponding packet is examined. A list of blocked IP addresses often is referred to as blacklist, while a list of always-allowed IP addresses often is referred to as a whitelist. A packet from a blocked IP address is discarded immediately, and no additional processing is performed on any portion of its contents. Similarly, a packet from an IP address that is to be allowed always is forwarded immediately, with no additional processing.

Note: For either a blacklist or a whitelist, administratively enabling/disabling the list affects the *rule* associated with the list, and any corresponding notification. Once an address is added to a list, traffic from that address is *always* blocked (blacklist) or allowed (whitelist), even if the list itself is administratively disabled.

You can type a range as a spread of IP addresses (200.1.1.1 - 200.1.1.255), with a subnet mask (200.1.1.0 255.255.255.0) or as an offset (200.1.1.0/24).

A range of blocked or always allowed addresses is allowed to contain another range completely, but partial overlaps are not allowed. As long as each subset of addresses is completely enclosed within its superset, it is possible to nest address ranges as you require. The example that follows shows a range of whitelisted addresses that is completely enclosed within a range of blacklisted addresses:

- 1.1.1.1 - 1.1.1.200 is blacklisted (blocking)
- 1.1.1.101 - 1.1.1.199 is whitelisted (always allow)

This is valid, since 1.1.1.1 - 1.1.1.200 completely contains 1.1.1.101 - 1.1.1.199, and addresses in the smaller range (1.1.1.101 - 1.1.1.199) are always allowed. If necessary, you could even blacklist a range of addresses within that whitelist; say, 1.1.1.151 - 1.1.1.159. Priority is given to the smallest (most specific) range that contains a particular IP address; therefore, 1.1.1.153 would be blocked, even though it is contained also by a superset address range that is always allowed.

Click **Defense Configuration** > **IP Access Control** to display the IP Access Control screen for adding one or more IP addresses to the list. Alternatively, type `set defense ip-access-control` in the CLI.

Add an IP address to the list to be blocked or always allowed:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Add one or more IP addresses to the list to be blocked.	Web UI: **Defense Configuration** > **IP Access Control** to display the IP Access Control screen, then click **Add** to specify the addresses to blacklist or whitelist. CLI: `set defense ip-access-`	Specify one or more IP addresses or address ranges. In the web UI, make sure to click **Create** to add the

Action	Where to perform it	What to do
	`control access-control-entry` **`listname`** `source-ip` **`ipaddress`** `access [always-allow | block] admin-state [disabled | enabled]` REST API: `curl -k -v -u admin:smartwall-X POST https://`**`cmsipaddress`**`/api/running/defense/blocking:ip-blocking/ -d '{"range": [{"ip-address": "`**`ipaddress`**`", "access": "always-allow"} ] } ' --header "Content-Type:application/vnd.yang.data+json"`	addresses to the list. Choose **Always Allow** or **Block** from the **Access** menu to indicate how the addresses will be handled. You can type a range as a spread of IP addresses (200.1.1.1 - 200.1.1.255), with a subnet mask (200.1.1.0 255.255.255.0) or as an offset (200.1.1.0/24).
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**/`commit` occurs automatically upon execution of the cURL command.		

Configuring Server Protection

Server protection comprises the set of responses that occur when incoming traffic matches one or more profiles.

Server protection is configured per server group; a server protection instance combines a server group (and the service protocol and profiles that it comprises) with default responses, called treatments, for traffic coming inbound to the servers and going back outbound to the clients. Due to the nature of their components, server protection instances incorporate all aspects of CMS protection configuration, including client groups, client profiles, and rule sets, each of which is a component of a server protection instance at some level.

Click **Defense Configuration** > **Server Groups and Protection** in the web UI tool bar to display the Server Groups and Protection screen. Alternatively, type the CLI command `set defense server-protection` to access the same functionality.

The CMS provides a single built-in server group named, "Default network-wide server protection". This is a default, catch-all server group that implicitly contains all server IP addresses that are not explicitly included in another server group. To be clear: a server IP address that is not explicitly included in a user-defined server group is implicitly included in the built-in server group named, "Default network-wide server protection". You cannot specify explicit IP addresses for this server group, and it has treatments associated with it only, not server profiles. It is necessary to define a server group explicitly to be able to use a server profile.

Modify these settings to configure server protection:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Create a server group.	Web UI: **Defense Configuration** >	Create the new server

Action	Where to perform it	What to do
	Server Groups and Protection in the web UI tool bar to display the Server Groups and Protection screen. Click **Add** to display the Add New Group dialog. CLI: `set defense server-protection server-group` **`customername servergroupname`** REST API: `curl -k -v -u admin:smartwall -X POST https://` **`cmsipaddress`** `/api/running/defense/server-protection:server-protection -d '{ "server-group": {"customer": "`**`customername`**`", "server-group-name": "`**`servergroupname`**`" } }' --header "Content-Type:application/vnd.yang.data+json"`	group by specifying an existing customer to associate with it (choose **DEFAULT** if you do not have other customers associate with your site) and typing a descriptive name. In the web UI, make sure to click **Add** in the Add New Group dialog to finish creating the server group.
Specify the addresses of the servers that belong to the server group, or members of any IP address range subsets that you want to define.	Web UI: Select the name of an existing server group in the Server Groups and Protection screen, then click **Add** in the **Address Ranges** tab to display the Add IP Address Range dialog. CLI: `set defense server-protection server-group DEFAULT` **`servergroupname`** `server-addresses` **`ipaddressrange`** REST API: `curl -k -v -u admin:smartwall -X POST https://` **`cmsipaddress`** `/api/running/defense/server-protection:server-protection -d '{ "server-group": {"customer": "`**`customername`**`", "server-group-name": "`**`servergroupname`**`" {"server-addresses": "`**`ipaddressrange`**`" } } }' --header "Content-Type:application/vnd.yang.data+json"`	Specify one or more server IP addresses to include in the server group. You can type individual addresses or a range of addresses (e.g., 2.2.2.1-2.2.2.100, or 2.2.2.1/24).
Configure the SYN flood threshold and connection	Web UI: Select the name of an existing server group in the Server Groups and	Choose an existing

Action	Where to perform it	What to do
limits for the server group. This is the server profile.	Protection screen, then click the **Thresholds & Limits** tab. CLI: `set defense profiles server-profiles [connection-limit-profile | syn-flood-profile]`	profile or specify custom values as necessary. The values are explained in the Server Group and Protection screen. Specify an existing profile by name or specify custom values as necessary. *Note that the CLI exposes that the threshold and limit values are actually divided between different types of profiles.* When using the REST API, note that both profile types must be created using the same profile name in order for the server profile to be associated with a server protection instance.
Specify the client group and protocol group from which the server group is being protected, and specify the treatment to be used for traffic from those clients. This involves client groups and service protocols that are configured separately.	Web UI: Select the name of an existing server group in the Server Groups and Protection screen, then click the **Server Protection** tab. Click **Add** to display the Add Traffic Treatment dialog.	Add Traffic Treatment dialog box: • Choose an existing protocol group • Choose an existing client group • Click **Next** to display the Protocol Group/Client Group dialog box Protocol Group/Client Group dialog box: • Choose an action for the server protection instance (**Allow** or **Block**). • Select an existing security rule set • Select an existing

Action	Where to perform it	What to do
		reputation rule set • Click **Issue Security Event** if you want to generate an event • Choose an event severity if you are generating an event • Click **OK** to finish defining the server protection instance
	CLI: `set defense server-protection server-group DEFAULT` **`servergroupname`** `service-protocol-group` **`serviceprotocolname`** `client-group` **`clientgroupname`** `treatment [action | admin-state | comments | event-severity | security-ruleset | issue-security-event | reputation-ruleset | traffic-pattern]`	It's possible to define the server protection instance in a single command line, though it's not necessary to do so. You can break up the configuration over multiple command lines if you find that easier. Specify the server group, then specify the service protocol and client group of interest, along with the treatment to be applied for traffic of that type originating from those clients.
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**/`commit` occurs automatically upon execution of the cURL command.		

Return to the Basic Configuration Checklist if you're configuring your site for the first time.

Configure Client Groups

A client group is a list of IP addresses or countries from which incoming traffic is handled the same way by the appliances.

Click **Defense Configuration** > **Client Groups** in the web UI tool bar to display the Client Groups screen for defining and managing client groups. Alternatively, you can configure client groups using the CLI command `set defense profiles client-group`.

The CMS provides a single built-in client group named, "IP addresses not in a specific group". This is a default, catch-all client group that implicitly contains all client IP addresses that are not explicitly included in another client group. To be clear: a client IP address that is not explicitly included in a user-defined client group is implicitly included in the built-in client group named, "IP addresses not in a specific group".

Modify these settings to configure a client group:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Create the client group.	Web UI: Click **Add** above the list of client group names to display the Add New Group dialog.	Type the client group name, specify its type (**IP Addresses** or **Geographical Locations**), and click **Add** to create the group.
	CLI: `set defense profiles client-group` **`clientgroupname`** `group-type [ip | geographic]`	Type the name of the client group and specify whether it is an IP group or a geographic group.
	REST API: `curl -k -v -u admin:smartwall -X POST https://`**`cmsipaddress`**`/api/running/defense/profiles/client-groups -d '{"client-group": {"name": "`**`clientgroupname`**`"} }' --header "Content-Type:application/vnd.yang.data+json"`	Specify the name of the client group.
Specify which IP addresses or countries are members of the client group, or members of any IP address range subsets that you want to define.	Web UI: Select the client group name and click the **Client Addresses and Ranges** tab, then click **Add** to display the Add IP Address Range dialog or the Add Country dialog (as appropriate for the client group type).	For an IP address client group, type the address or address range and click **Add** to include it in the group. For a geographic client group, choose one or more countries to include, then click **Add**.
	CLI: `set defense profiles client-group` **`clientgroupname`** `addresses` **`ipaddressrange`** OR `set defense profiles client-group` **`clientgroupname`** `countries` **`countryname`**	Type the IP addresses or country names to include as members of the client group.
	REST API: `curl -k -v -u admin:smartwall -X POST https://`**`cmsipaddress`**`/api/running/defense/profiles/client-groups/client-group/`**`clientgroupname`** `-d '`	

Action	Where to perform it	What to do
	`{"client-addresses": {"ip-address": "ipaddressrange"} } ' --header "Content-Type:application/vnd.yang.data+json"`	
Specify the threshold and limit values that make up the client profile.	Web UI: Select the client group name and click the **Thresholds & Limits** tab.	Choose an existing profile or specify custom values as necessary. The values are explained in the Client Groups screen.

	CLI:
	Specify an existing profile:
	`set defense profiles client-group clientgroupname assigned-profiles [connection-limit | packet-rate-limit | syn-flood]`
	OR create a custom profile:
	`set defense profiles client-profiles [connection-limit | packet-rate-limit | syn-flood] profilename`
		Specify an existing profile by name or specify custom values as necessary. *Note that the CLI exposes that the various threshold and limit values are actually divided between different types of profiles. Note also that profiles are assigned using a different command than that used to define them.*

	REST API: `curl -k -v -u admin:smartwall -X POST https://cmsipaddress /api/running/defense/profiles /client-profiles -d '{"syn-flood-profile": {"name": "profilename", "trusted-threshold": "60", "suspicious-threshold": "70", "suspicious-exit-threshold": "65", "malicious-threshold": "90", "proxy-requests-from-unknown-ip": "true"} }' --header "Content-Type:application/vnd.yang.data+json"`	
Optional: Export the client profile for re-use later.	Web UI: Click the link at the bottom of the **Thresholds & Limits** tab to display the Add New Shared Profile dialog. CLI: The CLI handles client profiles separately from client groups, so there's no	Type the name of the file to which you want to export the profile settings.

Action	Where to perform it	What to do
	reason to export the profile explicitly.	
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**/`commit` occurs automatically upon execution of the cURL command.		

Return to the Basic Configuration Checklist if you're configuring your site for the first time.

Configure Client Profiles

Client profiles define sets of characteristics that constitute acceptable traffic behavior from clients. There are three types of client profiles: SYN flood profiles, connection limit profiles, and packet rate limit profiles. A number of profiles of each type are built-in, and you can create new ones as you see necessary.

In the web UI, click **Manage Profiles** in the **Thresholds & Limits** tab of the Client Groups screen to display the Client Profiles screen for creating new profiles and modifying existing ones. Alternatively, you can configure client profiles using the CLI command `set defense profiles client-profiles`.

A client profile is assigned to a client group as part of client group configuration.

Follow these steps to configure client profiles:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Access the controls for defining client profiles.	Web UI: **Defense Configuration** > **Client Groups** to display the Client Groups screen. Select a client group, click the **Thresholds & Limits** tab, and click **Manage Profiles** to display the Client Profiles screen. CLI: `set defense profiles client-profiles` REST API: `curl -k -v -u admin:smartwall -X POST https://`**`cmsipaddress`**`/api/running/defense/profiles/client-profiles -d '{"syn-flood-profile": {"name": "`**`profilename`**`", "trusted-threshold": "60", "suspicious-threshold": "70", "suspicious-exit-threshold": "65", "malicious-threshold": "90", "proxy-requests-from-unknown-ip": "true"} }' --header "Content-Type:application/vnd.yang.dat`	This provides access to all the controls for defining client profiles. Note how the web UI combines this into client group configuration.

Action	Where to perform it	What to do
	`a+json"`	
Define the client profile that you need.	As above.	There are three types of client profile: • SYN flood • Connection limit • Packet rate limit Refer to the Client Groups screen for descriptions of the relevant controls. When using the REST API, note that all three profile types must be created using the same profile name in order for the shared profile to be associated with a client group.
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**/`commit` occurs automatically upon execution of the cURL command.		

Return to the Basic Configuration Checklist if you're configuring your site for the first time.

Block IP Addresses By Geography

It is possible to block IP addresses from one or more specific countries, or, otherwise, handle the packets originating from those countries in some special way. This combines elements of client group configuration with server protection instance configuration.

To block IP addresses based on their country of origin:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Define a geographic client group that includes the countries from which you want to block traffic.	Web UI: **Defense Configuration** > **Client Groups** to display the Client Group screen. CLI: `set defense profiles client-group` **`clientgroupname`** `group-type geographic countries` **`countryname`** REST API: `curl -k -v -u admin:smartwall -X POST`	Create a client group. Make sure it's a geographic client group that includes the countries you want to block.

Action	Where to perform it	What to do
	https://**cmsipaddress**/api/running/defense/profiles/client-groups -d '{"client-group": {"name": **"clientgroupname**", "group-type": "geographic", "countries": {"name": **"countryname**" } } }' --header "Content-Type:application/vnd.yang.data+json"	
Include that geographical client group in a server protection instance to block traffic from those countries to those servers.	Web UI: **Defense Configuration** > **Server Groups and Protection** to display the Server Groups and Protection screen. CLI: `set defense server-protection server-group DEFAULT` **`servergroupname`** `service-protocol-group` **`serviceprotocolgroupname`** `client-group` **`clientgroupname`** `treatment action drop` REST API: `curl -k -v -u admin:smartwall -X POST https://` **`cmsipaddress`** `/api/running/defense/server-protection:server-protection/server-group/` **`customername`** `%2C` **`servergroupname`** `/service-protocol-group/` **`serviceprotocolname`** `-d '{"client-group":{"name": "` **`clientgroupname`** `"}}' --header "Content-Type:application/vnd.yang.data+json"` Followed by: `curl -k -v -u admin:smartwall -X PATCH https://` **`cmsipaddress`** `/api/running/defense/server-protection:server-`	Configure a server protection instance that uses the geographic client group. Make sure the treatment Action is configured to drop packets from the client group.

Action	Where to perform it	What to do
	`protection/server-group/` **`customername`** `%2C` **`servergroupname`**`/service-protocol-group/` **`serviceprotocolname`**`/client-group/` **`clientgroupname`**`/treatment -d '{"treatment": {"action": "block", "admin-state": "enabled", "event-severity": "minimal", "security-ruleset": "All Rules Off", "reputation-ruleset": "All Reputation Rules Off", } }' --header "Content-Type:application/vnd.yang.data+json"`	
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**`/commit` occurs automatically upon execution of the cURL command.		

Return to the Basic Configuration Checklist if you're configuring your site for the first time.

The blocking of IP addresses by geography is closely related to the reputation data furnished by the Threat Update Service. Refer to the Threat Update screen description for information about configuring the Threat Update Service.

Configure User Accounts and Authentication

Specify user settings for your configuration by clicking **Site Configuration** > **Users & Authentication** in the web UI tool bar to display the AAA screen, or by typing `set aaa` in the CLI. Refer to AAA (Authentication, Authorization, and Accounting) Overview for a general description of the CMS's user account management.

Follow these steps to configure a user account on the CMS:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Specify an existing user account to modify or begin defining a new user account. Define the user's privileges and other information.	Web UI: **CMS** > **Users & Authentication** to display the AAA screen. Click **Authentication** > **Users** to display the Users screen. CLI: `set aaa authentication users user` **`username`** `set aaa authentication users`	Web UI: Click an existing user name, or click **Add** to begin defining a new user. CLI: Specify the username to access its properties, whether you are modifying an

Action	Where to perform it	What to do
	`user` **`username`** `[gid | homedir | password | ssh_keydir | uid]` REST API: `curl -k -v -u admin:smartwall -X POST https://`**`cmsipaddress`**`/api/running/aaa/authentication/users -d '{"user": {"name": "`**`username`**`", "uid":` **`useridvalue`**`, "gid":` **`groupidvalue`**`, "password": "`**`passwordstring`**`", "ssh_keydir": "/var/corero/homes/`**`username`**`/.ssh", "homedir": "/var/corero/homes/`**`username`**`"} }' --header "Content-Type:application/vnd.yang.data+json"`	existing user or creating a new one. Specify values for username, UID, GID, Password, SSH Keydir, and Homedir. These parameters are described in the User screen topic.
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**`/commit` occurs automatically upon execution of the cURL command.		

Return to the Basic Configuration Checklist if you're configuring your site for the first time.

Upgrade Corero Appliance Software

This topic describes the process of upgrading the software on any Corero appliance:

- SmartWall Network Threat Defense Appliance
- SmartWall Network Bypass Appliance
- SmartWall Network Forensics Appliance

Click **Site Configuration** > **Push files/images** in the web UI tool bar to display the Push Files/Images screen for upgrading the software of one or more of the Corero appliances in your site.

(The screen and the procedure are the same in the event that you ever decide to roll your appliance software back to an earlier version, as well.)

To upgrade Corero appliance software:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Upload the new software image to the CMS if you have not done so already.	Web UI: **Site Configuration** > **Push files/images**, then click the **Device Firmware** tab. Click **Add** to display the	Specify the location where the image you want to upload resides, then specify the file name you want the image to

Action	Where to perform it	What to do
	Upload Device Firmware dialog.	have on the CMS if you don't want to use the image's original name. Click **Upload** to execute the transfer. The software image is added to the list in the Device Firmware tab.
Push the software image to one or more Corero appliances in the site.	Web UI: In the Device Firmware tab, select the software image you want to distribute to one or more appliances.	Click Push selected file **to a single device** and select it from the pulldown menu, or click **all devices** to push it to every appliance in the site. *Recommended:* Click **Install and activate this image after transferring** if you want the new image to be loaded and used immediately. Click **Push File(s)** to execute the distribution of the selected software image.
Save your changes when you're finished. Web UI: **Save**		

NOTE: Make sure to **Save** your configuration after making any changes using the Push Files/Images screen to ensure that information about those changes is retained properly. For example, if you upload a file to the CMS for distribution to Corero appliances, but exit the CMS before saving the configuration change, the file will be present in the directory to which it was uploaded, but will not be shown in CMS configuration.

Configure Packet Capture

Using packet capture as part of your SmartWall Threat Defense System involves the addition of a Forensics appliance and configuration of network storage for collecting the capture data, as well as configuration of one or more queries for searching that data. This topic summarizes the overall process of configuring packet capture support for your site.

Add a Forensics Appliance To The Site

Follow the procedure for Adding a Corero Appliance to Manage, making sure that it is a Forensics appliance. This is done in the web UI by clicking **Site Configuration** > **Cluster Configuration**. When you add the Forensics appliance, there are some additional things you need to do to configure the appliance. As you add the appliance, click **Forensics Settings** above the image of the Forensics appliance to display a set of pop-up fields for specifying the IP address and subnet mask for port 3 and port 4 on the Forensics appliance. One of these ports will connect to the storage infrastructure used for collecting data, and the other will connect to a Defense appliance.

Configure Network Storage

Follow the procedure for configuring the network storage that will collect the packet capture data. This is done in the web UI by clicking **Site Configuration** > **Network Storage** to display the Network Storage screen.

Add the portal. Click **Add** to display the Portal screen. Type a user-defined name for the storage entity and click **Create**. In the Host field, type the IP address or DNS hostname for the device. Type a port as well, or accept the default of 3260. Click **Packet Capture** in the tool bar path to return to the Packet Capture screen [same as the Network Storage screen].

Add the storage volume. Click **Add** in the Storage section to display the Storage screen. Type the volume name; this needs to be the formal volume name, as obtained via a system command. Then click **Create** to add the volume and begin configuring it. Specify the Device name; this is the Forensics appliance from which the network storage device receives data. Choose the Type of the storage device; this is Iscsi or RAM. Type the size of the device, expressed in GB. For Target, type the name of the PCAP file to which the data will be written. Specify an Admin state for the device, either **Enabled** or **Disabled**.

Configure Queries

Follow the procedure for configuring one or more queries for searching the data. This is done in the web UI by clicking **Defense Monitoring** > **Capture Admin & Queries** to display the Capture Admin & Queries screen. Choose **Bypass** or **Capture** to set the Capture mode; **Bypass** is the default. Click **Add** to display the New Packet Capture Query dialog box.

Return to the Basic Configuration Checklist if you're configuring your site for the first time.

Configure a Service Protocol

A service protocol specifies how traffic corresponding to a particular application is handled by the appliance. A large number of service protocols are built-in to the CMS and provided by default. You can modify these built-in service protocols or define your own custom ones, as necessary. Some of these built-in service protocols correspond to legitimate applications, or to applications that otherwise are not originally intended to be attack vectors, while others correspond to specific well-known attack vectors.

Click **Defense Configuration** > **Service Protocols** in the web UI tool bar to display the Service Protocol screen for defining service protocols. Alternatively, type `set defense profiles service-protocol` in the CLI.

To configure a service protocol:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Name a new service protocol.	Web UI: **Defense Configuration** > **Service Protocols** to display the Service Protocol screen, then click **Add**. CLI: `set defense profiles service-protocol` **`serviceprotocolname`** REST API: `curl -k -v -u admin:smartwall -X POST`	In the web UI, click **Create** to add the service protocol to the list. Note the large number of predefined service protocols. You may find the one

Action	Where to perform it	What to do
	`https://`**`cmsipaddress`**`/api/running/defense/profiles:profiles/service-protocols -d '{"service-protocol": [ {"name": "`**`serviceprotocolname`**`" } ] }' --header "Content-Type:application/vnd.yang.data+json"`	that you need in the list already, without having to create a new one.
Specify the transport protocol and transport port, as well as a connection timeout.	Web UI: Click the service protocol name to edit it. CLI: `set defense profiles service-protocol` **`serviceprotocolname`** `transport-protocol` **`protocolname`** `transport-port` **`portnumber`** REST API: `curl -k -v -u admin:smartwall -X PATCH https://`**`cmsipaddress`**`/api/running/defense/profiles:profiles/service-protocols/service-protocol -d '{"service-protocol": {"name": "`**`serviceprotocolname`**`", "connection-timeout": `**`value`**`, "transport-protocol": "`**`value`**`", "transport-port": "`**`value`**`" } }' --header "Content-Type:application/vnd.yang.data`	Specify a transport protocol, connection timeout value, and transport port.
Configure any other service protocol parameters you want to change from their defaults.	Web UI: Click the service protocol name to edit it. CLI: `set defense profiles service-protocol` **`serviceprotocolname`**	See the parameter descriptions in the Service Protocol screen topic for details.
Save your changes when you're finished. Web UI: **Save** CLI: `commit`		

Configure a Service Protocol Group

A service protocol group collects one or more related service protocols so that they can be used conveniently with a server group in a server protection instance. Click **Defense Configuration** > **Service Protocol Groups** in the web UI tool bar to display the Service Protocol Group screen. Alternatively, type `set defense profiles service-protocol-group` in the CLI.

To configure a service protocol group:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Name the new service protocol group.	Web UI: **Defense Configuration** > **Service Protocol Groups** in the web UI tool bar to display the Service Protocol Group screen. CLI: `set defense profiles service-protocol-group` **`serviceprotocolgroupname`** REST API: `curl -k -v -u admin:smartwall -X POST https://`**`cmsipaddress`**`/api/running/defense/profiles/service-protocols -d '{"service-protocol-group":{"name":"`**`serviceprotocolgroupname`**`"}}' --header "Content-Type: application/vnd.yang.data+json"`	Type the name of the service protocol group to create it. In the web UI, click **Create** to finalize the addition of the service protocol group.
Include one or more service protocols in the group.	Web UI: Click the service protocol group name to edit it. The Service Protocol screen appears. CLI: `set defense profiles service-protocol-group` **`serviceprotocolgroupname`** `service-protocol` **`serviceprotocolname`**	Specify one or more existing service protocols from the dropdown menu. You can create a new service protocol if you need to.
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**/`commit` occurs automatically upon execution of the cURL command.		

Threat Update screen

The Threat Update screen enables you to configure the means by which the CMS will obtain updates to its threat definitions via Protection Packs and Intelligence Packs. You must have a valid Threat Update Service license to be able to obtain these updates.

Click **Defense Configuration** > **Threat Update Service** in the web UI tool bar to display the Threat Update screen. Access the same functionality in the CLI with the `set threat-update` command.

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

The Threat Update screen provides these controls:

- **License Key** – Type your Threat Update Service license key. The CLI equivalent for this is `set threat-update license-key`.
- **Threat Update Server Address** – This is set explicitly to "tus.corero-cns.com", and cannot be modified.

- **Threat Update Schedule** – Click this link to display the Threat Update Schedule screen for specifying the frequency with which the CMS will seek Threat Update Service updates from the server and push them to managed Defense appliances. You can choose to query and push updates every hour (the default), every day at a certain time, or at some custom interval. The CLI equivalent for this is `set threat-update threat-update-schedule`.
- License Information – This section displays information about the Threat Update Service license associated with your license key. This includes the license's expiration date, current status, and the number of days remaining until it expires.

Configure Customers

Customers are associated with server groups and server protection instances. Although it is necessary to have a customer as part of each server group definition, customers are not binding in any other respect, nor do they contribute to any other aspect of the configuration. There are Rank and Status parameters associated with customers, but these are provided for use at the user's discretion. The Rank and Status parameters provide another means for customers to be sorted, but they do not contribute to the definition of any other aspects of the configuration.

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Specify the name of the customer.	Web UI:**CMS** > **Customers**, then click **Add** to display the Customers screen. CLI:`set customers customer` **`customername`** REST API:`curl -k -v -u admin:smartwall -X POST https://` **`cmsipaddress`** `/api/running/customers -d '{"customer": {"id": "`**`customername`**`"} }' --header "Content-Type:application/vnd.yang.data+json"`	Specify the name of the customer.
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**/`commit` occurs automatically upon execution of the cURL command.		

Accessing Bypass Mode and Defense Mode Controls

Controls for configuring Bypass mode and Defense mode reside in different places for different contexts. Individual clusters can have their Bypass mode and Defense mode configured independently of the Global Bypass mode and Global Defense mode that are configured as the default behavior for the entire site.

Remember that Bypass mode is available only when a Bypass appliance is in use.

Global Bypass mode configures the default Bypass behavior for the entire site. Global Bypass mode supersedes Global Defense mode; if Global Bypass mode is set to Bypass, it doesn't matter how Global Defense mode is configured, because the attached Defense appliances won't receive any traffic. If

Global Bypass mode is set to Inline or Monitor, then Defense and Forensics appliances will still receive packets and handle them as they are configured to do so.

Bypass mode/Defense mode can be configured in the following contexts:

- *To configure the default Bypass mode for the entire site:* Choose **Site Configuration** > **Cluster Configuration** from the web UI tool bar and choose the desired option from the Global Bypass menu. The CLI equivalent is `set site topology-settings global-bypass-mode`.
  For the REST API, type: `curl -k -v -u admin:smartwall -X POST https://cmsipaddress/api/running/site/topology-settings -d '{"global-bypass-mode": "bypassmode" }' --header "Content-Type:application/vnd.yang.data+json"`
- *To configure the default Defense mode for the entire site:* Choose **Defense Configuration** > **Operations Settings** from the web UI tool bar, and choose the desired option from the Global Defense Mode menu. The CLI equivalent is `set defense operations-settings global-defense-mode`.
  For the REST API, type: `curl -k -v -u admin:smartwall -X POST https://cmsipaddress/api/running/defense/operations-settings -d '{"global-defense-mode": "defensemode" }' --header "Content-Type:application/vnd.yang.data+json"`
- *To configure the Bypass mode for a specific cluster:* Choose **Site Configuration** > **Cluster Configuration** from the web UI tool bar and select the cluster for which you want to configure the Bypass mode. Choose the desired option from the Bypass menu in the Edit Protection Group section of the screen. The CLI equivalent is `set site topology-settings cluster clustername protection-group protectiongroupname bypass-mode`.
  For the REST API, type: `curl -k -v -u admin:smartwall -X POST https://cmsipaddress/api/running/site/topology-settings/cluster/clustername/protection-group/protectiongroupname -d '{"bypass-mode": "bypassmode" }' --header "Content-Type:application/vnd.yang.data+json"`

Configure Network Storage

If you are using a Forensics appliance to support packet capture activities, you must configure the network storage resource that will collect the packet capture data. This is done in the web UI by clicking **Site Configuration** > **Network Storage** in the web UI tool bar to display the Network Storage screen. Alternatively, type `set site packet-capture storage` in the CLI.

Configure the storage resource:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Add the portal.	Web UI: **Site Configuration** > **Network Storage** to display the Network Storage screen. Click **Add** to display the Portal screen. CLI: `set site packet-capture portal portalname host ipaddress port portnumber`	Type a user-defined name for the storage entity and click **Create**. In the Host field, type the IP address or DNS hostname for the device. Type a port on which to

Action	Where to perform it	What to do
	REST API: `curl -k -v -u admin:smartwall -X POST https://` **`cmsipaddress`** `/api/running/site/packet-capture -d '{"portal": {"name": "portalname", "host": "hostipaddress", "port": "portnumber" } }' --header "Content-Type:application/vnd.yang.data+json"`	communicate with it, or accept the default of 3260.
Add the storage volume.	Web UI: Click **Add** in the Storage section of the Network Storage screen to display the Storage screen. CLI: `set site packet-capture storage` **`volumename`** `type iscsi size` **`value`** `device` **`devicename`** `target` **`formaltargetname`** `admin-state enabled` REST API: `curl -k -v -u admin:smartwall -X POST https://` **`cmsipaddress`** `/api/running/site/packet-capture/ -d '{"storage": ["volume-name": "`**`volumename`**`", "type": "iscsi", "size":` **`value`**`, "device": "`**`forensicsappliancename`**`", "target": "`**`formaltargetname`**`", "admin-state": "enabled" ] }' --header "Content-Type:application/vnd.yang.data+json"`	Type the volume name; this needs to be the formal volume name, as obtained via a system command. Then click **Create** to add the volume and begin configuring it. Specify the Device name; this is the Forensics appliance from which the network storage device receives data. Choose the Type of the storage device; this is **Iscsi** or **RAM**. Type the size of the device, expressed in GB. For Target, type the name of the PCAP file to which the data will be written. Specify an Admin state for the device, either **Enabled** or **Disabled**.
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: Save/commit occurs automatically upon execution of the cURL command.		

Configure Packet Capture Queries

Queries enable you to filter the set of packet capture data that is collected by the Forensics appliance. This is done in the web UI by clicking **Defense Monitoring** > **Capture Admin & Queries** to display the Capture Admin & Queries screen. Alternatively, type the `set forensics packet-capture` CLI command.

Define a packet capture query:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Set the packet capture mode.	Web UI: **Defense Monitoring** > **Capture Admin & Queries** to display the Capture Admin & Queries screen. CLI: `set forensics packet-capture administration capture-mode capture` REST API: `curl -k -v -u admin:smartwall -X POST https://`**`cmsipaddress`**`/api/running/forensics/packet-capture -d ' {"capture:packet-capture": {"administration": {"capture-mode": "capture" }' --header "Content-Type:application/vnd.yang.data+json"`	Set the Capture mode to **Capture**. The default setting is **Bypass**.
Define a query.	Web UI: Click **Add** to display the New Packet Capture Query dialog box. CLI: `set forensics packet-capture query-management query` **`queryname`** REST API: `curl -k -v -u admin:smartwall -X POST https://` **`cmsipaddress`** `/api/running/forensics -d ' {"capture:packet-capture": {query-management {"query": [ {"query-name": "`**`queryname`**`", "query-start-time": "`**`time`**`", "query-end-time": "`**`time`**`", "packet-length-min": "`**`value`**`", "packet-length-max": "`**`value`**`", "ip-protocol": "`**`protocol`**`", "flow-direction": "`**`direction`**`", "ethertype": "`**`type`**`", "query-batch-size": `**`value`**` } ] } } } ' --header "Content-Type:application/vnd.yang.data+json"`	Name the query. User-defined name. Specify the start and end times of the query. query-start-time and query-end-time Specify minimum and maximum packet lengths. packet-length-min, packet-length-max Choose the IP protocol. ip-protocol Choose the flow direction. flow-direction Choose the Ethertype. ethertype Specify the batch size. Specify a range of source IP addresses. Specify a range of destination IP addresses. Specify a range of source port numbers. Specify a range of

Action	Where to perform it	What to do
		destination port numbers.
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**/`commit` occurs automatically upon execution of the cURL command.		

If the storage device associated with a Forensics appliance lacks sufficient space to collect a new query, the query fails and an alarm is generated. If this happens, delete some old queries to free space for the new query.

Backing Up and Restoring a CMS Configuration

Clicking **CMS** > **CMS Backup/Restore** in the web UI tool bar displays the CMS Backup/Restore screen, providing controls for saving and reloading the CMS configuration. Configuration backups (backup bundles) are created for a specific CMS version.

Note: No more than one backup is retained on the CMS file system(s) at any one time.

Action	Where to perform it	What to do
Back up the current CMS configuration.	Web UI: Click **CMS** > **CMS Backup/Restore** in the web UI tool bar. CLI: `request backup create file` **`filename`**	Click **Create Backup** to execute the backup operation. The web UI will prompt you for confirmation that you really want to back up the configuration.
Restore the backed up CMS configuration.	Web UI: Click **CMS** > **CMS Backup/Restore** in the web UI tool bar. CLI: `request backup restore file` **`filename`**	Click **Restore Backup** to execute the restore operation. The web UI will prompt you for confirmation that you really want to restore the configuration. **NOTE:** The CMS will restart as soon as the restoration is complete.
Save your changes when you're finished. Web UI: **Save** CLI: `commit`		

Optional and Advanced CMS Configuration

This section comprises instructions for configuration tasks that are outside of the basic, typical configuration.

Configure Packet Rate Limiting

Packet rate limiting restricts the number of packets that servers can accept from one or more clients in a time interval. It is based on an upper limit on the number of packets that can be accepted from clients in a particular client group with a particular client profile. This is a credit-based mechanism in which the credits are replenished over time, so a client is not restricted perpetually if its traffic diminishes and falls within the configured limit.

In previous releases of Corero software, packet rate limiting was known as *client request rate limiting*, or CRRL.

Configuring packet rate limiting combines elements of client group configuration and service protocol configuration.

To configure packet rate limiting:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Assign client IP addresses to a client group.	Web UI: **Defense Configuration** > **Client Groups** to display the Client Group screen, then click the **Address Ranges** tab.	Select an existing client group or create a new one. Specify one or more singleton addresses or specify an address range to associate with the group.
	CLI: `set defense profiles client-groups client-group` **`clientgroupname`** `client-addresses` **`ipaddressrange`**	
	REST API: `curl -k -v -u admin:smartwall -X POST https://`**`cmsipaddress`**`/api/running/defense/profiles/client-groups/client-group/`**`clientgroupname`** `-d '{"client-addresses": {"ip-address": "`**`ipaddressrange`**`"} }' --header "Content-Type:application/vnd.yang.data+json"`	
Configure the packet rate limit profile.	Web UI: **Defense Configuration** > **Client Profile** to display the Client Group screen, then click the **Thresholds & Limits** tab.	The important parameter is **Maximum packet rate for any client in this group**. You can choose an existing shared profile from the pulldown menu and use the value that is configured there, or click **Use a custom profile** to set a new explicit value for the parameter.
	CLI: `set defense profiles client-profiles packet-rate-limit-profile` **`clientprofilename`** `limit` **`limitvalue`**	Specify an existing packet rate limit profile and use the limit value that is configured there, or

Action	Where to perform it	What to do
	REST API: `curl -k -v -u admin:smartwall -X PATCH https://`**`cmsipaddress`**`/api/running/defense/profiles:profiles/client-profiles/packet-rate-limit-profile -d '{"packet-rate-limit-profile": {"name": "`**`profilename`**`", "limit": `**`value`**`} }' --header "Content-Type:application/vnd.yang.data+json"`	define a new packet rate limit profile with a custom limit value.
Optional, for a custom client profile: Export the custom client profile for re-use later.	Web UI: Click the link at the bottom of the **Thresholds & Limits** tab to display the Add New Shared Profile dialog.	Type the name of the file to which you want to export the profile settings.
Enable request limiting.	Web UI: **Defense Configuration** > **Service Protocols**, and select an existing service protocol to edit it, or create a new one.	Enable **Request Limiting**. In the web UI, click the option in the Service Protocol screen; in the CLI, set the value of `request-limiting` to true.
	CLI: `set defense profiles service-protocol` **`serviceprotocolname`** `request-limiting true`	
Optional: Modify the costs for packet types. **NOTE:** Use caution when increasing the per-packet cost of any packet type. Increasing the cost of a specific packet type can increase the packet cost of a session significantly, and when you multiply that by many sessions, the total cost can accumulate quickly. This may result in legitimate traffic being blocked unintentionally.	Web UI: **Defense Configuration** > **Advanced Security Controls** to display the Advanced Controls screen.	Type costs for individual packet types.
	CLI: `set defense profiles advanced-controls packet-costs`	
Save your changes when you're finished. Web UI: **Save**CLI: `commit` REST API: **Save**/`commit` occurs automatically upon execution of the cURL command.		

Configure Client Connection Limiting

This procedure configures an absolute limit on the number of connections that servers can accept from a client group or from all clients.

Configuring client connection limiting is an element of client group configuration. Click **Defense Configuration** > **Client Groups** in the web UI tool bar to display the Client Group screen. Alternatively, type the CLI command `set defense profiles connection-limit-profile`.

To configure client connection limiting:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Configure the connection limit values for an existing client group.	Web UI: **Defense Configuration** > **Client Groups** in the web UI tool bar to display the Client Group screen. Select an existing client group and click the **Thresholds & Limits** tab.	The important parameters are **Maximum connections for this entire group** and **Maximum connections for any client in this group**. You can choose an existing shared profile from the pulldown menu and use the values that are configured there, or click **Use a custom profile** to set new explicit values for the parameters.
	CLI: `set defense profiles client-profiles connection-limit-profile` **`profilename`** `max-group-connections` **`value`** `max-member-connections` **`value`**	Specify an existing connection limit profile to use its values, or define a new connection limit profile and configure explicit values for `max-group-connections` and `max-member-connections`.
	REST API: `curl -k -v -u admin:smartwall -X PATCH https://`**`cmsipaddress`**`/api/running/defense/profiles:profiles/client-profiles/connection-limit-profile -d '{"connection-limit-profile": {"name":` **`"profilename"`**`, "max-group-connections":` **`value`**`, "max-member-connections":` **`value`** `} }' --header "Content-Type:application/vnd.yang.data+js`	

Action	Where to perform it	What to do
	`on"`	
Optional, for a custom client profile: Export the custom client profile for re-use later.	Web UI: Click the link at the bottom of the **Thresholds & Limits** tab to display the Add New Shared Profile dialog.	Type the name of the file to which you want to export the profile settings.
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API:**Save**/`commit` occurs automatically upon execution of the cURL command.		

Configure SYN Flood Rate Limiting

This procedure configures a limit on the number of SYN packets that servers can accept from one or more clients in a time interval.

Click **Defense Configuration** > **Client Groups** in the web UI tool bar to display the Client Group screen. Alternatively, type the CLI command `set defense profiles client-profiles syn-flood-profile`.

To configure SYN flood rate limiting:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Configure the SYN flood limit values for an existing client group.	Web UI: > **Client Groups** in the web UI tool bar to display the Client Group screen. Select an existing client group and click the **Thresholds & Limits** tab.	The important parameters are **Trusted client connection threshold**, **Suspicious client connection threshold**, **Suspicious client connection exit threshold**, and **Malicious client connection threshold**, as well as **Initial treatment of clients in this group**. These parameters indicate the counts of SYN packets at which clients are characterized as trusted, suspicious, or malicious. You can choose an existing shared profile from the pulldown menu and use the values that are

Action	Where to perform it	What to do
		configured there, or click **Use a custom profile** to set new explicit values for the parameters.
	CLI: `set defense profiles client-profiles syn-flood-profile profilename [malicious-threshold | suspicious-exit-threshold | suspicious-threshold | trusted-threshold | proxy-requests-from-unknown-ip].`	Specify an existing SYN flood profile to use its values, or define a new SYN flood profile and configure explicit values for these parameters as needed.
	REST API: `curl -k -v -u admin:smartwall -X PATCH https://`**`cmsipaddress`**`/api/running/defense/profiles:profiles/client-profiles/syn-flood-profile -d '{"syn-flood-profile": {"name": "`**`profilename`**`", "trusted-threshold": `**`value`**`, "suspicious-threshold": `**`value`**`, "suspicious-exit-threshold": `**`value`**`, "malicious-threshold": `**`value`**`, "proxy-requests-from-unknown-ip": "`**`booleanvalue`**`"} }' --header "Content-Type:application/vnd.yang.data+json"`	
Optional, for a custom client profile: Export the custom client profile for re-use later.	Web UI: Click the link at the bottom of the **Thresholds & Limits** tab to display the Add New Shared Profile dialog.	Type the name of the file to which you want to export the profile settings.
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API:**Save**/`commit` occurs automatically upon execution of the cURL command.		

Configure Server SYN Flood Protection

SYN flood protection can be configured for server protection instances using the SYN flood threshold parameter. Click **Defense Configuration** > **Server Groups and Protection** in the web UI tool bar to display the Server Groups and Protection screen. Alternatively, type the CLI command `set defense profiles server-profiles syn-flood-profile`.

Configure server SYN flood protection:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Configure the SYN flood limit for the server protection instance.	Web UI: **Defense Configuration** > **Server Groups and Protection** in the web UI tool bar to display the Server Groups and Protection screen. Select an existing server protection instance and click the **Thresholds & Limits** tab.	The important parameter is **SYN flood threshold**, which is the maximum allowable number of open SYN requests to servers in this protection instance. You can choose an existing profile from the pulldown menu and use the value that is configured there, or click **Use a custom profile** to set a new explicit value for this parameter.
	CLI: `set defense profiles server-profiles syn-flood-profile` **`profilename`** `connection-threshold` **`thresholdvalue`**	Specify the value of the SYN flood threshold via a SYN flood profile and the `connection-threshold` parameter.
	REST API: `curl -k -v -u admin:smartwall -X PATCH https://`**`cmsipaddress`**`/api/running/defense/profiles:profiles/server-profiles/syn-flood-profile -d '{"syn-flood-profile": {"name": "`**`synfloodprofilename`**`", "connection-threshold": `**`value`**` } } ' --header "Content-Type:application/vnd.yang.data+json"`	
Associate the SYN flood limit with an existing server group.	CLI: `set defense server-protection DEFAULT` **`servergroupname`** `assigned-profiles syn-flood` **`profilename`**	Associate the SYN flood profile with the corresponding server protection instance by specifying the server group name and the SYN flood profile name. This is not necessary when using the web UI, since the profile is associated in context.
	REST API: `curl -k -v -u admin:smartwall -X PATCH https://`**`cmsipaddress`**`/api/running/defense/server-protection:server-protection/server-group/`**`customername`**	

Action	Where to perform it	What to do
	`%2C`**`servergroupname`**`/assigned-profiles -d '{"assigned-profiles": {"syn-flood": "`**`profilename`**`" } }' --header "Content-Type:application/vnd.yang.data+json"`	
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**`/commit` occurs automatically upon execution of the cURL command.		

Configure Server Connection Limiting

Click **Defense Configuration** > **Server Groups & Protection** in the web UI tool bar to display the Server Groups and Protection screen. Alternatively, type the CLI command `set defense profiles server-profiles connection-limit-profile`.

Configure server connection limiting:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Configure the maximum number of connections allowable for servers in the server protection instance.	Web UI: **Defense Configuration** > **Server Groups & Protection** in the web UI tool bar to display the Server Groups and Protection screen. Select an existing server protection instance and click the **Thresholds & Limits** tab.	The important parameters are **Maximum connections for this entire group** and **Maximum connections for any server in this group**. You can choose an existing profile from the pulldown menu and use the values that are configured there, or click **Use a custom profile** to set new explicit values for these parameters.
	CLI: `set defense profiles server-profiles connection-limit-profile` **`connectionlimitprofilename`** `max-group-connections` **`limitvalue`** `max-member-connections` **`limitvalue`**	Specify an existing connection limit profile or define a new one to set the values of the `max-group-connections` and `max-member-connections`

Action	Where to perform it	What to do
	REST API: `curl -k -v -u admin:smartwall -X PATCH https://`**`cmsipaddress`**`/api/running/defense/profiles:profiles/server-profiles/connection-limit-profile -d '{"connection-limit-profile": {"name": "`**`connectionlimitprofilename`**`", "max-group-connections": `**`value`**`, "max-member-connections": `**`value`**` } }' --header "Content-Type:application/vnd.yang.data+json"`	parameters.
Associate the connection limits with an existing server group.	CLI: `set defense server-protection DEFAULT `**`servergroupname`**` assigned-profiles connection-limit-profile `**`connectionlimitprofilename`**	This command associates the connection limit profile with the server protection instance. This is not necessary when using the web UI, since the profile is associated in context.
	REST API: `curl -k -v -u admin:smartwall -X PATCH https://`**`cmsipaddress`**`/api/running/defense/server-protection:server-protection/server-group/`**`customername`**`%2C`**`servergroupname`**`/assigned-profiles -d '{"assigned-profiles": {"connection-limit": "`**`connectionlimitprofilename`**`" } }' --header "Content-Type:application/vnd.yang.data+json"`	
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**`/commit` occurs automatically upon execution of the cURL command.		

Block IP Headers

You can block traffic from entering your network based on its IP header type.

Click **Defense Configuration** > **Advanced Security Controls** to display the Advanced Security Controls screen. Alternatively, configure header blocking using the CLI command `set defense header-blocking`.

To block by IP header type:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Specify the header types that you want to block.	Web UI: **Defense Configuration** > **Advanced Security Controls** to access the packet header types. CLI: `set defense header-blocking [ip-tunnels | ipv4-packets | ipv6-packets] [allow | block]` REST API: `curl -v -k -u admin:smartwall -X POST https://` **`cmsipaddress`** `/api/running/defense/header-blocking -d '{"ipv6-packets": "block"}' --header "Content-Type:application/vnd.yang.data+json"`	Specify **Block** for any header type that you want to exclude. Note that all header types are allowed by default. The REST API example shows how to set a header type to "block"; IPv6 in this case.
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**/`commit` occurs automatically upon execution of the cURL command.		

Configure ICMP Rate Limiting

ICMP rate limiting is configured as a packet cost.

To configure ICMP rate limiting:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Specify the cost for ICMP packets.	Web UI: **Defense Configuration** > **Advanced Security Controls** in the web UI tool bar to display the Advanced Security Controls screen. CLI: `set defense profiles advanced-controls packet-costs icmp-packets` **`value`** REST API: `curl -k -v -u admin:smartwall -X PATCH https://` **`cmsipaddress`** `/api/running/defense/profiles:profiles/advanced-controls/packet-costs -d '{"packet-costs": {"icmp-packets":` **`value`** `} }' --header "Content-Type:application/vnd.yang.data+json"`	Type a packet cost value in the **ICMP Packets** field.
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: **Save**/`commit` occurs automatically upon execution of the cURL command.		

Configure Rule Sets

A rule set is a named collection of rules that is configured to identify and address traffic for a particular purpose. Rule sets provide basic responses to traffic that matches certain characteristics. User-defined rule sets can be created from built-in rule sets, and existing rule sets can be modified. The CMS divides rule sets broadly into two categories: security rule sets and reputation rule sets.

You can configure the treatment for a particular rule to be different for different rule sets, and you can modify the treatment applied to individual rules within a set. A treatment modification applies only to the instance of the rule associated with that named rule set.

Follow this procedure to configure security rule sets or reputation rule sets. Note that instructions for creating new rule sets are included, also.

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Access the type of rule set you want to work with: Security or Reputation.	Web UI: **Defense Configuration** > **Security Rule Sets** OR **Defense Configuration** > **Reputation Rule Sets** to display the corresponding Rule Set screen. CLI: `set defense profiles rule-groups [security-rule-set | ip-reputation-rule-set] rulesetname`	Web UI: Click the rule set name to display its parameters in their own screen. CLI: Specify the rule set name to access its parameters.
Modify the rule set parameters as necessary.	Web UI: Use the appropriate Rule Set screen as described above. CLI: `set defense profiles rule-groups [security-rule-set | ip-reputation-rule-set] rulesetname`	Modify parameters as necessary. A subset of rule set parameters is available for modification: • Block Active • Default Log • Description • Rules
Optional: Create a new rule set. You can create the new rule set as a copy of an existing one.	Web UI: Click **Add** in the Rule Set screen to display controls for creating a new rule set. CLI: Type the name of a new rule set to create it.	Configure the various rule set parameters. See the Security Rule Set screen or the IP Reputation Rule Set screen for the complete list.
Optional: Specify which rules should be enabled or disabled in the rule set.	Web UI: Click **Rules** in the screen for a particular rule set to display the Rule screen.	Specify a rule in the rule set and change its admin state to **Enabled** or **Disabled** as necessary.

Action	Where to perform it	What to do
	CLI: `set defense profiles rule-groups security-rule-set` **`rulesetname`** `rules rule` **`rulenumber`** `admin-state [enabled | disabled]`	
Save your changes when you're finished. Web UI: **Save** CLI: `commit`		

Configure Redundancy (HA) For Corero appliances

Execute these steps to configure CMS high availability.

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Specify the virtual IP address that will float across the two CMS instances.	Web UI: **CMS** > **High Availability** to display the High Availability screen. CLI: `set high-availability virtual-ip-address` **`virtualipaddress`** REST API: `curl -k -v -u admin:smartwall -X POST https://` **`cmsipaddress`** `/api/running/high-availability -d '{"virtual-ip-address":` **`virtualipaddress`** `}' --header "Content-Type:application/vnd.yang.data+json"`	Specify the virtual IP address that will float across the two CMS instances.
Save your changes when you're finished. Web UI: **Save** CLI: `commit` REST API: Save/commit occurs automatically upon execution of the cURL command.		

Configuring an External Authentication Regime

To access the controls for configuring an external authentication regime, click **CMS** > **Users** in the web UI tool bar to display the Users screen, then click **External Authentication** to display the External Authentication screen. The same functionality is accessible in the CLI by using the `set aaa external-authentication` command in configuration mode.

Configure an external authentication regime:

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

Action	Where to perform it	What to do
Specify the authentication regime to use.	Web UI: In the **Authentication Type** section, choose **Active Directory** or **RADIUS** from the pulldown menu. CLI: `set aaa external-authentication [active-directory | radius]`	If you specified **Active Directory**, there is no additional configuration. If you specified **RADIUS**, type the shared secret to use.
Specify the authentication server to use.	Web UI: In the **Server** section,type the server name and .choose its administrative state, **Enabled** or **Disabled**. CLI: `set aaa external-authentication server` **`servername`** `admin-state [enabled | disabled]`	Specify the DNS name or IP address of the server and its administrative state.
Map external user groups to as CMS user groups as necessary.	Web UI: Use the **Group Mapping** section and the **Reject If No Group** option. CLI: `set aaa external-authentication reject-if-no-group [false | true]` `set aaa external-authentication group-mapping [admin | defense | monitor | no-access | support]` **`externalgroupname`**	Select **Reject If No Group** if you want to deny all access to users if their profile does not belong to one of the groups mapped here. This is selected by default. Specify group name equivalents.
Save your changes when you're finished. Web UI: **Save** CLI: `commit`		

View CMS Version Information

To display software version information for the CMS:

- Web UI: Click **CMS** > **System Info** in the web UI tool bar.
- CLI: Type the CLI command `show system`
- REST API: Type `curl -k -v -u admin:smartwall -X GET https://`**`cmsipaddress`**`/api/operational/system --header "Accept: application/vnd.yang.data+json"`

Reference Information

Reference information comprises a variety of detailed descriptions and definitions that give in-depth background for the overviews and step-by-step instructions that appear in other sections.

- The Concepts and Explanations section provides detailed explanations of how the appliance and CMS work.
- The Screen and Command Descriptions section describes the controls that appear in each CMS screen.
- The Glossary provides brief definitions of terms that have special meaning in the behavior of the Corero appliance and CMS.
- The Configuration Data Models section provides detailed descriptions of the sets of elements that enable you to configure the appliance and the CMS.
- The Command Line Interface section provides an introduction to the CMS CLI.

Descriptions of how to execute tasks in the CMS and configure the appliance appear in non-reference sections such as Configuring Basic Behavior, Analyzing Traffic, and Blocking Threats.

Concepts and Explanations

Topics in this section describe in detail how the appliance and CMS work.

Descriptions of the controls that appear in each CMS screen are available in Screen and Command Descriptions.

Descriptions of how to execute tasks in the CMS and configure the appliance appear in sections such as Configuring Basic Behavior, Analyzing Traffic, and Blocking Threats.

AAA (Authentication, Authorization, and Accounting) Overview

The closely related functions authentication, authorization, and accounting are referred to collectively as "AAA." AAA configuration is described in Configuring User Accounts and Authentication; the controls used to perform AAA administration are described in the Users & Authentication screen.

AAA configuration defines how individual users and groups of users are enabled to access the CMS and managed devices, as well as which actions they are allowed to execute once they gain access.

The authentication described here is local authentication; that is, authentication based on credentials specified in the CMS configuration itself, rather than credentials specified in an external authentication regime such as Active Directory or RADIUS. By default, the CMS looks for authentication credentials locally, first, before looking for credentials at an external server.

In any case, whether the authentication regime in use is local or external, once a user authenticates successfully, that user is matched to a group. The user's access privileges are determined from the authorizations defined for the group.

The CMS supports a set of groups that correspond to essential roles:

- admin – Access all aspects of the data models required to manage users, devices, data, etc. This group is intended for administrators who require privileges to define and change all aspects of the configuration. Note that admin users can change passwords for all accounts, including support users.
- defense – Configure server defense policies and view configurations, statistics, and reports.

- monitor – View configurations, statistics, and reports. No configuration actions are allowed.
- support – This group is reserved for use by Corero customer support. Similarly, screens and commands that are available to support users are reserved for use by Corero customer support, and are not described in this help set. Note that admin users can change passwords for support users.

The CMS comes with two predefined users; admin and support, each corresponding to the role of the same name.

The privileges of the various user roles are summarized here:

Operation	Admin	Defense	Monitor
Manage user accounts and roles: create, delete, and disable user accounts.	Yes	No	No
Perform software upgrades	Yes	Yes	No
Device and topology configuration: network level setup, load balancing, HA	Yes	Yes	No
Defense configuration: server protection instances, PCAP filters, bypass mode changes	Yes	Yes	No
View configuration: statistics/dashboards, generate reports, create support packages. Change own password.	Yes	Yes	Yes
Linux shell access	Yes	No	No

Bypass Mode and Defense Mode

This topic describes the options that can be configured for Bypass mode/Defense mode.

Refer to Network Bypass Behavior for a full description of how the network bypass feature works.

Refer to Accessing Bypass Controls for instructions on where to access the Bypass mode and Defense mode controls in different configuration contexts.

The SmartWall Threat Defense System provides two configurable modes for controlling how individual clusters or the site as a whole handle serverbound traffic as they encounter it. *Defense mode* controls the behavior of Defense appliances, and *Bypass mode* controls the behavior of a Bypass appliance, if one is in use. The two modes comprise settings that control whether incoming traffic should be inspected and treated according to configured rules (Mitigate/Inline), inspected and logged according to configured rules without being treated (Monitor), or simply passed through without being touched (Bypass).

The Bypass/Defense mode can be set to one of the following states. By default, the mode is set to **Monitor** for newly added appliances and clusters. You must change the mode explicitly for appliances to operate in Mitigate/Inline mode.

- Mitigate – Inspect all traffic and treat it as specified by the server protection instances that are in effect. Specifically, for each appliance type:
  - Bypass appliance – Mitigate mode is called Inline mode for the Bypass appliance. The Bypass appliance passes all traffic through, leaving it to be handled by the attached devices as they are configured to do.

  - Forensics appliance - Apply filters and record activity before passing the traffic through to attached devices.
  - Defense appliance - Inspect all traffic and pass through only good traffic, as defined by the server protection instances in effect.
- Monitor – Pass traffic through to the servers without treating it, but log the way that server protection instances would respond to the traffic if the operational mode was set to Mitigate. Specifically, for each appliance type:
  - Bypass appliance – Copy traffic to attached devices.
  - Forensics appliance - Apply filters without recording activity, and pass the traffic through to attached devices.
  - Defense appliance – Inspect all traffic and pass it through.
- Bypass – Pass traffic through to the servers without treating it, completely untouched, without any logging or event reporting. *This mode is available only if a Bypass appliance is in use.* Specifically, for each appliance type:
  - Bypass appliance – Pass through all traffic. In the event of power loss, this happens automatically as a result of the Bypass appliance's passive optical passthrough capability.
  - Forensics appliance - Pass through all traffic without applying any filters or recording any activity.
  - Defense appliance – Pass through all traffic without inspecting or treating any of it.

Depending on the context in which you are configuring the mode, two other options may be available:

- **Inherit Global** – This option causes a cluster to use the Global Bypass/Global Defense setting, whatever that is configured to be.
- **Auto** – The Bypass appliance determines whether it can be in Inline mode or Monitor mode, or needs to be in Bypass mode, depending on whether it is receiving heartbeat from the devices attached to it.

Network Bypass Behavior

The SmartWall Threat Defense System includes the Corero SmartWall™ Network Bypass Appliance, which provides the site with full network bypass capability. The Bypass appliance and the Bypass operating mode that it supports ensure that your server infrastructure will be accessible to users at all times, without interruption, even in the event of power failure, unpredictable equipment behavior, or planned maintenance interval.

Automatic Bypass vs. Manual Bypass

Bypass mode can be activated manually, by a user configuring Bypass mode explicitly through the CMS. A site or cluster also can be configured for automatic bypass; this occurs when the Bypass appliance stops receiving heartbeat from attached devices, and it switches from Inline or Monitor mode to Bypass mode, passing all incoming traffic along to the server infrastructure unimpeded, without it being inspected or treated in any way.

Bypass Behaviors

The Bypass appliance supports the following behaviors:

- Inspection/Inline – This is Mitigate mode. Traffic is bridged between the Bypass appliance's inbound network port and its inbound inspection port. The traffic is handed off to the attached appliances, and, once their handling of the traffic is complete, the treated traffic re-enters the bypass appliance at its outbound inspection port, where it is bridged to the outbound network port for being passed to the server infrastructure.
- Pure Bypass – This is Bypass mode. Traffic passes through the Bypass appliance directly from one network port to the other, without being passed to attached appliances, or being handled or treated in any other way.
- Monitored Bypass (Tap) – This is Monitor mode. Traffic passes straight through the Bypass appliance, but is copied also to the inbound inspection port so that it can be handled by attached appliances. This is the auto Bypass mode.

Accessing Bypass Mode and Defense Mode Controls

Controls for configuring Bypass mode and Defense mode reside in different places for different contexts. Individual clusters can have their Bypass mode and Defense mode configured independently of the Global Bypass mode and Global Defense mode that are configured as the default behavior for the entire site.

Remember that Bypass mode is available only when a Bypass appliance is in use.

Global Bypass mode configures the default Bypass behavior for the entire site. Global Bypass mode supersedes Global Defense mode; if Global Bypass mode is set to Bypass, it doesn't matter how Global Defense mode is configured, because the attached Defense appliances won't receive any traffic. If Global Bypass mode is set to Inline or Monitor, then Defense and Forensics appliances will still receive packets and handle them as they are configured to do so.

Bypass mode/Defense mode can be configured in the following contexts:

- *To configure the default Bypass mode for the entire site:* Choose **Site Configuration** > **Cluster Configuration** from the web UI tool bar and choose the desired option from the Global Bypass menu. The CLI equivalent is `set site topology-settings global-bypass-mode`.
  For the REST API, type: `curl -k -v -u admin:smartwall -X POST https://cmsipaddress/api/running/site/topology-settings -d '{"global-bypass-mode": "bypassmode" }' --header "Content-Type:application/vnd.yang.data+json"`
- *To configure the default Defense mode for the entire site:* Choose **Defense Configuration** > **Operations Settings** from the web UI tool bar, and choose the desired option from the Global Defense Mode menu. The CLI equivalent is `set defense operations-settings global-defense-mode`.
  For the REST API, type: `curl -k -v -u admin:smartwall -X POST https://cmsipaddress/api/running/defense/operations-settings -d '{"global-defense-mode": "defensemode" }' --header "Content-Type:application/vnd.yang.data+json"`
- *To configure the Bypass mode for a specific cluster:* Choose **Site Configuration** > **Cluster Configuration** from the web UI tool bar and select the cluster for which you want to configure the

Bypass mode. Choose the desired option from the Bypass menu in the Edit Protection Group section of the screen. The CLI equivalent is `set site topology-settings cluster` **`cluster-name`** `protection-group` **`protectiongroupname`** `bypass-mode`.
For the REST API, type: `curl -k -v -u admin:smartwall -X POST https://`**`cmsipaddress`**`/api/running/site/topology-settings/cluster/`**`clustername`**`/protection-group/`**`protectiongroupname`** `-d '{"bypass-mode": "`**`bypassmode`**`" }' --header "Content-Type:application/vnd.yang.data+json"`

Bypass Appliance Best Practices

This topic summarizes some best practices to employ and some ideas to consider when using a SmartWall Network Bypass Appliance in your site.

Modes of operation:

- Bypass mode: Bypass mode provides connectivity protection to your site in the event of a power outage, or the need to perform maintenance or experiments on other SmartWall appliances. Traffic passes through the Bypass appliance directly from one network port to the other, without being passed to attached appliances, or being handled or treated in any other way.
- Inline mode: Traffic is bridged between the Bypass appliance's inbound network port and its inbound inspection port. The traffic is handed off to the attached Corero appliances, and, once their handling of the traffic is complete, the treated traffic re-enters the bypass appliance at its outbound inspection port, where it is bridged to the outbound network port for being passed to the server infrastructure.
- Monitor mode: Traffic passes straight through the Bypass appliance, but is copied also to the inbound inspection port so that it can be handled by attached Corero appliances. The Bypass appliance drops the treated traffic at its outbound network port, preventing the treated traffic from entering the network. This mode is useful for non-intrusive evaluation and experiments for which you want to have visibility into the behavior of the SmartWall Threat Defense System without allowing the treated traffic into your network.

You can upgrade the software for any Corero appliance in any of these modes.

The optical bypass mode should used solely in the event of a power failure. It is possible to test the optical bypass mode by setting the **Optical Bypass** option in the Cluster Configuration screen's Bypass Mode menu, but this option should not be used for any purpose other than verifying that the optical bypass mode works as it should.

After a power failure has occurred and power has been restored, the Bypass appliance will come back up in Bypass mode (switched bypass, not optical bypass). You will need to change the mode explicitly to Inline or Monitor once you have determined that the rest of your network has recovered properly.

Threat Update Service

The CMS includes access to the Threat Update Service. Threat Update Service is an automated protection update program that provides Corero customers with advanced security services to maximize security, availability, and performance of their network. It offers proactive protection from zero-day threats and resolution to security issues.

Server Protection Overview

All SmartWall Threat Defense System functionality exists to protect your servers, but the term "server protection," used explicitly within the CMS, refers to the set of responses (treatments) that are configured to occur when incoming traffic matches one or more profiles associated with a service protocol for a server group. Server protection is configured per server group, and a server protection instance combines a server group (and the service protocol and profiles that it comprises) with default responses, called treatments, for traffic coming inbound to the servers and going back outbound to the clients. As a result of the nature of their components, server protection instances incorporate all aspects of CMS protection configuration, including client groups, client profiles, and rule sets, each of which is a component of a server protection instance at some level.

The default traffic treatment is **Default network-wide server protection**, and it is used when no other traffic treatment has been configured. This default treatment handles traffic as being Random (asymmetric), with all rules off for both security rule sets and reputation rule sets.

Server protection is configured using the Server Protection screen.

How Traffic is Evaluated Against a Server Protection Instance

Server protection instances specify the criteria by which traffic coming in to your servers from clients outside your firewall is evaluated for being treated by the Defense appliance. A server protection instance comprises a server group (a set of server IP addresses collected together for common traffic treatment), one or more service protocols, and one or more client groups. As traffic is encountered, it is evaluated against the definition of the server protection instance in this order:

1. server IP address
2. service protocol
3. client group

A server IP address can belong to a single server group only, and this guarantees that inbound traffic will match some permutation of the server protection instance conclusively. The matching traffic will be allowed, dropped, or blocked, as specified by the server protection instance.

The table below describes the order in which incoming traffic is evaluated against the server protection instance to identify a match with a treatment. The order of evaluation goes left-to-right, then top-to-bottom.

Order	Server IP	Service Prot	Client Group
1	Specific	Specific	Specific
2	Specific	Specific	Any
3	Specific	Any	Specific
4	Specific	Any	Any
5	Any	Specific	Specific
6	Any	Specific	Any
7	Any	Any	Specific
8	Any	Any	Any

"Any" functions as a catch-all or wildcard. For any component of the server protection instance (server IP address, service protocol, or client group) it is evaluated for a match *last*, after every explicit definition has been tried.For server groups, it corresponds to "Default network-wide server protection", and for client groups, it corresponds to "IP addresses not in a specific group". In the table above, incoming traffic that matches a specific server IP address, a specific service protocol, and a specific client group is matched with a traffic treatment immediately. Incoming traffic that does not match any specific server IP address, service protocol, or client group receives the default, last-resort treatment that is specified for traffic that matches no specific component of the server protection instance. Other incoming traffic will match criteria somewhere between those two extremes, and will be treated accordingly.

Some examples follow, with the corresponding traffic treatments shown.

This example shows a simple case of a single server group. a single service protocol group, and three client groups, each with a different treatment. DNS traffic to DNS servers from Trusted Clients is allowed through, DNS traffic to DNS servers traffic from Known Bad Clients is blocked outright, and all other DNS traffic to DNS servers is detected, meaning that it is logged.

Server Group	Service Protocol Group	Client Group	Treatment
All DNS Servers	DNS	Trusted DNS Clients	Allow
		Known Bad Clients	Block
		Any	Detect

The next example expands on the previous one, with the addition of the "Any" server group (Default network-wide server protection) and an ICMP service protocol group. The example is largely the same as the previous one, except that ICMP traffic [e.g., ping] is blocked, regardless of its destination server or its originating client.

Server Group	Service Protocol Group	Client Group	Treatment
All DNS Servers	DNS	Trusted DNS Clients	Allow
		Known Bad Clients	Block
		Any	Detect
Any	ICMP	Any	Block

The next example is a little more elaborate, adding a server group for HTTP servers, with a few "Any" entries for service protocol group and client group. Stepping through the three server groups from top to bottom:

- For All HTTP Servers:
  - HTTP traffic from clients in undesired countries of origin is blocked completely.
  - ICMP traffic from any client is allowed.
  - Any other traffic bound for the HTTP servers is blocked.
- For ALL DNS Servers: see the first example.

- For Any server not included in All HTTP Servers or All DNS Servers:ICMP traffic from any client is blocked completely.

Remember that an IP address, whether for a server or a client, can be included in a single server group or client group only. In this example, if a server IP address is not included explicitly in All HTTP Servers or All DNS Servers, it defaults to being in the Any server group.

Server Group	Service Protocol Group	Client Group	Treatment
All HTTP Servers	HTTP	Geo Bad Clients	Block
	ICMP	Any	Allow
	Any	Any	Block
All DNS Servers	DNS	Trusted DNS Clients	Allow
		Known Bad Clients	Block
		Any	Detect
Any	ICMP	Any	Block

Catch-All

There is a catch-all server group for all server addresses not included in any other group. This server group cannot be disabled, and all rules associated with it are disabled. Packets arriving on the external port for which the destination address is not part of any server group are silently allowed to transit through the appliance. This catch-all server group is not shown in the CMS.

Traffic Patterns

A traffic pattern describes the anticipated behavior of the traffic flows the Defense appliance encounters. Asymmetric traffic occurs when multiple flows between a client and server use different routes to reach each other.

Some protection features are not supported for asymmetric traffic, or are supported with variations, due to the fact that the Defense appliance may not see all of the significant packets in the flow.

Traffic patterns are components of traffic treatments. The traffic pattern is configured as part of defining a server protection instance. The traffic pattern is Symmetric by default.

The Defense appliance supports these traffic patterns:

- **Symmetric** – A single Defense appliance receives all of the packets of any flow it observes. This is the default option. This mode supports all protection features.

  The client request and server response for the flows traverse through the same Defense appliance. In this mode, the Defense appliance can behave as a proxy and act upon each new SYN connection request coming into the network for clients not in a fully trusted state. The complete three-way handshake is between the client and the Defense appliance. Once the three-way handshake is established, the three-way handshake for the TCP connection is then set up between the appliance and the server.

- **One-way** – [Asymmetric] If the network configuration around the Defense appliance is such that there is more than one path between a client and server, and it is possible to predict, with a low margin of error, which path through the network a flow's packets will follow, after the initial packet

of the flow is observed.

In this traffic pattern, the client requests always take a predictable path, but the responses may traverse a different path. The Defense appliance will force a non-trusted client to re-initiate a SYN connection request after the client has already sent in the first SYN connection request by forcing a reset of the TCP connection (the appliance responds back initially with a SYN ACK with an invalid sequence number).

The client state of untrusted to trusted or untrusted to malicious is still based on the number of outstanding open SYNs awaiting completion of the three-way handshake from the client, since the client ACK must always be received by the Defense appliance.

For this mode to work, all the client requests must go through the Defense appliance.

- **Random** – [Asymmetric] There is no way to predict the path that a flow's packets will take. Load-balancing is one possible cause of this situation.

  In this traffic pattern, there is no way to predict the path for a given request or response. This mode assumes everything is already a mid-flow much like a power-up state. It is possible to classify a client as malicious (no suspicious classification possible). Fragmentation exploit inspection is not possible (automatically disables IP Fragmentation assembly for system when enable random asymmetric mode for a server group)

  No TCP proxy is supported in this mode.

IP Address Range Subsets

When defining host groups, it is possible to specify ranges of IP addresses that are subsets of other address ranges, as long as each subset is completely enclosed within its superset, and no address is a member of more than one range. Subsets can exist within the same host group, or they can be used to define other host groups. A range can consist of a single address. Overlapping address ranges are not allowed. This is useful if you need to create a specialized profile for a subset of clients or servers.

Address Range Rules and Examples

An address range can be defined as a numerical range (e.g., 10.1.1.1 - 10.1.1.255) or as a block (10.1.1/24). A single address can constitute a range, as well. A single IP address needs to be removed explicitly from one host group in order to be added to another.

Some examples follow, using two host groups (Hg1 and Hg2):

- Hg1: 10.1.1.100; Hg2: 10.1.1.100 - 10.1.1.125 *This does not overlap; the single address gets moved from the smaller range to the larger range.*
- Hg1: 10.1.1.125; Hg2: 10.1.1.100 – 10.1.1.150 *This does not overlap; the single address gets moved from the smaller range to the larger range.*
- Hg1: 10.1.1.125; Hg2: 10.1.1.100 – 10.1.1.125 *This does not overlap; the single address gets moved from the smaller range to the larger range.*
- Hg1: 10.0.0.0 – 10.255.255.255; Hg2 – 10.1.0.1 10.1.0.100 *This does not overlap; the second range is completely enclosed within the first range.*
- Hg1: 10.1.0.10 – 10.1.1.20; Hg2: 10.1.1.15 10.1.1.25 *The second range overlaps with the first, and this is not allowed.*

- Hg1: 10.1.0.10 - 10.1.1.20; Hg2: 10.1.1.20 10.1.1.25 *The second range overlaps with the first, and this is not allowed.*

Viewing IP Address Range Definitions in the CMS UI

In the CMS web UI, each server group and client group has a column next to its name titled **Has Subsets**, which indicates whether one or more of the address ranges that make up that host group is a superset of any other address ranges. In the Server Addresses & Ranges tab (for server groups) and the Client Addresses & Ranges tab (for client groups), there is one column for the IP Address/Range, and another column, Subset Ranges Overview, that provides a graphical representation of any subsets that have been defined. Floating the cursor over one of these bars displays a balloon summarizing the subset ranges that have been defined, as well as the names of the associated host groups.

The CLI command `show defense address-summary host-group` summarizes in text, for each server group and each client group (including the default server and client groups), the following address range information:

- **Configured Ranges** – The address ranges that have been explicitly configured.
- **Enclosing Ranges** – The address ranges that enclose other ranges.
- **Enclosing Hosts** – The server or client groups that enclose the address ranges
- **Actual Ranges** – The effective address ranges.

Backup Bundle

Backed-up CMS configurations are stored in "backup bundles", each of which is a gzipped tar file. When the backup action is invoked, a backup bundle is created and written to "/var/www/html/backup", using a user-specified file name. If a backup bundle already exists when the backup action is invoked, the existing file will be removed before the new bundle is created. The latest backup bundle will always remain until the next backup is made. After the backup has been created, the backup bundle is available for retrieval using HTTP GET.

A backup bundle comprises the following files, all stored at the top-level of the tar container:

- /opt/corero/cms/app/ncs-cdb/A.cdb
- /opt/corero/cms/app/ncs-cdb/C.cdb
- /opt/corero/cms/app/ncs-cdb/O.cdb
- /opt/corero/cms/app/ncs.conf
- /opt/corero/cms/app/log4j.xml
- backup-info, a file containing information about the backup itself

The "backup-info" file is in JSON format. For example:

```
{
"backup-date": "2014-09-24T14:25:01-0400",
"ncs-version": "ncs-3.2",
"cms-base-rpm": "cms-base-V8.10.084-0.x86_64.rpm",
"cms-pkg-rpm": "cms-pkg-V8.10.084-0.noarch.rpm"
}
```

Traffic Scaling

The Traffic Scaling parameter, accessed in the Cluster Configuration screen, enables you to apply a multiple to global traffic parameters for the associated protection group. This is useful when the SmartWall appliances protecting your network use interface speeds other than 10 Gbps, for which the global defaults are intended. Traffic Scale is set to 1.0 by default; you can specify a value in *x.y* format, with one digit on each side of the decimal point.

Traffic Scaling is effective per protection group, and is displayed per protection group accordingly.

These parameters are affected by the Traffic Scale parameter:

- Advanced Security Controls
  - UDP Global Rate
  - ICMP Global Rate
  - Non-TCP Rate
- Threat Awareness parameters
  - Activation Parameters [New IP Addresses, TCP Setup Rate]
  - Exit Parameters [New IP Addresses, TCP Setup Rate]
- Smart Rule parameters

Traffic Scale does not change the configured values of the affected parameters, but is applied to them as specified per protection group. Traffic Scale does not affect SYN limits and connection limits.

Smart Rule

The Smart Rule feature enables you to configure threshold values and rate limits for protecting servers from various types of flood traffic. For a relevant traffic type, if the average rate exceeds the configured threshold, a rate limit is enabled that restricts the amount of that type of traffic until the average rate falls below the threshold. The Smart Rule functionality is able to distinguish one or more servers at risk of being flooded, and executes the rate limiting action on the traffic destined for that server, not indiscriminately on all traffic of that type that is entering the network. Smart Rule functionality acts on traffic that is not blocked by other broader, less discriminating rules. The average rate is determined over a moderate interval length that has been chosen to provide a reasonable amount of protection against not only sudden surges in traffic, but also slower, more gradual escalations.

The relevant traffic categories are:

- *Service*: A service at risk of being flooded by targeted traffic, where a service is defined as a protocol and a destination port. Smart Rule takes into account a number of factors, including the destination IP address, destination port, and protocol, among others, to determine whether the service is under attack.
- *Reflection*: Incoming traffic bearing the characteristics of a reflection attack.
- *UDP*: An excess of UDP traffic destined for a protected server.

Smart Rule examines both the bit rate and the packet rate of the relevant traffic types, and, for each traffic category, there are four parameters used to identify and reduce problematic traffic:

- bit threshold
- bit rate limit
- packet threshold
- packet rate limit

Use the bit threshold and bit rate limit parameters to control traffic to the destination based on the rate of data, after the threshold has been crossed. Use the packet threshold and packet rate limit parameters to control traffic to the destination based on the packet rate, after the threshold has been crossed.

These parameters are provided in the Early Access Capabilities screen.

Screen and Command Descriptions

Topics in this section list the controls that appear in each CMS screen along with a description of what each one does.

Descriptions of how to execute tasks in the CMS and configure the appliance appear in sections such as Configuring Basic Behavior, Analyzing Traffic, and Blocking Threats.

Detailed explanations of how the appliance and CMS work are provided in the Concepts and Explanations section.

Advanced Security Controls screen

Clicking **Defense Configuration** > **Advanced Security Controls** in the web UI tool bar displays the Advanced Security Controls screen. This screen enables you to specify custom costs for individual packet types, as well as rate limits for ICMP traffic.

These advanced security controls are available through the CLI using the `set defense profiles advanced-controls` command, except for the Header Blocking controls, which are available through the `set defense header-blocking` command.

Header Blocking:

Use these controls for blocking traffic from entering your network based on its IP header type. Each type of traffic is allowed by default, though you should note that MPLS traffic is affected by default by the administrative state of rule CNS-000071 (which is enabled by default).

- **IPv4 Packets** – Choose **Allow** or **Block**. **Allow** is the default.
- **IPv6 Packets** – Choose **Allow** or **Block**. **Allow** is the default.
- **MPLS Packets** – Choose **Inspect** or **Per Rule CNS-000071**; the latter is the default.
- **IP Tunnels** – Choose **Allow** or **Block**. **Allow** is the default.

Service Controls:

- **Outbound Packet Inspection** – This is **Disabled** by default. Setting it to **Enabled** activates the inspection of traffic destined for the outside of the network.

ICMP Rate Limits:

- **ICMP V4 Rate**
- **ICMP V6 Rate**

- **ICMP V6 Priority Rate**

Connection Rate Limits:

- **Non TCP Setup Rate** – The rate at which non-TCP traffic will be allowed to pass through before triggering rule CNS-001022.

Fragmentation:

- **Max Age** – Maximum fragment age in milliseconds before triggering CNS-100019.
- **Max Rate** – Total number of fragments allowed per second.
- **UDP Fragmentation Ratio** – Approximate number of fragmented packets allowed for every 100 non-fragmented packets.

Slow Client Checks:

- **Small Packet** – A client packet with a TCP payload smaller than this is considered suspicious.
- **Small Packet Threshold** – This number of consecutive small packets triggers CNS-003016.
- **Small Window** – A client packet advertising a TCP window smaller than this is considered suspicious.
- **Small Window Threshold** – This number of consecutive small windows triggers CNS-003017.

Probation Controls:

- **IP Address Timeout** – When the Defense appliance recognizes that a DDoS attack is in progress and a new IP address is seen for the first time, that address is put "on probation" for the length of time (in seconds) specified by this parameter. If the IP address is seen again during this interval, it is taken off probation and promoted to being handled normally. Otherwise, if the IP address is not seen again during the interval, the original packet is dropped, and the next packet seen from that address (if any) will start the probation interval again.

Advanced Site Controls screen

Clicking **Site Configuration** > **Advanced Site Controls** in the web UI tool bar displays the Advanced Site Controls screen. These controls specify a variety of global defaults for the CMS and the site that it manages.

The Advanced Controls screen provides these controls:

Packet Processing

- **Base Sampling Period** – Choose **Short**, **Medium**, or **Long**. **Medium** is the default.

Device Communications

- **Security Event Port** – The default is 6343.
- **Sampled Packet Port** – The default is 6343.
- **Syslog Port** – The default is 11222.

- **SNMP Trap Port** – The default is 162.
- **SNMP Community** – The default string is smartwall.

Alarm Center screen

Click the alarm clock icon in the web UI status bar to display the Alarm Center screen. Alternatively, use the CLI command `show alarms summary` or `show alarms alarm-list` to see current alarm information.

Alarms indicate noteworthy occurrences related to the CMS itself or the Corero appliances that it manages. Alarms are characterized with severities ranging from Warning, indicating notifications of which you should be aware, up to Critical, indicating serious problems with the function of your server defense infrastructure. Specifically, the Alarm Center screen shows *uncleared* alarms; these are notifications which have not yet been addressed, whether by an action executed by a human operator, or, in some cases, by self-clearing, if the circumstance that triggered the alarm has corrected itself.

Alarm severities are assigned explicitly to the definitions of specific known types of occurrences, and are not assigned dynamically based on numerical values or the crossing of predefined thresholds. The supported alarm severities are:

- Critical – Critical alarms are not self-cleared.
- Major
- Minor
- Warning
- Indeterminate

Alarms can be exported from the CMS via SNMP and syslog. The relevant configuration parameters are available in the Advanced Site Controls screen.

You can perform these actions on selected alarms (click the checkbox to the left of the alarm entry):

- **Compress** – Show only the latest state change associated with the alarm.
- **Purge** – Delete entries associated with the alarm.
- **Handle** – Choose a handling method from the pulldown menu: **Acknowledged**, **Investigation**, **Observation**, **Closed**.

Clicking an alarm entry opens a panel that shows information about the alarm in greater detail:

- **Alarm Handling** – Choose a handling method from the pulldown menu: **Acknowledged**, **Investigation**, **Observation**, **Closed**.
- **Status Change** – This section lists the individual changes in status for the alarm.

If the storage device associated with a Forensics appliance lacks sufficient space to collect a new query, the query fails and an alarm is generated. If this happens, delete some old queries to free space for the new query.

Blocked & Detected Events Screen

Click **Defense Monitoring** > **Blocked & Detected Events** in the web UI tool bar to display the Blocked & Detected Events screen. This screen is a basic tool for monitoring suspicious traffic as it

reaches your network's perimeter; it displays information about current attacks dynamically in real time, sorted by rule name.

Some of this information can be displayed in the CLI using the command `show defense statistics rule-statistics rule-counts`.

The following controls are available:

- **Clear counters for selected rules** – Reset the event and packet count for each rule you've selected in the table.
- **Clear counters for all rules** – Reset the event and packet count for every rule.
- **Search** panel – Type a string to find its match in the contents of the Blocked & Detected Events screen.

The Blocked & Detected screen provides these controls:

- **Rule Name** – The ID of a rule that was triggered when incoming traffic matched the characteristics specified in the rule.
- **Rule Description** – The short description of the triggered rule.
- **Block Event Count** – The number of times traffic was blocked as a result of triggering the rule.
- **Block Packet Count** – The packet that causes the Blocked Event is also counted as one Dropped Packet. Any further packets that arrive for that flow are also dropped, but it still only counts as one event. This means the Dropped Packet Count is usually a larger number than the Blocked Event Count.
- **Detect Event Count** – The number of times traffic was detected as a result of triggering the rule.
- **Detect Packet Count** – The number of packets detected associated with this rule since the last time the counter for this attack was reset.

You can sort the contents of the Blocked & Detected Events screen by any column. The primary sort order is indicated by a caret (∧ or ∨, depending on the desired direction), and the secondary sort order is indicated by a plus sign (+).Once you choose a column by which to sort, you must refresh the Blocked & Detected Events screen to clear the current sort order and return to the default sort order, which is by rule number.

Capture Admin & Queries screen

Clicking **Defense Monitoring** > **Capture Admin & Queries** in the web UI tool bar displays the Capture Admin & Queries screen for administering queries of packet capture data acquired by a Forensics appliance. Alternatively, type the CLI command `set forensics packet-capture query-management` to access the same functionality.

Clicking on a query definition in the list shows its details, which can be edited.

The Capture Admin & Queries screen provides these controls:

- **Capture Mode** – Choose **Bypass**, **Capture**, or **Capture Until Full** to indicate whether the Forensics appliance is acquiring data or not.
- **Add** – Click this to display the Add New Query screen for defining a new query of existing packet capture data.

- **Query Name** – A user-defined name for the query.
- **Query Start Time** – Specify the time from which you want to begin filtering captured packets.
- **Query End Time** – Specify the time at which you want to stop filtering captured packets.
- **Packet Length Min** – Specify the smallest packet for which you want to filter.
- **Packet Length Max** – Specify the largest packet for which you want to filter.
- **IP Protocol** – Choose a protocol for which you want to filter.
- **Flow Direction** – Choose **Unidirectional** or **Bidirectional**.
- **Ethertype** – Choose the type of packet encapsulated by Ethernet frames.
- **Query Batch Size** – Specify the number of bytes of data to include in each portion of the query you retrieve at one time.
- **Source IP Begin** and **Source IP End** – Define a range of source IP addresses for which you want to filter.
- **Dest IP Begin** and **Dest IP End** – Define a range of destination IP addresses for which you want to filter.
- **Source Port Begin** and **Source Port End** – Define a range of source port numbers for which you want to filter.
- **Dest Port Begin** and **Dest Port End**– Define a range of destination port numbers for which you want to filter.

- **Get Next Batch** – Click this to retrieve the next portion of packet capture data included in the query.
- **Delete** – Select an existing query definition to remove it. If the storage device associated with a Forensics appliance lacks sufficient space to collect a new query, the query fails and an alarm is generated. If this happens, delete some old queries to free space for the new query.
- **Status** – This indicates the number of batches from the query that have been retrieved.
- **Batch Statistics** – This indicates the number of packets processed and filtered for the most recent batch.
- **Query Statistics** – This indicates the number of packets processed and filtered across all retrieved batches for the query.

Client Groups screen

Click **Defense Configuration** > **Client Groups** to display the Client Groups screen for defining a list of IP addresses from which incoming traffic is handled the same way by the appliances. The same controls are available in the CLI by using the `set defense profiles client-group` command.

The Configure Client Groups topic describes how to define and modify client groups.

The Client Groups screen provides these controls:

Name – A descriptive identifier for the client group. This is all that is required to create the client group, though it isn't useful until the rest of fields are populated.

Add/Delete buttons – Click **Add** to display the Add New Group dialog box and define a new client group. The client group can be based on IP addresses or geographical locations (countries of origin). Click **Delete** to remove a selected client group from the site.

Client Addresses and Ranges tab – Select an existing client group and click this tab to specify one or more IP addresses to include as members of the client group. For a geographical client group, select one or more countries to include in the client group.

- **IP Address/Range** – For an IP address range client group, this shows the ranges of IP addresses that define the client group.
- **Subset Ranges Overview** – If there are subsets of the IP address ranges that define the client group, this provides a graphical representation of where the subsets exist within the overall range. If there are no subsets, this field will say that there are no subranges.

Thresholds & Limits tab – Select an existing client group and click this tab to select or define a client profile that specifies what constitutes acceptable traffic from these clients.

- **Manage Profiles** – Display the Client Profiles screen for working with client profiles independently of client groups.
- **Use a shared profile** – Choose an existing client profile. The profiles **High Limits**, **Low Limits**, and **Megaproxy Limits** are built-in and provided by default.
- **Use a custom profile** – Specify explicit values for the profile parameters:
  - **Trusted client connection threshold (minimum)** – The number of successful and well-executed connections required to establish this IP address as a trusted IP address. Requests from trusted clients are forwarded to the destination address.
  - **Suspicious client connection threshold** – The source IP address has reached a user-specified threshold (Suspicious Threshold) due to the number of open SYNs. The device will proxy all connection requests to any device from that source IP address.
  - **Suspicious client connection exit threshold** – The device maintains a count of the number of incomplete TCP connections destined to a server group and allows the user to set a value for this threshold (known as the Connection Threshold). When a source Client Group is in the Unknown or Trusted states, and the destination's Server Connection Threshold has not been reached, then SYN packets received from this source to the given destination are passed directly on to the server group (subject to other security policy checks).
  - **Malicious client connection threshold** – The source IP address has reached a user-specified threshold (Malicious Threshold) due to the number of open SYNs. The device will drop all connection requests to any device from that source IP address.
  - **Initial treatment for clients in this group** – Click **Treat as trusted** or **Treat as suspicious** (the default).
  - **Maximum connections for this entire group**
  - **Maximum connections for any client in this group**
  - **Maximum packet rate for any client in this group** – Specify the packet rate limit for this client.
  - **Click here to export as a shared profile** – Save the custom profile to a file so it can be re-used.

Client Profiles screen

Click **Manage Profiles** in the Client Groups screen to display the Client Profiles screen. Client profiles define sets of characteristics that constitute acceptable traffic behavior from clients; this includes SYN floods, connection limits, and packet rate limits. Each is a separate type of profile which can be assigned independently to client groups. Each of these is configured in its own screen, which is nested in this Client Profiles screen.

For each profile type, click **Add** to display the corresponding screen and begin creating a new one, or click the name of an existing profile to modify it.

All of these controls are available also through the corresponding CLI commands, as noted in each section below.

SYN Flood Profile

A SYN flood profile enables you to define a set of thresholds for identifying and responding to excessive numbers of SYN packets. There are two built-in SYN flood profiles: **megaproxy-client-behavior** and **normal-client-behavior**.

You can access these controls also through the CLI command `set defense profiles client-profiles syn-flood-profile`.

- **Name** Type a descriptive name for the profile.
- **Trusted Threshold** – The number of successful and well-executed connections required to establish this IP address as a trusted IP address. Requests from trusted clients are forwarded to the destination address. The default is 5.
- **Suspicious Threshold** – The source IP address has reached a user specified threshold (Suspicious Threshold) due to the number of open SYNs. The device will proxy all connection requests to any device from that source IP address. The default is 5.
- **Suspicious Exit Threshold** – The device maintains a count of the number of incomplete TCP connections destined to a server group and allows the user to set a value for this threshold (known as the Connection Threshold). When a source Client Group is in the Unknown or Trusted states, and the destination's Server Connection Threshold has not been reached, then SYN packets received from this source to the given destination are passed directly on to the server group (subject to other security policy checks). The default is 45.
- **Malicious Threshold** – The default is 100. The source IP address has reached an even higher user-specified threshold (Malicious Threshold) because of the number of open SYNs. The appliance blocks any connection request from this address until one of two things happens:
  - The address stops initiating requests for a given period of time called the timeout period.
  - The number of new, open SYNs drops enough over time (due to a decay mechanism) to allow the total open SYNs to decay to the suspicious level.

  If this limit is exceeded, the following rule is triggered and the action as defined for this rule is followed: cns-001007 'Connection From Malicious Source IP Address'
- **Proxy Requests From Unknown IP** – This is unselected by default. Used for TCP connection traffic patterns for this address that are not yet identified. For IP addresses with an unknown threat level, you can configure the device to either:

- Forward connection requests from this address to the destination device.
- Proxy connection requests until the IP address becomes trusted.

Connection Limit Profile

Specify a maximum number of group connections (the default is 1048575) and a maximum number of member connections (the default is 32767). There are four built-in connection limit profiles: **low-limits**, **normal-limits**, **very-high-limits**, and **virtual-server-limits**.

You can access these controls also through the CLI command `set defense profiles client-profiles connection-limit-profile`.

Packet Rate Limit Profile

Specify the packet rate limit for this client. There are two built-in packet rate limit profiles: **20000-per-minute** and **request-limiting-disabled**.

You can access these controls also through the CLI command `set defense profiles client-profiles packet-rate-limit-profile`.

Cluster Configuration screen

Click **Site Configuration** > **Cluster Configuration** in the web UI tool bar to display the Cluster Configuration screen. This screen enables you to add Coreroappliances to the site and define the clusters and protection groups that are essential components of server defense.

The Cluster Configuration screen provides these controls:

- **Add/Delete** – Clicking **Add** displays the Add a New Cluster dialog, described in the lines that follow. Clicking **Delete** removes a selected protection group from the site. The Add a New Cluster dialog provides these controls:
  - **The Cluster Name Is** – Type a unique descriptive name for a set of Coreroappliances that work together to handle inbound traffic on one network interface at a site.
  - **The Cluster Has A** – Indicate whether your Coreroappliances operate individually or have redundant counterparts.
    - **Single Protection Group**– Each Coreroappliance in the cluster operates individually, and does not have a redundant counterpart. This can include two load-balanced Defense appliances.
    - **High Availability Pair of Protection Groups** – Each Coreroappliance has a redundant counterpart in a different protection group. Both protection groups belong to the same cluster.
  - **Each Protection Group Looks Like** – Choose a representation of the Coreroappliances in your cluster. If you selected HA Pair, the CMS doubles the representation for you, with the redundant counterparts as members of a separate protection group.
  - **Traffic Scale Parameter** – Traffic Scale enables you to apply a multiple to global traffic parameters for the associated protection group. This is useful when the SmartWall appliances protecting your network use interface speeds other than 10 Gbps, for which the global defaults are intended. Traffic Scale is set to 1.0 by default; you can specify a value in *x.y* format, with one digit on each side of the decimal point.

  - **The Device Settings Are** – Type the IP address and a descriptive name for the appliance you are adding. The CMS will provide a default name based on the IP address that you provide and the appliance type.
  - **Forensics Settings** – Specify IP addresses and network masks for a Forensics appliance's interfaces.
  - **Add** – Click this button to execute the creation of the cluster.
- **Protection Group** – The protection group name is derived automatically from the cluster name you specify.
- **Cluster Name** – The cluster name you specified when you added the appliances to the site.
- **Config Preview** – A graphical representation of the cluster you've defined.

CMS Backup/Restore screen

Clicking **CMS** > **CMS Backup/Restore** in the web UI tool bar displays the CMS Backup/Restore screen, providing controls for saving and reloading the CMS configuration. Configuration backups (backup bundles) are created for a specific CMS version.

Note: No more than one backup is retained on the CMS file system(s) at any one time.

The CMS Backup/Restore screen provides these controls:

- Time the most recent backup was performed.
- **Create Backup** – Save the current CMS configuration to a file.
- **Browse** – Navigate the filesystem to locate a CMS backup bundle from which to restore the configuration.
- **Restore Backup** – Reload the selected CMS configuration file. Note that the CMS will restart as soon as the restoration is complete.

The same functionality is available through the CLI by using the command `request backup [create | restore] file` **`filename`**.

Configuration Actions screen

Click **Site Configuration** > **Configuration Actions** in the web UI tool bar to display the Configuration Actions screen. This screen provides controls for coordinating the transfer of the site configuration between the CMS and the managed devices.

The Configuration Actions screen provides these controls:

- **Sync All Devices** – For all devices in the site, default the configuration on each appliance and push the new configuration from the CMS.
- **Deploy** – Push recent configuration changes to the site's devices. This is the same as using the CLI command `request config deploy`.
- **Force Sync** – For an individual device, default its configuration and push the new configuration from the CMS.
- **Reset Configuration** – For a selected appliance, choose. **Reset Policy Configuration** or **Reset All Configuration**.

- **Restart Device** – For a selected appliance, choose **Application Only**, **Operating System**, or **Hardware** to indicate the level at which you want to restart the appliance.

The **Deploy** and **Force Sync** commands may be necessary when appliances are down at the time the configuration is saved, or when new appliances are added to a running site. If there is a need to perform one of these actions, the **Sync Status** icon below the CMS menu bar will be lit. Unless the **Sync Status** icon is lit and indicates that a **Deploy** or **Force Sync** action is necessary, you should use the **Save** command as the primary mechanism for pushing a configuration to appliances.

Customers screen

Clicking **CMS** > **Customers** in the web UI tool bar displays the Customers screen for defining the names of customers to be associated with server groups and server protection instances. The same functionality is available using the CLI command `set customers customer` **`customername`** in Configuration mode. Each server group must have a customer name as part of its definition. There is a built-in customer named DEFAULT, so it is not necessary to define a new customer if you do not have multiple customers using your site.

Although it is necessary to have a customer as part of each server group definition, customers are not binding in any other respect, nor do they contribute to any other definition. There are Rank and Status parameters associated with customers, but these are provided for use at the user's discretion. The Rank and Status parameters provide another means for customers to be sorted, but they do not contribute to the definition of any other aspects of the configuration.

CMS Dashboard

The CMS dashboard appears by default when you log in to the CMS, displaying a number of graphs and tables that, together, summarize the overall threat status of the traffic seeking entry to your network. You can re-display the dashboard at any time by clicking **Home** in the path in the web UI tool bar.

Security Rule Block Rates (per second)

This tool graphs the rates at which rules of the different major categories are triggered on your Defense appliances by incoming traffic. This provides a broad sense of the type of unwanted traffic that your network is encountering, providing insight into possible attack methods. The graph is refreshed every 60 seconds.

Every 60 seconds the CMS creates a report that summarizes rate of packet blocking in the system for each of the overall classifications for security rules on the Defense appliances.

This tool graphs these statistics:

- **All Rules** – The total rate of packet blocking for all rules in the system.
- **Access Restriction Rules (Network Threats)** – The rate of packet blocking for rules of the ARNET classification.
- **Rate Limit Rules (Network Threats)** – The rate of packet blocking for rules of the RLNET classification.
- **Protocol Validation Rules (Network Threats)** – The rate of packet blocking for rules of the PVNET classification.

- **Integrity Analysis Rules (Network Threats)** – The rate of packet blocking for rules of the IANET classification.
- **Integrity Analysis Rules (Application Threats)** – The rate of packet blocking for rules of the IAAPP classification.

This information is carried in Block Rate Statistics Type syslog messages.

Excessive Rate Indications (per second)

This tool graphs these statistics:

- **TCP Setup Rate** – The current TCP flow setup rate.
- **Non-TCP Setup Rate** – The current non-TCP flow setup rate.
- **Protocol Validation Block Rates** – The rate of packet blocking for rules of the PVNET classification.

This information is carried in Setup Rate Statistics Type and Block Rate Statistics Type syslog messages.

IP Address Classifications (counts)

This tool graphs these statistics:

- **In Use Trusted Addresses** – IP addresses in use classified as trusted.
- **In Use Suspicious Addresses** – IP addresses in use classified as suspicious.
- **In Use Malicious Addresses** – IP addresses in use classified as malicious.
- **In Use Unclassified Addresses** – IP addresses in use not yet classified.

This information is carried in IP Address Classifications Type syslog messages.

Top Recent Service Ports

This lists the service ports most used recently by incoming traffic. They are listed by protocol and their percentage of overall use.

Top Recent Clients

This lists the clients to have sent the most traffic to your network recently. They are listed by IP address and their percentage of the overall traffic.

Top Recent Servers

This lists the servers in your network to have received the most traffic recently. They are listed by IP address and their percentage of the overall traffic they've received.

Top Recent Security Events

This lists the rules most triggered recently by incoming traffic. They are listed by rule and their percentage of the overall traffic.

Active Flows by Protocol (counts)

This tool graphs these statistics:

- **Total Flows**
- **Total TCP Flows**
- **Total UDP Flows**
- **Total ICMP Flows**
- **Total Other Flows**

Mission Port Rates (Mbps)

This tool graphs these statistics:

- **External Port Bit Receive Rate** – External mission port(s) aggregrate rate.
- **External Port Bit Transmit Rate** – External mission port(s) aggregrate rate.
- **Internal Port Bit Receive Rate** – Internal mission port(s) aggregrate rate.
- **Internal Port Bit Transmit Rate** – Internal mission port(s) aggregrate rate.

Summary Packet Rates (per second)

This tool graphs these statistics:

- **External Port Packet Receive Rate** – External mission port(s) aggregrate rate.
- **All Rules Block Rate** – The total rate of packet blocking for all rules in the system.
- **Internal Port Packet Transmit Rate** – Internal mission port(s) aggregrate rate.

Defense Summary screen

Clicking **Defense Monitoring** > **Defense Summary** displays the Defense Summary screen, summarizing the high-level health of your site

The screen shows this information:

Cluster

Protection Group

Protection Status

Deployment State

Bypass

- **Mode**
- **State**

Defense

- **Mode**
- **State**

Network Statistics

- **Devices**
- **Mission in Packet Rate**
- **All Rules Block Rate**
- **Mission Out Packet Rate**

Device Status screen

Display the Device Status screen by clicking **Site Configuration** > **Device Status** in the CMS tool bar. This screen enables you add, delete, and manage Corero appliances in the current site.

The Device Status screen provides these controls, which are available also when using the CLI command `set devices device`:

- **Status indicator icon** – This icon shows the overall health of the appliance.
- **Alarms** – If any alarms have been triggered on the appliance, this field shows the number of each category of alarm that is in effect currently.
- **Name** – A descriptive, user-defined name for the appliance.
- **Address** – The IP address or host name associated with the appliance's management interface.
- **Software Version** – This identifies the build of software running on the appliance currently.
- **State** – Indicates whether the appliance is in Mitigate, Monitor, or Bypass mode.
- **Uptime** – The amount of time that has passed since the last time the appliance was rebooted.
- **Reachable** – Indicates whether the device is reachable from the CMS or not.
- **Deployment State** – Indicates whether the device's configuration is in-sync with the CMS, or whether a synchronization action is required to bring the appliance into accord with the CMS.

Early Access Capabilities screen

Clicking **Defense Configuration** > **Early Access Capabilities** in the web UI tool bar displays the Early Access Capabilities screen. This screen provides recently introduced threat defense commands that are made available on a provisional basis for evaluation in your configuration. Commands that prove over time to be used widely or frequently may be moved later to permanent homes with other related commands. Commands in this screen that do not prove to be used frequently over time may be deprecated in some future version.

Smart Rule Parameters – These parameters configure the Smart Rule feature, and specify thresholds and rate limits for protecting servers from flood attacks and reflection attacks. The CLI equivalent is `set defense early-access-capabilities smart-rule-parameters`.

- **Service Bit Threshold** – The bit rate threshold over which a flood attack is determined to be occurring.
- **Service Bit Rate Limit** – The maximum bit rate for traffic determined to constitute a probable flood against a service protocol.

- **Service Packet Threshold** – The packet rate threshold over which a flood attack is determined to be occurring.
- **Service Packet Rate Limit** – The maximum packet rate for traffic determined to constitute a probable flood against a service protocol.
- **Reflection Bit Threshold** – The bit rate threshold over which a reflection attack is determined to be occurring.
- **Reflection Bit Rate Limit** – The maximum bit rate for traffic determined to constitute a probable reflection attack.
- **Reflection Packet Threshold** – The packet rate threshold over which a reflection attack is determined to be occurring.
- **Reflection Packet Rate Limit** – The maximum packet rate for traffic determined to constitute a probable reflection attack.
- **UDP Server Bit Threshold** – The bit rate threshold over which a UDP flood is determined to be occurring.
- **UDP Server Bit Rate Limit** – The maximum bit rate for traffic determined to constitute a probable UDP flood attack.
- **UDP Server Packet Threshold** – The packet rate threshold over which a UDP flood attack is determined to be occurring.
- **UDP Server Packet Rate Limit** – The maximum packet rate for traffic determined to constitute a probable UDP flood attack.

External Authentication screen

Clicking **External Authentication** on the Users screen displays the External Authentication screen for configuring an external authentication regime for use with the CMS. You can access the same functionality using the CLI command `set aaa external-authentication`.

The External Authentication screen provides these controls:

- **Authentication Type** – Choose **Active Directory** or **RADIUS** from the pulldown menu.
- **Server** – Click **Add** to display the Server screen and define a new authentication server for the Authentication Type you selected.Type the server name, and choose its Admin state, **Enabled** or **Disabled**.
- **Reject If No Group** – Select this if you want to deny all access to users if their profile does not belong to one of the groups mapped here. This is selected by default.
- **Group Mapping** – Specify an external group to match to each of these built-in CMS groups:
  - **Support**
  - **Admin**
  - **Defense**
  - **Monitor**
  - **No Access** – Users and groups that are explicitly denied access to the CMS.

Flex Rule Blocking screen

The CMS provides the Flex Rule utility for defining custom packet-blocking filters. These custom filters enable you to respond quickly to new threats you encounter that are not yet addressed by formal rules in Corero's SmartWall software, or, otherwise, to threats for which you want a very specific or nuanced response that is not provided to your liking by the built-in software. All custom packet filters defined using the Flex Rule utility are associated with a single SmartWall rule (CNS-001037) that indicates that traffic matching one of the filter definitions was recognized.

Clicking **Defense Configuration** > **Flex Rule Blocking** in the web UI tool bar displays the Flex Rule Blocking screen for defining these custom filters. The equivalent CLI command is `set defense flex-rule-blocking filter` **`filtername`**.

The Flex Rule Blocking screen provides the following controls:

- **Filter Description** – Type a brief explanation for one or more filters.
- **Match Rate Limit** – Rule CNS-002010 is triggered when the filtered traffic exceeds this limit. (**Note:** CNS-002010 will trigger *only* if rule CNS-001037, the rule associated with matches to the custom filters, is *not* set to Block, which it is by default.)
- **Add** – Click to display the Filter screen to name the filter and define it using tcpdump pcap-filter syntax.
- **Delete** – Remove a selected filter from the configuration.
- **Name** – The filter's user-defined name. Filters are executed in the alphanumeric order of their names. A filter name cannot be changed after the filter has been created; if you want to change the name of an existing filter, you must delete it and re-create it with the new name.
- **Definition** – The pcap-filter expression that defines the custom filter.

These filters block only inbound IP packets which either are not fragmented or are the first fragment of the IP datagram. Non-IP packets are never blocked. Individual filters are combined into a single filter expression by surrounding each expression with parentheses and inserting “or” between them, and rule CNS-001037 is triggered any time one of the filter conditions is met. CNS-001037 is designed to detect and block attacks with extremely high packet rates, and security events are logged only for packets which are also already selected for Sflow sampling. The corresponding blocked and detected counts are accurate, but many fewer events will be shown for CNS-001037 than for comparable packet-based rules.

The filter syntax and behavior is that of the publicly available tcpdump pcap-filter. Refer to http://www.tcpdump.org/manpages/pcap-filter.7.html for additional information. The version supported by the CMS is not necessarily the latest version available from tcpdump.org.

Note these differences between this filter utility and the tcpdump pcap-filter:

- Hosts must be specified by number rather than by name.
- The IP broadcast condition is not supported. (There is no means of specifying the subnet mask of an interface.)

Examples

As stated previously, this filter behaves in a way that is very similar to tcpdump pcap-filter, and the keyword syntax and examples provided at http://www.tcpdump.org/manpages/pcap-filter.7.html are very informative if you are unfamiliar with its use.

A few different examples follow here:

- Filter all packets being sent to port 1024: `dst port 1024`
- Filter all UDP packets that are 71 bytes in length and that have a TTL field of 237, 238, or 239 (note the byte offset value [8] specified for the TTL field): `udp and len == 71 and (ip[8] >= 237 and ip[8] <= 239)`
- Block traffic from 123.123.123.12 if the source port is 8080 or the source port < 1024,and the destination is 221.211.111.0: `src host 123.123.123.12 && (src port 8080 || src port <1024) && dst host 221.211.111.0`
- Block single-tag packets: `vlan && dst port 1024`
- Block stacked vlan tagged packets (8100/8100): `(vlan && vlan && dst port 1024)`
- Block untagged, single tagged and double tagged traffic (8100/8100 or 9100/8100): `(dst port 1024) || (vlan && dst port 1024) || (vlan && dst port 1024)`

Note: If you are filtering for VLAN tagged traffic, you must include the `vlan` keyword in your filter definition explicitly. In addition, when filtering for VLAN tagged traffic, you should be very conscious of how the `vlan` keyword will affect the decoding offset of the expression; this may not be immediately obvious to you, unless you are already familiar with the nuances of the `vlan` keyword's behavior. The manpage for the tcpdump pcap-filter command does note this in its description of the `vlan` keyword, though the statement is embedded in the body of the description without being highlighted in any way:

> "Note that the first `vlan` keyword encountered in expression changes the decoding offsets for the remainder of expression on the assumption that the packet is a VLAN packet. The `vlan [vlan_id]` expression may be used more than once, to filter on VLAN hierarchies. Each use of that expression increments the filter offsets by 4."

As such, be conscious also of the order in which the `vlan` keyword appears in the expression, particularly when filtering for some mix of tagged and untagged traffic.

High Availability screen

Click **CMS** > **High Availability** in the web UI tool bar to display the High Availability screen and configure a virtual IP address to be shared across both CMS instances in a redundant CMS pair. The same functionality is available in the CLI using the `set high-availability` command.

The configuration that is required for each CMS instance in the pair to recognize the other is done with the assistance of Corero customer support at installation time.

Save your changes when you're finished. Web UI: **Save** CLI: `commit`

IP Access Control screen

Clicking **Defense Configuration** > **IP Access Control** in the web UI tool bar displays the IP Access Control screen. This screen enables you to define access control instances to always block (blacklist) or always allow (whitelist) traffic from a specified IP address or range of IP addresses. Blocking/always allowing an IP address adds that IP address to a list of addresses that is checked before any other part of the corresponding packet is examined. A packet from a blocked IP address is discarded immediately, and no additional processing is performed on any portion of its contents. Similarly, a packet from an IP address that is to be allowed always is forwarded immediately, with no additional processing.

The IP Access Control screen provides these controls:

Add/Delete – Create or remove an entry. Click **Add** to display the Add New IP Access Control screen.

- **Name** – Type a descriptive name for the blacklist or whitelist.
- **Source IP** – Type the source IP address or range to include in the access control instance. You can type a range as a spread of IP addresses (200.1.1.1 - 200.1.1.255), with a subnet mask (200.1.1.0 255.255.255.0) or as an offset (200.1.1.0/24). Refer to Creating a Blacklist or Whitelist of IP Addresses for instructions about nesting address ranges, if you need to do that.
- **Admin State** – Choose **Enabled** or **Disabled**.
  Note: For a whitelist, this enables/disables the *rule* associated with the list, and any corresponding notification. Once an address is added to a whitelist, traffic from that address is *always* allowed, even if the whitelist itself is administratively disabled.
- **Access** – Choose **Block** for a blacklist or **Always Allow** for a whitelist.

The equivalent CLI command is:

```
set defense ip-access-control access-control-entry name access [always-allow | block] admin-state [enabled | disabled] source-ip ipaddress
```

The equivalent REST API command is:

```
curl -k -v -u admin:smartwall -X PATCH https://cmsipaddress:443/api/running/defense/ip-access-control/access-control-entry/ -d '{"access-control-entry": {"name": "listname", "access": "block", "admin-state": "enabled", "source-ip": "ipaddress" } }' --header "Content-Type:application/vnd.yang.data+json"
```

IP Reputation Rule Set screen

The IP Reputation Rule Set screen enables you to manage IP reputation rule sets for the site. IP reputation rules are triggered when traffic is encountered that has originated from an IP address that is characterized currently as suspicious or malicious.

Click **Defense Configuration** > **IP Reputation Rule Sets** in the web UI tool bar to display the IP Reputation Rule Set screen, or use the CLI command `set defense profiles ip-reputation-rule-set` to access the same functionality.

Refer to Configuring Server Protection to learn how rule sets are associated with server protection instances.

The IP Reputation Rule Set screen provides these controls:

- **Add/Delete** buttons – Define a new IP reputation rule set to add it to the collection of rule sets that are available to the site, or remove an existing rule set.
- **Name** – A descriptive name for the rule set. The names of built-in rule sets provide some indication of each rule set's purpose.
- **Description** – An optional, more verbose explanation of the rule set's purpose.
- **Block Active** – **True** or **False**. Traffic that triggered the rule is stopped with no indication given to the originating client.
- **Default Log** – This indicates whether a log entry should be made when a rule in the rule set is triggered.

- **Copy Of** – If this rule set was created initially by copying another, this tells which rule set was used as the seed for this one.

If you decide to change the configuration of a rule set, you should do that using a copy, preserving the original rule set for reference.

Network Storage screen

Clicking **Site Configuration** > **Network Storage** in the web UI tool bar displays the Network Storage screen for specifying storage resources used by Forensics appliances. Alternatively, type the CLI command `set site packet-capture storage` to access the same functionality.

The Network Storage screen provides these controls:

Storage section – Obtain the name of the target from your SAN.

- **Add** – Click this to display the Storage screen for specifying the target of the packet capture data.
- **Delete** – Select an existing target in the list and click this to remove it from the site.
- **Volume Name** – The user-defined name of the target.
- **Device** – Choose the Forensics appliance from which the target will receive packet capture data.
- **Type** – Choose **Iscsi** or **RAM**.
- **Size** – Specify the target size, expressed in GB.
- **Target** – The formal name of the target, as obtained from your SAN.
- **Status** – The current state of the target.
- **Admin State** – Choose **Enabled** or **Disabled** to activate or de-activate the target.

Portal section – This is the IP address of the storage resource.

- **Add** – Click this to display the Portal screen for specifying the name, IP address, and port of the storage resource.
- **Delete** – Select an existing storage resource in the list and click this to remove it from the site.
- **Name** – The user-defined name of the storage resource.
- **Host** – The IP address or DNS name that identifies the storage resource.
- **Port** – The port on which the Forensics appliance communicates with the storage resource. The default is 3260.

Operations Settings screen

Click **Defense Configuration** > **Operations Settings** in the web UI tool bar to display the Operations Settings screen.

The Operations Settings screen provides these controls:

- **Global Defense Mode** – Choose the operational mode for the shelf: **Monitor**, **Mitigate**, or **Bypass**.
- **Cluster** – Specify the devices included in a high availability cluster.

These controls are available also through the CLI commands `set defense operations-settings global-defense-mode` and `set defense operations-settings cluster`.

Packet-Based Rules screen

Click **Defense Configuration** > **Packet-Based Rules** in the web UI tool bar to display the Packet-Based Rules screen, or use the CLI command `set defense profiles rule-groups packet-rules` to access the same functionality.

This screen provides a catalog of all the packet-based rules used in the SmartWall Threat Defense System, displaying a summary of basic information for each, including its Admin State, Action, whether it is logged or not, its Severity, and short and long descriptions of the rule. Clicking a rule number displays a screen that enables you to customize the rule's configuration.

Pull Files From CMS screen

Clicking .**CMS** > **Pull Files From CMS** in the web UI tool bar displays the Pull Files From CMS screen. This provides access to several different CMS files that are useful for debugging problems or for replicating configurations.

Specify the type of file that you want to download:

- **Application Log** – The downloaded file is "cms-app.log", which is useful for debugging some CMS-related problems.
- **Audit Log** – The downloaded file is "audit.log", which is useful for debugging some CMS-related problems.
- **Configuration Checkpoint** – The downloaded file is "config.audit.LATEST". This file lists configuration commands in CLI format, and is useful for reconstructing the CMS configuration.

Click **Pull File From CMS** to download the specified file type.

Pull Files From Device screen

Click **Site Configuration** > **Pull Files From Device** in the web UI tool bar to display the Pull Files From Device screen. This screen enables you to download management logs and diagnostics bundles from a selected Corero appliance.

Select the appliance, the type of file you want to download, and specify the individual file that you want. Click the **Pull file from selected device** button to download that file to the CMS and your local client.

Push Images To Device screen

Clicking **Site Configuration** > **Push Images To Device** in the web UI tool bar displays the Push Images To Device screen. This screen enables you to distribute a variety of different image file types to the appliances in your site, including software upgrades.

This screen enables you to upgrade the software on any Corero appliance:

- SmartWall Network Threat Defense Appliance
- SmartWall Network Bypass Appliance
- SmartWall Network Forensics Appliance

NOTE: Make sure to **Save** your configuration after using this screen to make any changes to ensure that information about those changes is retained properly. For example, if you upload a file to the CMS for distribution to Corero appliances, but exit the CMS before saving the configuration change, the file will be present in the directory to which it was uploaded, but will not be shown in CMS configuration.

The following information is shown for each image file, regardless of its type:

- **Name**
- **URL**
- **Date**
- **Status**

The following controls are provided for handling the images:

- **Add** button – Display the Upload dialog to acquire a file that can be distributed to one or more appliances.
  In the current release, you need to browse for the image in your local file system, rather than specifying its URL.
- **Delete** button – Select a file to remove it from the configuration and the directory in which it is stored.
- **Push selected file to** – Indicate whether the selected file will be pushed to every appliance in the site or to a single specific appliance only. Activate this button by selecting an image in the list.
- **Push File** button – Click this to execute the transfer of the selected file to the specified appliances. Activate this button by selecting an image in the list.

Upon being pushed to an appliance, the image is transferred, and it takes some time until it runs actively on the appliance. Go to the Device Status screen to verify that the applaince is back up and running the expected version of software.

Query screen

Click **Add** in the Query Management screen to display the Query screen.

- **Command** –
- **Query Type** –
- **Start Time** –
- **End Time** –
- **Min Packet Length** –
- **Max Packet Length** –
- **IP Protocol** –
- **Flow Direction** –
- **Ethertype** –
- **Query Batch Size** –
- **Source IP Begin** –
- **Source IP End** –
- **Dest IP Begin** –

- **Dest IP End** –
- **Source Port Begin** –
- **Source Port End** –
- **Dest Port Begin** –
- **Dest Port End** –

Query Management screen

Click **Query Management** in the Packet Capture screen to display the Query Management screen. The CLI equivalent is `set forensics packet-capture query-management query` **`queryname`**.

The Query Management screen provides the following controls:

Query

- **Add/Delete** – Click **Add** to display the Query screen for defining a new capture query.
- **Name** –
- **Command** –
- **Query Type** –
- **Start Time** –
- **End Time** –
- **Min Packet Length** –
- **Max Packet Length** –

Status

- **Query Result** –
- **Query Status** –
- **Query Result File** –
- **Batch Packets Processed** –
- **Batch Packets Filtered** –
- **Batch Process Delta Time** –
- **Query Result Repository** –

Rate-Based Rules screen

Click **Defense Configuration** > **Rate-Based Rules** in the web UI tool bar to display the Rate-Based Rules screen, or use the CLI command `set defense profiles rule-groups rate-limit-rules` to access the same functionality.

This screen provides a catalog of all the rate-based rules used in the SmartWall Threat Defense System, displaying a summary of basic information for each, including its Admin State, Action, whether it is logged or not, its Severity, and short and long descriptions of the rule. Clicking a rule number displays a screen that enables you to customize the rule's configuration for that rule set: you can configure the treatment for a particular rule differently from one rule set to another, and you can modify the individual treatments, as well. Treatment modifications apply only to the instance of that rule associated with the named rule set.

Recording Filter screen

Clicking **Security Monitoring** > **Capture Filters** in the web UI tool bar displays the Recording Filter screen.

Rule screen

This screen lists packet rules or rate limit rules, depending on which you specified.

Click **Defense Configuration** > **Security Rule Sets** or **Defense Configuration** > **IP Reputation Rule Sets** to display this screen with a list of the corresponding rules. The CLI equivalents are `set defense profiles rule-groups packet-rules rule` and `set defense profiles rule-groups rate-limit-rules rule`.

The Rule screen provides these controls:

- **Rule Name** – The rule identifier.
- **Admin State** – Choose **Enabled** or **Disabled** for the rule. Mostly, rules are enabled by default, with a few exceptions.
- **Action** – Choose what to do with traffic that triggers the rule:
  - **Allow** – Traffic that triggered the rule passes to its destination without restriction.
  - **Detect** – Traffic that triggered the rule is identified and logged, but is otherwise passed to its destination without restriction.
  - **Block Active** – Traffic that triggered the rule is stopped with no indication given to the originating client.
  - **Block Passive** – Traffic that triggered the rule is stopped and a specific response is executed, such as resetting the connection.
- **Log** – Choose **True** or **False**, indicating whether to log the event or not when the rule is triggered.
- **Log Severity** – The importance of the log message for the rule. Choose **Minimal**, **Moderate**, or **Critical**.
- **Description** – The rule's short description.
- **Long Description** – The rule's long description.

Security Event Viewer

Click **Defense Monitoring** > **Security Event Viewer** in the web UI tool bar to display the Security Event Viewer.

The Security Event Viewer is a security monitoring and management tool in the web UI that enables you to easily examine and react to traffic that triggers security rules. The viewer includes features that enable you to filter and focus on security event details and take action immediately to address them.

The following controls are available for controlling the display of security events:

- **Pause** – Suspend the display of new security events.
- **Resume** – Show new security events again as they occur.

The Security Event Viewer lists the following information per security event:

- **Event Timestamp** – The time at which the event was triggered.
- **Source IP Address** – The IP address from which the traffic flow originated.
- **Source Port** – The TCP port from which the traffic flow originated.
- **Destination IP Address** – The IP address to which the traffic flow is directed.
- **Destination Port** – The TCP port to which the traffic flow is directed.
- **Action** – The response that the appliance takes for the traffic flow. This can be Allow, Block Passive, or Block Active.
- **Rule(s) Triggered** – The specific rule(s) triggered by this traffic flow.

Security Rule Set screen

The Security Rule Set screen enables you to manage security rule sets for the site. Security rule sets describe behaviors for identifying and addressing traffic based on the nature of the traffic itself, such as its packet formation or its content.

Click **Defense Configuration** > **Security Rule Sets** in the web UI tool bar to display the Security Rule Set screen, or use the CLI command `set defense profiles rule-groups security-rule-set` to access the same functionality.

Refer to Configuring Server Protection to learn how rule sets are associated with server protection instances.

The Security Rule Set screen provides these controls:

- **Add/Delete** buttons – Define a new Security rule set to add it to the collection of rule sets that are available to the site, or remove an existing rule set.
- **Name** – A descriptive name for the rule set. The names of built-in rule sets provide some indication of each rule set's purpose.
- **Description** – An optional, more verbose explanation of the rule set's purpose.
- **Block Active** – **True** or **False**. Traffic that triggered the rule is stopped with no indication given to the originating client.
- **Default Log** – This indicates whether a log entry should be made when a rule in the rule set is triggered.
- **Copy Of** – If this rule set was created initially by copying another, this tells which rule set was used as the seed for this one.

If you decide to change the configuration of a rule set, you should do that using a copy, preserving the original rule set for reference.

Server Groups and Protection screen

Clicking **Defense Configuration** > **Server Groups and Protection** in the web UI tool bar displays the Server Groups and Protection screen for defining the set of responses that protect your server infrastructure when incoming traffic matches one or more profiles.

Optionally, you can access this functionality through several CLI commands:

- For server protection instance configuration: `set defense server-protection server-group`
- For server profile configuration: `set defense profiles server-profiles`
- For client group configuration: `set defense profiles client-group`

Refer to Configuring Server Protection for instructions for configuring a server protection instance.

The Server Groups and Protection screen provides these controls:

- **Customer** – This indicates the name of the customer associated with each server group. If no customer has been associated with the server group explicitly, the server group will be associated with the Default customer.
- **Server Group Name** – This lists the server groups that have been defined for the site. Select a server group in this list to configure it or modify it.
- **Has Subsets** – This indicates whether the IP address ranges that define the server group enclose any IP address range subsets.
- **Add**/**Delete** buttons – Click **Add** to display the Add New Group dialog box for creating a new server group.
  Select an existing server group and click **Delete** to remove it from the site.
- **Server Addresses & Ranges** tab – Select an existing server group and click this tab to specify one or more IP addresses to include in the server group.
  - **Add**/**Delete** buttons – Click **Add** to display the Add IP Address Range dialog box for specifying one or more IP addresses to include in the server group, or to define a subset of an existing address range.
    Select one or more IP addresses and click **Delete** to remove the addresses from the server group.
  - **IP Address/Range** – This shows the IP address ranges that define the server group.
  - **Subset Ranges Overview** – If there are subsets of the IP address ranges that define the server group, this provides a graphical representation of where the subsets exist within the overall range. If there are no subsets, this field will say that there are no subranges.
- **Thresholds & Limits** tab – Select an existing server group and click this tab to select or define a server profile that controls how the server group responds to incoming flows. Each profile sets a value for **SYN flood threshold**, **Maximum connections for this entire group**, and **Maximum connections for any server in this group**.
  - **Manage Profiles** – Display the Server Profiles screen for working with server profiles explicitly.
  - **Use a shared profile** – Choose an existing server profile. The profiles **High Limits**, **Low Limits**, and **Megaproxy Limits** are built-in and provided by default.

- **Use a custom profile** – Specify explicit values for the profile parameters.
  - **SYN flood threshold** – Specify a value for the maximum number of open SYN requests.
  - **Maximum connections for this entire group** – Specify the maximum total number of connections allowed for the server group.
  - **Maximum connections for any server in this group** – Specify the maximum number of connections allowed per individual member of the server group.
- **Server Protection** tab – Select an existing server group and click this tab to configure the details of the server protection instance.
  - **Protocol Group** – Displays the protocol group associated with the server protection instance.
  - **Client Group** – Displays the client group associated with the server protection instance.
  - **Traffic Treatment** – Summarizes the set of responses the server group uses for flows from the protocol group and client group.
  - **Add**/**Delete** buttons – Click **Add** to display the Add Traffic Treatment dialog box and begin configuring the server protection instance. Click **Delete** to remove a selected server protection instance.
    Choose an existing Protocol Group and an existing Client Group and click **Next** to display the Protocol Group/Client Group dialog box for configuring the traffic treatment.

Server Profiles screen

Clicking **Manage Profiles** in the Thresholds & Limits tab of the Server Groups and Protection screen displays the Server Profiles screen for defining sets of traffic characteristics that the appliance uses to identify traffic that is unacceptable to one or more servers. There are two separate profile types, each of which can be assigned to server groups independently. Each profile type is configured in its own screen, which is nested in this Server Profiles screen.

For each profile type, click **Add** to display the corresponding screen and begin creating a new one, or click the name of an existing profile to modify it.

All of these controls are available also through the corresponding CLI commands, as noted in each section below.

The Server Profiles screen provides these controls:

SYN Flood Profile

Specify a threshold value for the maximum number of connections. Three profiles are built-in: **megaproxy-server-load**, **normal-server-load**, and **virtual-server-load**.

You can access these controls also through the CLI command `set defense profiles server-profiles syn-flood-profile`.

Connection Limit Profile

Specify a maximum number of group connections (the default is 1048575) and a maximum number of member connections (the default is 32767). Four profiles are built-in: **low-limits**, **normal-limits**, **very-high-limits**, and **virtual-server-limits**.

You can access these controls also through the CLI command `set defense profiles server-profiles connection-limit-profile`.

Service Protocol screen

Service protocols specify how traffic corresponding to a particular application is handled by the appliance, and are an essential component of server protection. Click **Defense Configuration** > **Service Protocols** to display the Service Protocol screen. Alternatively, type the CLI command `set defense profiles service-protocol` to access the same functionality.

A large number of service protocols are built-in to the CMS and provided by default. You can modify these built-in service protocols or define your own custom ones, as necessary. Some of these built-in service protocols correspond to legitimate applications, or to applications that otherwise are not originally intended to be attack vectors, while others correspond to specific well-known attack vectors.

Refer to Configuring Server Protection to see how service protocols are associated with server protection instances.

Each service protocol is defined by the following parameters:

- **Name** – A descriptive name for the service. Built-in service protocols include the transport protocol and port in their names.
- **Description** – A description that further identifies the service. This field is examined by the Search function, so it is important to add a description that will help you pick out the new service from the list of services.
- **Connection Timeout** – The number of seconds without traffic for this service that the appliance should wait before timing out the connection.
- **Transport Protocol** – The protocol (IP, ICMP, TCP, or UDP) used by the service.
- **Transport Port** – The port number associated with the service.
- **Kind** – This can be **Any**, **Address Specification**, or **Host Group**.
- **Address Specification** – Define the server or group of servers to associate with this service. Enter a single IP address, range of IP addresses (e.g., 2.2.2.1-2.2.2.100, or 2.2.2.1/24), IP address range and port range (e.g., 221-225, 230), or combinations of these. Separate entries with commas.
  - **Discard Priority** – This value indicates how likely it is that the appliance will discard packets for this service during periods of extremely heavy traffic. Selecting a higher priority specifies that you want the traffic associated with this service to be more likely to be discarded. Choose **High**, **Medium** (the default), or **Low**.
  - **Flow Setup Threshold** – *Modifying this setting can adversely affect system operation. Contact* Corero Network Security *before changing this setting.* Choose **Always Create** (the default), **Never Create**, **Less Than 25 Percent**, **Less Than 50 Percent**, **Less Than 75 Percent**.
    This parameter indicates how critical it is to record connection information for this service. The appliance uses a flow table to record certain state information about each traffic connection. Under extreme traffic conditions, the flow table could become full.
    This parameter enables you to specify how critical it is to record connection information for a specific service. By not creating an entry in the flow table, you preserve the table for more critical services during periods of heavy traffic. For each service, you can choose whether you never want to create an entry in the flow table, or whether you always do. Alternatively, you can specify that the appliance create an entry in the table only if the table is less than a specific percentage full. This way, as the table fills, fewer flows are set up for lower priority traffic.

- **Request Limiting** – Click to enable this and restrict the number of requests that can be made by this service protocol.

Service Protocol Group screen

A service protocol group specifies the set of service protocols originating from a client group that will be handled with a particular treatment if encountered by the corresponding server group. Clicking **Defense Configuration** > **Service Protocol Groups** in the web UI tool bar displays the Service Protocol Group screen for defining collections of service protocols to be used in server protection instances. Alternatively, type the CLI command `set defense profiles service-protocol-group` to access the same functionality.

A number of service protocol groups are built-in and provided by default; these are basic groupings of commonly used protocols. You can define your own custom service protocol groups to collect service protocols for convenient use by your server protection instances.

Click **Add** in the Service Protocol Group screen to type a name for the group and create it. Choose one or more existing service protocols to include in the service protocol group.

Refer to Configuring Server Protection to see how service protocol groups are associated with server protection instances.

Status screen [Site Configuration]

Clicking **Site Configuration** > **General Status** in the web UI tool bar displays a Status screen .providing an overview of software version usage, CPU utilization, and maximum temperature across the site. The same information can be accessed using the CLI command `show site status`.

The following information is available:

- **Summary Status** – Normal.
- **Min Software Version** – Software loaded on an appliance in the site.
- **Max Software Version** – Software loaded on an appliance in the site.

Status screen [Defense Monitoring]

The Status screen summarizes the current behavior of the site's Defense, Bypass, and Forensics appliances, as well as very high-level summaries of excessive activity counts that may indicate an attack on your servers.

Click **Defense Monitoring** > **General Status** in the web UI tool bar to display the Status screen, or type the CLI command `show defense status` to display the same information.

Summary

- **Defense State** – The default value is Normal.
- **Bypass State** – The default value is Normal.
- **Forensics State** – The default value is Normal.

Threat Awareness

- **Excessive Packets From Bad Hosts** – The default value is False.
- **Excessive Proxy Setup Rate** – The default value is False.

- **Excessive New IP Addresses** – The default value is False.
- **Excessive Non-TCP Setup Rate** – The default value is False.
- **Excessive Failed Proxy Rate** – The default value is False.
- **Excessive TCP Setup Rate** – The default value is False.

SNMP screen

Click **CMS** > **SNMP** in the web UI tool bar to display the SNMP screen for configuring the site's SNMP behavior. The same functionality is accessible via the CLI command `set snmp`.

The SNMP screen provides these controls:

- Agent – Display the Agent screen for configuring the SNMP agent used by the CMS.
  - Enabled – Click this to activate the SNMP agent.
  - IP – Specify the IP address on which the SNMP agent will listen for notifications. Keep the default.
  - UDP Port – Specify the UDP port that the SNMP agent will use for communicating. The default is 161.
- System – Type descriptive information for the person responsible for administering SNMP for this CMS.
  - Contact
  - Name
  - Location
- Target – Display the Target screen to define a destination for the SNMP traps that the CMS sends.
  - Name – A descriptive name for the trap recipient.
  - IP – The trap recipient's IP address.
  - UDP Port – The UDP port on which the recipient listens for traps. The default is 162.
  - Tag – Any specific tag values used to identify the recipient address.
  - Timeout – This is for SNMP v3 only.
  - Retries – This is for SNMP v3 only.
  - Engine ID – This is for SNMP v3 only.
- Community – Display the Community screen for defining a new SNMP community.
  - Index
  - Name
  - Sec Name
  - Target Tag
- Notify – Display the Notify screen for defining additional notification targets.

Network Storage screen

Clicking **Site Configuration** > **Network Storage** in the web UI tool bar displays the Network Storage screen for specifying storage resources used by Forensics appliances. Alternatively, type the CLI command `set site packet-capture storage` to access the same functionality.

The Network Storage screen provides these controls:

Storage section – Obtain the name of the target from your SAN.

- **Add** – Click this to display the Storage screen for specifying the target of the packet capture data.
- **Delete** – Select an existing target in the list and click this to remove it from the site.
- **Volume Name** – The user-defined name of the target.
- **Device** – Choose the Forensics appliance from which the target will receive packet capture data.
- **Type** – Choose **Iscsi** or **RAM**.
- **Size** – Specify the target size, expressed in GB.
- **Target** – The formal name of the target, as obtained from your SAN.
- **Status** – The current state of the target.
- **Admin State** – Choose **Enabled** or **Disabled** to activate or de-activate the target.

Portal section – This is the IP address of the storage resource.

- **Add** – Click this to display the Portal screen for specifying the name, IP address, and port of the storage resource.
- **Delete** – Select an existing storage resource in the list and click this to remove it from the site.
- **Name** – The user-defined name of the storage resource.
- **Host** – The IP address or DNS name that identifies the storage resource.
- **Port** – The port on which the Forensics appliance communicates with the storage resource. The default is 3260.

Syslog and Splunk Configuration screen

Clicking **CMS** > **Syslog & Splunk Configuration** in the web UI tool bar displays the Syslog and Splunk Configuration screen. This screen provides controls for specifying recipients of the CMS's event and log data, such as a Splunk indexer that will be used with Corero's SecureWatch Analytics application. The same functionality is available through the CLI commands `set syslog remote-server` and `set syslog splunk-forwarder`.

The Syslog and Splunk Configuration screen provides these controls:

Syslog Servers – Click the **Add** button to specify a new syslog server, and provide the following information to define it.

- **Name** – A descriptive name for the syslog server.
- **Host** – The IP address or domain name of the syslog server.

- **Port** – The port on which the syslog server will listen for events. By default, this is **514**, the commonly used syslog port.
- **Protocol** – The Transport layer protocol used to carry the syslog messages to the server; choose **UDP** (the default) or **TCP**.

Splunk Forwarder - This controls the forwarding of event and log data from the CMS to listening Splunk indexers. The sending of events and logs is disabled by default; choose **Enabled** for each, as you require.

Splunk Indexers - Specify the **Name**, **Host** [IP address or domain name] and **Port** of one or more Splunk instances that will receive event and log data from the CMS and index it. The port defaults to **9997**, the commonly used Splunk indexing port.

Syslog Message Controls – These controls enable you to configure the behavior of some of the message types sent from the CMS via Syslog.

- **Type Event Message Limit** – Maximum number of security event syslog messages sent per second per rule.
- **Type Event Send Detected** – Enable sending of syslog messages for detected events. (This is **Enabled** by default.)
- **Type Sflow Message Limit** – Maximum number of Sflow syslog messages sent per second per direction.

System screen

Clicking **CMS** > **System Info** in the web UI tool bar displays the System screen, showing a summary of configuration information for your site:

- **Version** – The version of Corero Management Server software that is running currently.
- **OVA Version** – The version number of the virtual application software associated with the CMS software.
- **Config State** – By default, this is **Immediate**, meaning that configuration changes are sent to the appliances directly and without delay when **Save**/`commit` is executed.
- **Last Defense Config Change** – The time of the most recent saved change to the Defense appliance configuration.
- **Last Bypass Config Change** – The time of the most recent saved change to the Bypass appliance configuration.
- **Last Forensics Config Change** – The time of the most recent saved change to the Forensics appliance configuration.
- **Timezone** – This pulldown enables you to specify a timezone for the CMS explicitly, rather than simply inheriting the timezone of the CMS host.

System Actions screen

Clicking **CMS** > **CMS Actions** in the web UI tool bar displays the System Actions screen. The CLI equivalent is `request system-actions`.

The following commands are available:

- **Restart** – Exit the current CMS software process and start it again.
- **Reset** – Restore the CMS to its factory default configuration and restart it.

Threat Awareness screen

Clicking **Defense Monitoring** > **Threat Awareness** in the web UI tool bar displays the Threat Awareness screen, providing controls for adjusting the broad, high-level behavior of the Defense appliance's DDoS identification functionality. When one or more of these thresholds is crossed, the Defense appliance uses functionality that is optimized to respond to massive amounts of incoming traffic.

The equivalent CLI command is:`set defense threat-awareness [activation-thresholds | exit-thresholds]`

The Threat Awareness controls are:

Activation Thresholds – These parameters specify thresholds above which the Defense appliance characterizes incoming traffic as constituting a DDoS attack.

- **New IP Addresses** – The maximum number of new IP addresses allowed per second.
- **TCP Setup Rate** – If a DDoS attack is not under way, already, this is the rate at which TCP traffic will be allowed to pass before reporting an interval as an attack.

Exit Thresholds – These parameters specify thresholds below which the Defense appliance determines that it is not encountering DDoS attack traffic.

- **New IP Addresses** – The number of new IP addresses allowed per second.
- **TCP Setup Rate** – The TCP Setup Rate exit threshold; this should be lower than the TCP Setup Rate activation threshold.

Threat Update screen

The Threat Update screen enables you to configure the means by which the CMS will obtain updates to its threat definitions via Protection Packs and Intelligence Packs. You must have a valid Threat Update Service license to be able to obtain these updates.

Click **Defense Configuration** > **Threat Update Service** in the web UI tool bar to display the Threat Update screen. Access the same functionality in the CLI with the `set threat-update` command.

NOTE: If you use the CLI, remember to enter configuration mode prior to attempting any other configuration tasks. Do this by typing `configure` at the CLI command prompt.

The Threat Update screen provides these controls:

- **License Key** – Type your Threat Update Service license key. The CLI equivalent for this is `set threat-update license-key`.
- **Threat Update Server Address** – This is set explicitly to "tus.corero-cns.com", and cannot be modified.
- **Threat Update Schedule** – Click this link to display the Threat Update Schedule screen for specifying the frequency with which the CMS will seek Threat Update Service updates from the server and push them to managed Defense appliances. You can choose to query and push updates every hour (the default), every day at a certain time, or at some custom interval. The CLI equivalent for this is `set threat-update threat-update-schedule`.

- License Information – This section displays information about the Threat Update Service license associated with your license key. This includes the license's expiration date, current status, and the number of days remaining until it expires.

Threat Update Schedule screen

Clicking the **Threat UpdateSchedule** link in the Threat Update Service screen displays the Threat Update Schedule screen for specifying the frequency with which the CMS will seek Threat Update Service updates from the server and push them to managed Defense appliances. The CLI equivalent for this is `set threat-update threat-update-schedule`.

The Threat Update Schedule screen shows the update method that is in effect, currently. Click the **Choices** dropdown menu to change the update method:

- **Set Interval** – Specify a custom interval, expressed in minutes, at which the CMS will seek and then push updates.
- **Every Day At** – Specify a single time at which the CMS will seek and push updates every day.
- **Every Hour** – This is the default method. Seek and push updates every hour. You must select this option to activate it.

CMS Upgrade screen

Clicking **CMS** > **CMS Upgrade** in the web UI tool bar displays the Upgrade screen, enabling you to change the version of CMS software that you're running.

Click the **Browse** buttons to select the package upgrade file and, separately, the base upgrade file.

Click **Perform Upgrade** when you're ready to proceed. Note that, when you do this, the CMS will restart and you will be logged out of your session.

Upload CMS SSL Certificate screen

Clicking **Upload** CMS **SSL Certificate** in the CMS menu in the web UI tool bar displays the Upload CMS SSL Certificate screen for installing your own SSL certificate.

- **Certificate file (.crt)** – Browse to the location of the file you want to install to the CMS.
- **Certificate key file (.key)** – Browse to the location of the file you want to install to the CMS.

User screen

Click an existing user account to modify it, or click **Add** in the Users screen to display the User screen and begin defining a new user.

Refer to AAA (Authentication, Authorization, and Accounting) Overview for a general description of the CMS's user account management.

The User screen provides these controls, which are available also through the CLI command `set aaa authentication users user` **`username`**:

- **UID** – The numerical user ID value associated with this account.
- **GID** – The numerical group ID value associated with this account.
- **Password** – The password string for the account. This is displayed as a hashed value, but you can type a new password in clear text.

- **SSH Keydir** – The path to where this account's SSH key resides, if SSH is used during the authentication process.
- **Homedir** – The absolute path to this account's home directory.

Users & Authentication screen

Clicking **CMS** > **Users & Authentication** in the web UI tool bar displays the Users & Authentication screen. This screen lists all user accounts defined locally in the CMS (this does not include user accounts defined externally), but only users with admin or support-level privileges can see users other than themselves. This screen provides controls for adding users and managing their access; click **Add**, **Edit**, or **Delete** to manage the set of existing user accounts. The same controls provided here are available using the CLI command `set aaa`.

The user accounts "admin" and "support" are provided by default.

Refer to AAA (Authentication, Authorization, and Accounting) Overview for a detailed description of AAA for the CMS.

NOTE: It is possible to delete the default user accounts, "admin" and "support". Selecting one of these accounts and clicking **Delete** will remove that account immediately, without prompting for confirmation first.

Click **External Authentication** to display the External Authentication screen for configuring the CMS to use a third-party authentication regime, such as Active Directory or RADIUS.

Glossary

This section provides brief definitions of terms that have special meaning in the behavior of the Corero appliance and CMS. Explanations in greater detail are provided in the Concepts and Explanations section.

Action

When traffic triggers a security rule, the appliance responds with an *action*, which is part of the treatment of a security policy. The supported actions are:

- **Allow**: Pass the traffic to its destination.
- **Drop**: Block the traffic passively by stopping the traffic with no indication to the originating client.
- **Reject**: Block the traffic actively by stopping the traffic and executing a specific response, such as resetting the connection.

Each rule in a rule set has a user-modifiable action associated with it.

Allow

When a rule's action is set to Allow, the traffic that triggered the rule passes to its destination without restriction.

Block Active

When a rule's action is set to Block Active, the traffic that triggered the rule is stopped and a specific response is executed, such as resetting the connection.

Block Passive

When a rule's action is set to Block Passive, the traffic that triggered the rule is stopped with no indication given to the originating client.

Bypass Mode

When a Corero appliance or site's Defense mode is set to Bypass, incoming traffic is allowed to pass through the appliance and into the network without being inspected or treated in any way. In this case, the appliance behaves simply as a wire.

Client

For the purposes of the CMS, a client is the origin of a connection or flow.

Client Group

A client group is a list of IP addresses or countries from which incoming traffic is handled the same way by the appliances. Traffic from a client group that matches one or more client profiles will be responded to with a treatment that is defined as part of the server protection for the site.

Client Profile

Client profiles define sets of characteristics that constitute acceptable traffic behavior from clients.

Cluster

A cluster is a set of Corero appliances working together to handle inbound traffic on one network interface at a site. A cluster comprises one or two protection groups, depending on whether the cluster is configured for HA redundancy or not.

Customer

Each server group must have a customer name as part of its definition. There is a built-in customer named DEFAULT, so it is not necessary to define a new customer if you do not have multiple customers using your site. Although it is necessary to have a customer as part of each server group definition, customers are not binding in any other respect, nor do they contribute to any other definition. There are Rank and Status parameters associated with customers, but these are provided for use at the user's discretion. The Rank and Status parameters provide another means for customers to be sorted, but they do not contribute to the definition of any other aspects of the configuration.

Data Model

A data model describes how data is represented and accessed in the CMS, providing a structured summary of the elements that need to be specified to configure a functional component.

DDoS Attack

A distributed denial of service (DDoS) attack is an event in which multiple sources assault a destination server with a volume of traffic that overwhelms that server's ability to respond to any traffic at all, making it unavailable to the legitimate traffic for which it is intended.

DDoS attacks can take many different forms and exploit a wide variety of protocol behaviors and design features to achieve their goal of making a server unavailable.

Defense Mode

The Defense mode sets the overall traffic handling behavior for a site and for a protection group. The Defense mode can be set to one of the following:

- **Bypass** – Incoming traffic is allowed to pass through the Corero appliance and into the network without being inspected or treated in any way. In this case, the appliance behaves simply as a wire.
- **Mitigate** – Incoming traffic is inspected, logged, and treated as required by an applicable server protection instance.
- **Monitor** – Incoming traffic is inspected and logged according to an applicable server protection instance, but is allowed to pass through to the network without being treated. This is the default Defense mode, and new Defense appliances are set to Monitor mode until they are configured explicitly to behave differently.

For a protection group, you can also choose **Inherit Global**, which causes the protection group to use the same setting specified for the site's Global Defense Mode.

Defense Pack

Defense pack is a collective term for intelligence packs and protection packs that are available through Corero's Threat Update Service.

Detect

When a rule's action is set to Detect, the traffic that triggered the rule is identified and logged, but is otherwise passed to its destination without restriction.

Device

A device is any purpose-built security appliance provided by Corero Network Security. A device can be a physical hardware unit, such as a appliance, or it can be a software-only virtual appliance, such as a vDDP. Functionally, the two forms are similar, and each qualifies as a device for the management purposes of the CMS.

Header Blocking

Header blocking enables you to define a set of conditions for refusing entry to your network by specific packet header types.

Host Group

"Host group" is a generic term for client groups and server groups that can apply to either or both.

Inline Mode

Inline mode is the Bypass appliance's equivalent of Mitigate mode. When the Bypass appliance is set to Inline mode, incoming traffic is passed to attached appliances for them to handle as they are configured to do.

Intelligence Pack

Threat Update Service subscribers receive Intelligence Packs that comprise IP address-specific reputation and geographic information, updated frequently.

Malicious IP Address

A host that the appliance has determined is behaving maliciously, based on criteria configured in the CMS. The appliance drops requests from malicious hosts.

Management Address

The IP address of the interface on a device that is used to handle administrative traffic .separately for "mission" (customer) traffic.

Mitigate Mode

When an appliance or cluster's Bypass mode is set to Mitigate, incoming traffic is inspected, logged, and treated as required an applicable server protection instance.

Monitor Mode

When an appliance or cluster's Bypass mode is set to Monitor, incoming traffic is inspected and logged according to an applicable server protection instance, but is allowed to pass through to the network without being treated.

Monitor mode is the default setting for newly added appliances and protection groups.

Profile

Profiles are definitions of acceptable traffic from clients and definitions of responses to traffic by servers.

Protection Group

A protection group is the unit of HA redundancy in the SmartWall Network Threat Defense Appliance. A protection group is associated with a cluster. If the appliances in the cluster operate individually, the cluster comprises a single protection group only. However, if the appliances in the cluster are configured for HA redundancy, then there will be two protection groups; if an appliance in one protection group fails, the second protection group will take over the handling of traffic.

Protection Pack

Protection packs are available through Corero's Threat Update Service. Protection packs are issued frequently, and provide frequent automated updates, technical support, security advisory and software subscription services, along with access to Corero's knowledge base and special delivery programs. These updates help to protect against zero-day threats and provide resolution to known security problems.

Request Limiting

Request limiting is a rate-based filter for UDP. It is based on a configured acceptable number of transactions per minute. If a client reaches that number, it is cut off until the next minute. If a pattern of overuse emerges, the client will be blocked completely until a configured hiatus period has expired.

Rule

Rules are building blocks of security policies; they identify traffic that may be suspicious in origin, malicious in intent, and harmful in its results.

Rule Set

A rule set is a named collection of rules that is configured to identify and treat traffic for a particular purpose.

Security Event

A security event is a record indicating that traffic entering the appliance has triggered a rule. A security event records the source and destination of a traffic flow, the rule(s) that the flow triggered, and the action that the appliance performed on that flow.

Security Policy

A *security policy* specifies a set of actions to be performed on incoming traffic that exhibits certain specific characteristics. Security policies are the logical constructions that inform the appliance's security decisions as it examines traffic flows to protect your network.

See Security Policies for a complete explanation.

The Security Policies screen provides the controls you need for configuring security policies.

Server

For the purposes of the CMS, a server is the destination of a connection or flow.

Server Group

A server group is a collection of servers that share common expectations of what constitutes acceptable traffic. These expectations comprise both access control (network protocols and client treatments) and rate control (SYN flood limits and client connection limits).

A server, as identified by its IP address, can be a member of only one server group at a time.

Server Protection

Server protection and server profiles comprise the set of responses that are configured to occur when incoming traffic matches one or more profiles.

Service Protocol

A service protocol specifies how traffic corresponding to a particular application is handled by the appliance. It comprises the transport protocol and port used by the application, and can include settings such as a connection timeout, a client group or another set of originating addresses, and characteristics such as discard priority, flow setup threshold, and request limiting. Service protocols are combined with server groups to make up server protection instances.

A large number of service protocols are built-in to the CMS and provided by default. You can modify these built-in service protocols or define your own custom ones, as necessary. Some of these built-in service protocols correspond to legitimate applications, or to applications that otherwise are not originally intended to be attack vectors, while others correspond to specific well-known attack vectors.

Shun Label

When shunning, you group IP addresses into collections called shun labels. You can assign IP addresses to a shun label based on any criteria you desire, including by common features of the IP source system, by date, or by attack type. The appliance supports up to 256 shun labels.

Shunning

appliances have an effective protection capability called *shunning* that can quickly block traffic from IP addresses, temporarily or permanently, that are suspected of originating or participating in an attack.

Site

A site is the set of appliances and policies that are managed by a single instance of the CMS.

Although the appliances and the servers that they protect should reside in the same location, the definition of a site, for the purposes of the appliance and the CMS, is not a geographical term. Also, while the CMS manages the protection of servers within your organization's firewall, the CMS does not manage the servers themselves; the servers can be associated with the site using elements such as server groups, but the constituent servers are not among the devices managed by the CMS.

Suspicious IP Address

A host that exhibits sufficient incorrect behavior that it does not merit being trusted. The appliance proxies their requests on behalf of the intended server.

SYN Flood

A SYN flood is a form of denial-of-service attack in which an attacker sends a succession of SYN requests to a target system in an attempt to consume enough server resources to make the system unresponsive to legitimate traffic. If an appliance receives an inappropriate number of SYN requests from one or more clients, these packets are dropped.

Treatment

A treatment is a response taken by the appliance .when traffic bound for a server from a client group matches one or more profiles. A treatment consists of an action and, possibly, a log entry at some severity that are triggered when inbound traffic matches one or more client or server profiles.

Treatments use rule sets, also, to determine the way in which the appliance should respond to incoming traffic.

Trusted IP Address

A host that exhibits behavior that is within the configured boundaries of acceptable traffic. The appliance sends requests from these hosts on to their destination servers to handle.

Unknown IP Address

The host IP address is identified, but the appliance has not yet determined its threat level. Depending on your configuration, the appliance may or may not proxy requests from hosts in this category.

Command Line Interface (CLI)

The CMS provides a JunOS-like command line interface (CLI) that can be used for all configuration and monitoring tasks.

Accessing the CMS CLI

Access the CMS CLI from the CMS web UI by clicking **CLI** in the web UI tool bar, or access it directly using a terminal emulator (such as PuTTY) by initiating an SSH session to the CMS's management interface on port 2024.

The CMS provides a command line interface (CLI) that can be used for all configuration tasks, including:

- Configuring all aspects of the CMS. Type `configure` at the CLI prompt to enter configuration mode.

- Inspecting the configuration and state of Corero appliances.
- Monitoring and inspecting log files.

The CLI is derived directly from the data models that describe the configuration of the objects managed by the CMS, and provides you with full access to those models for configuration and monitoring. The CLI supports these functions through two command modes: operational mode and configure mode.

The CLI provides a number of amenities for ease-of-use, including tab-completion of partial commands and full command line history.

Type `?` at any time to display a list of commands available in the current mode, or type `?` after a command name to display a brief help description for that command.

NOTE: The CLI requires use of the ASCII 0x34 double quotation mark (") symbol for both left and right quoting of strings when spaces or special characters are used in the input of a single field. Use of the left and right double quotation marks from the ISO and extended sets (8220 and 8221) is not supported, and will result in an error.

Operational Mode

The CMS CLI starts in operational mode. Use operational mode for displaying information about the CMS and the appliances that it manages. Use configure mode if you want to change some part of the configuration.

Operational mode provides the following commands:

- `compare` – Compare running configuration to another configuration or a file
- `configure` – Manipulate software configuration information
- `exit` – Exit the management session
- `file` – Perform file operations
- `help` – Provide help information
- `id` – Show user id information
- `quit` – Exit the management session
- `request` – Make system-level requests
- `script` – Script actions
- `set` – Set CLI properties
- `set-path` – Set relative show path
- `show` – Show information about the system
- `source` – File to source
- `top` – Exit to top level and optionally run command
- `up` – Exit one level of configuration

Configuration Mode

Type `configure` to enter the command mode for changing the configuration. Use operational mode for displaying information about the CMS and the appliances that it manages.

Configuration mode provides the following commands:

- `activate` – Activate a statement
- `annotate` – Add a comment to a statement
- `commit` – Commit current set of changes
- `compare` – Show configuration differences
- `copy` – Copy a list entry
- `deactivate` – Deactivate a statement
- `delete` – Delete a data element
- `edit` – Edit a sub-element
- `exit` – Exit from this level
- `help` – Provide help information
- `insert` – Insert a parameter
- `load` – Load configuration from an ASCII file
- `move` – Move a parameter
- `quit` – Exit from this level
- `rename` – Rename an identifier
- `request` – Make system-level requests
- `resolved` – Conflicts have been resolved
- `revert` – Copy configuration from running
- `rollback` – Roll back database to last committed version
- `run` – Run an operational-mode command
- `save` – Save configuration to an ASCII file
- `save-device-config` – Save device settings
- `set` – Set a parameter
- `show` – Show a parameter
- `status` – Display users currently editing the configuration
- `tag` – Manipulate statement tags
- `top` – Exit to top level and optionally run command
- `up` – Exit one level of configuration
- `validate` – Validate current configuration

Showing CLI Output

Use the `show` command in the CLI's operational mode to display information about the appliance and CMS configuration.

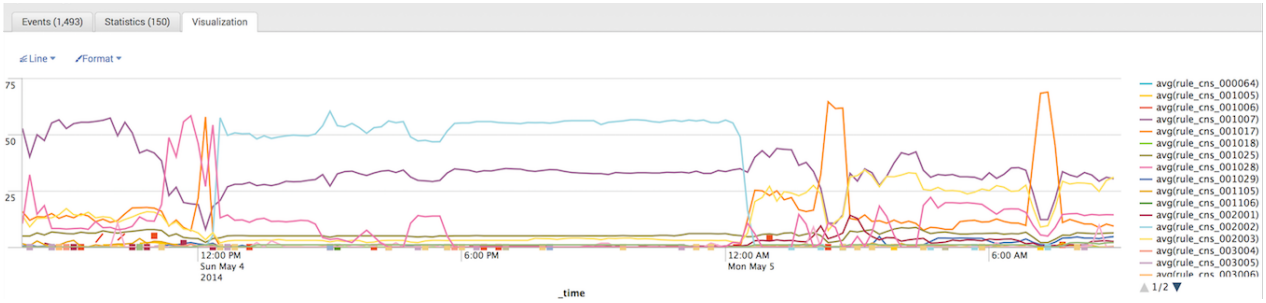
Syslog Overview

This section provides an overview of syslog support in the SmartWall Threat Defense System.

Introduction

This document describes methods and operations of how an external Security Information and Event Management System (SIEM) can receive security events and operational data from the SmartWall Corero Management Server (CMS) in a Syslog format. Corero is open to discussions with partners who are interested in receiving events through both mechanisms for a tighter integration.

The Corero CMS that manages a set of Network Bypass, Threat Defense, and Forensics appliances can send operational, forensics, and event data to SIEMs via syslog messages. The overall message is carried in a format specified by syslog reporting protocol (RFC 3164), but the details of the contents are provided in this document. The figure below shows an example of a Splunk dashboard created from the the CMS via syslog over a period of time. Based on the types of events, a dashboard like this can provide insight into the types of attacks and unwanted traffic that may be encroaching a customer network.



Example of Attack Analysis in a Customer Network

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Data Operations and Formats

Overview

With simple configuration of an enterprise's SIEM's IP addresses and ports, the Corero Management Server will start sending

1. Security events received from the SmartWall appliances
2. Periodic reports of recent activity including the top clients, servers, destination ports, and security events
3. Periodic reports of summarized key statistics across the site, including mission port traffic, resource usage, drop rates, and threat indications

Each of the events and reports are sent within a single syslog message with each element of security data represented by a simple name=value format designed for automatic parsing and manipulation by the leading SIEM tools in the field.

While the tools make collection and basic viewing of this data very easy, this document provides the background information that helps in the interpretation and deeper analysis of this valuable security information.

The data communicated via syslog to SIEM partners is based upon the same data that is available through the other CMS northbound interfaces including the CMS dashboard in the WebUI, the REST API, and CLI. In order to access this data through the other interfaces, please consult the Corero Management System User's Guide.

Syslog Message Format

The basic syslog message as defined in RFC 5424 has many different operations and paramters. However, because most common tools are built to understand the basic syslog structure, it is not necessary to define any more than the basic header contents that are populated when syslog messages are generated by the CMS. As received and presented by a common SIEM, a basic message structure looks like:

```
Jan 25 17:36:03 10.20.27.202 Jan 25 17:36:01 cms2-
global  < security data content >
```

Basic SIEM Data Format

In this example

- Jan 25 17:36:03 10.20.27.202 is the timestamp and source of the message as received by the SIEM.
- Jan 25 17:36:01 cms2-global is the time of message generation at the CMS and its hostname. This is standard syslog header data prepended to the security data as the message is formatted for transmission across the network.
- The security data content is the information described in this document.

Remembering that the date and time of these fields is the creation time of the syslog message, not the originating data, is important for proper understanding of the data. It is very useful information because much of the received data is periodically generated by the CMS as it does its analysis of the security information it is receiving from the devices under its control so the time of generation of this data corresponds to the time of creation of the report data.

Corero Syslog Information

Information Categories

There are 3 categories of information to provide a complete view of the operation of the system. These categories are:

- Security – Specific security events and summaries of the security statistics
- Network – Individual appliance and summarized network operations statistics
- System – Events and statistics relating to the health and operations of the appliances

Protection Groups and Clusters

A protection group is the unit of protection and security services unit for a single 10Gbps link. A cluster comprises one or two protection groups, depending on whether the cluster is configured for HA redundancy or not.

The status and statistics for a cluster and protection group are indicated in the syslog message with the cl and pg keywords. Every 60 seconds the status and statics for all protection groups are sent in syslog messages. Additionally, the status and statistics for the site as the aggregate of all of the protection groups are sent. In these messages, the cl and pg keywords contain the value of "all".

Common Types and Formats

All messages and values within the security data content of a syslog message have consistent formatting across similar information content and information structure (e.g. the representation of tabular information). By understanding some of the common representations as they are made visible in the receiving tools, most of the messages are relatively self-describing. This section provides a description of these common representations.

name = value

All data is represented as a sequence of name=value pairs, possibly within a JSON table representation based on the data. name is a short name optimized for syslog transmission and SIEM query construction, with consideration for ease of understanding from the full field name. value is the value of the named data. If names consist of more than one word, they are separated by a "-" (dash). There are no spaces in name=value.

Field separators

All fields (name=value) are separated by commas. There are no spaces before or after the comma.

Common keywords

There are several keywords that appear in the different messages. When these names appear, they can always be interpreted as defined below.

Field Name	Syslog Short Name	Definition
Category	cat	All messages start with the category of the message. The types are security, network, system, ops, diags, rt
Type	type	Type within the category
Version	v	Version of the Corero Syslog message format
Cluster	cl	The cluster of protection-groups in which the appliance is configured. The value "all" is used when data from all clusters are aggregated in the message

Protection Group	pg	The protection group in which the appliance is operating. The value “all” is used when data from all protection groups are aggregated in the message
Device/Appliance Name	aname	The name provisioned for a Corero Network device/appliance
IP	ip	An IP address. It could be a source or destination based on the conString of the message
Source IP address	sip	The source IP address of a data packet that was processed in some way to create this data
Destination IP address	dip	The destination IP address of a data packet that was processed in some way to create this data
Source Port	sprt	The source IP port of a data packet that was processed in some way to create this data
Destination Port	dprt	The destination IP port of a data packet that was processed in some way to create this data
Packet Length	plen	The length of the packet including the Ethernet header but not including the 4-byte CRC
IP Packet Length	iplen	The length of the IP packet including the IP header
Rule Name	rule	The Corero security event rule name (e.g. cns-001005)

Common Types

Multiple values or entries

When there are multiple entries associated with the data in a message, e.g. a top-sip report of the top 10 source IP addresses observed over the the last time period, the different entries are indicated by the index number appended directly to the name, e.g.

```
ip1=3.4.5.6,ip2=5.6.7.8,ip3=1.2.3.4,ip4=2.3.4.5.
```

This is less structured than is possible in other representations of tabular data such as through an XML or JSON, but this is designed to allow for quick summarization via name* wildcarding. More sophisticated structures can be parsed by common tools, but plugins and additional tools are often required to perform the processing.

Syslog Messages

This section defines the contents of each syslog message generated by the CMS by category

Security Category

Block Rate Statistics Type

Every 60 seconds the CMS creates a report that summarizes rate of packet blocking in the system for each of the overall classifications for security rules on the Defense appliances. There are several hundred security rules that may be protecting a specific network, and these rules are categorized into 8 categories so that the block rate per classification can be calculated. These classifications are provided below.

Layer	Access Restrictions (AR)	Rate Limits (RL)	Protocol Validations (PV)	Integrity Analysis (IA)
7 (APP)	Classification: ARAPP Example: WAF	Classification: RLAPP Example: RRB	Classification: PVAPP Example: PVMs	Classification: IAAPP Examples: The "unsolicited" checks
3/4 (NET)	Classification: ARNET Examples: Server Protection, Geo, Rep, Shunning, IP blocking, NTP Reflection	Classification: RLNET Examples: rate limits	Classification: PVNET Examples: Packet rules	Classification: IANET Examples: Fragment rules, tunnel rules

The following is an example how this report is sent to configured syslog receivers and a description of the contents.

```
Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-
cms.corero.com cat=security,v=1,cl=all,pg=all,
type=block-
rate,arbr=800343,narbr=600450,aarbr=0,nrlbr=0,arlbr
=0,npvbr=40003,apvbr=0,niabr=0, aiabr=0,
```

Example Block Rate Message

The specific content for the block rate message is:

Field Name	Short Name	Specific Value / Type	Description
Category	cat	security	Security event category
Type	type	block-rate	Block rate statistics per rule classification
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
all-rules-block-rate	arbr	Rate	The total rate of packet blocking for all rules in the system
network-access-restriction-block-rate	narbr	Rate	The rate of packet blocking for rules of the ARNET classification
application-access-restriction-block-rate	aarbr	Rate	The rate of packet blocking for rules of the ARAPP classification
network-rate-limit-block-rate	nrlbr	Rate	The rate of packet blocking for rules of the RLNET classification
application-rate-limit-block-rate	arlbr	Rate	The rate of packet blocking for rules of the RLAPP classification
network-protocol-validation-block-rate	npvbr	Rate	The rate of packet blocking for rules of the PVNET classification
application-protocol-validation-block-rate	apvbr	Rate	The rate of packet blocking for rules of the PVAPP classification
network-integrity-analysis-	niabr	Rate	The rate of packet blocking for rules of the

block-rate			IANET classification
application-integrity-analysis-block-rate	aiabr	Rate	The rate of packet blocking for rules of the IAAPP classification

Top Security Rules Type

Every 60 seconds the CMS creates a report that contains the security rules that had the highest blocked packet rates over the time period. There will be up to 10 entries in each message and each one contains the rule that fired and the percentage of the samples. The following is an example how this report is sent to configured syslog receivers and a description of the contents.

```
Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-
cms.corero.com cat=security, type=top-rules,
v=1,cl=all,pg=all,rule1=cns-
100027,pcent1=93,rule2=cns-
001025,pcent2=4,rule3=cns-001017, pcent3=2,
```

Example Top Security Rules Message

The specific content for the top-rules message is shown below:

Field Name	Short Name	Specific Value / Type	Description
Category	cat	security	Security event category
Type	type	top-rules	Most frequent occurences of security rules over the time period
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
Rule <rank>	rulex	Text: cns-xxxxxx	The Corero security event rule at the rank of <rank> for this time period
Percentage <rank>	pcentx	Number	The percent of the events at this rank that this rule represents

IP Address Classifications Type

Every 60 seconds the CMS creates a report that summarizes the classification of the IP addresses known across the Defense appliances. The following is an example how this report is sent to configured syslog receivers and a description of the contents.

```
Apr 6 12:56:34 12.12.12.5 Apr 6 08:58:03 server35-
cms cat=security,type=ip-class,
v=1,cl=all,pg=all,iua=6230138,iuta=56577,iusa=885,i
uma=48,iuua=6172628,
```

Example IP Address Classifications Message

The specific content for the IP address classification is defined below:

Field Name	Short Name	Specific Value / Type	Description
Category	cat	security	Security event category
Type	type	ip-class	IP address classification counts in the system
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
in-use-addresses	iua	Number	Total number of addresses in use
in-use-trusted-addresses	iuta	Number	IP addresses in use classified as trusted
in-use-suspicious-addresses	iusa	Number	IP addresses in use classified as suspicious
in-use-malicious-addresses	iuma	Number	IP addresses in use classified as malicious
In-use-unclassified-addresses	iuua	Number	IP addresses in use not yet classified

Rule Statistics Type

Every 60 seconds the CMS creates one message for each active security rule in the system that contains the counts of the number of packets blocked or just detected and the number of security events issued for these two situations for that rule. The following is an example how this report is sent to configured syslog receivers and a description of the contents.

```
Apr 6 12:56:34 12.12.12.5 Apr 6 08:58:03
cat=security,type=rule-stats,v=1,cl=all,pg=all,
name=cns-001005,rule=cns-001005, desc=PVNET:
Connection containing UDP port zero,
bec=1819,bpc=1819,dec=0,dpc=0,bpr=34,dpr=0,
```

Example Rule Statistics Content

The specific content for the security event data is defined below:

Field Name	Short Name	Specific Value / Type	Description
Category	cat	security	Security event category
Type	type	rule-stats	Security rule counts in the system
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
rule-name	name	Text: cns-xxxxxx	The Corero security event rule name Deprecated. To be removed in V8.10

rule-name	rule	Text: cns-xxxxxx	The Corero security event rule name (e.g. cns-001005)
rule-description	desc	Text	Short description that includes rule classification
block-event-count	bec	Number	Total number of block events for this rule
block-packet-count	bpc	Number	Total number of packets blocked by the rule
detect-event-count	dec	Number	Total number of detect events for this rule where the packet is not blocked
detect-packet-count	dpc	Number	Total number of packets detected by the rule
block-packet-rate	bpr	Rate	The rate of packets being blocked for the rule
detect-packet-rate	dpr	Rate	The rate of events being detected for the rule

Threat Awareness Type

Every 60 seconds the CMS creates a report that summarizes threat awareness status for a set of significant internal indicators across the Defense appliances. The following is an example how this report is sent to configured syslog receivers and a description of the contents.

```
Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-
cms.corero.com cat=security,type=threats,
v=1,cl=all,pg=all,epfbc=0,enia=0,eatu=0,entsr=0,ets
r=0,
```

Example Threat Awareness Message

The specific content for the security threat indicators is:

Field Name	Short Name	Specific Value / Type	Description
Category	cat	security	Security event category
Type	type	threats	Threat awareness indications status
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
excessive-packets-from-bad-clients	epfbc	Boolean (1/0)	Indicates the current status of this threat indication
excessive-new-ip-addresses	enia	Boolean (1/0)	Indicates the current status of this threat indication

excessive-address-table-usage	eatu	Boolean (1/0)	Indicates the current status of this threat indication
excessive-non-tcp-setup-rate	entsr	Boolean (1/0)	Indicates the current status of this threat indication
excessive-tcp-setup-rate	etsr	Boolean (1/0)	Indicates the current status of this threat indication

Network Category

Interface Statistics Type

Every 60 seconds the CMS creates a report that summarizes interface statistics for the external (from the Internet) and internal (toward the firewall and servers) ports of the protection groups on the Defense appliances. The following is an example how this report is sent to configured syslog receivers and a description of the contents.

```
Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-
cms.corero.com cat=network,type=interface,
v=1,cl=all,pg=all,eprp=230922943,eptp=181472787,epr
b=207772986979,eptb=61164066743,epbrr=4,epprr=670,e
pptr=443,epbtr=0,ipptr=670,ipbtr=4,ipprr=443,ipbrr=
0,iprp=181473093,iptp=230922664,
iprb=61164161900,iptb=207772844918,
```

Example Interface Statistics Message

The specific content for the interface statistics message is:

Field Name	Short Name	Specific Value / Type	Description
Category	cat	network	Network event category
Type	type	interface	Protection group interface input and output usage statistics
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
external-port-packet-receive-rate	epprr	Rate	External mission port(s) aggregrate rate in packets per second
external-port-bit-receive-rate	epbrr	Rate	External mission port(s) aggregrate rate in Mbits per second
external-port-packet-transmit-rate	epptr	Rate	External mission port(s) aggregrate rate in packets per second
external-port-bit-transmit-rate	epbtr	Rate	External mission port(s) aggregrate rate in Mbits per second

internal-port-packet-receive-rate	ipprr	Rate	Internal mission port(s) aggregrate rate in packets per second
internal-port-bit-receive-rate	ipbrr	Rate	Internal mission port(s) aggregrate rate in Mbits per second
internal-port-packet-transmit-rate	ipptr	Rate	Internal mission port(s) aggregrate rate in packets per second
internal-port-bit-transmit-rate	ipbtr	Rate	Internal mission port(s) aggregrate rate in Mbits per second
external-port-received-packets	eprp	Number	External mission port(s) aggregrate count
external-port-transmitted-packets	eptp	Number	External mission port(s) aggregrate count
external-port-received-bytes	eprb	Number	External mission port(s) aggregrate count
external-port-transmitted-bytes	eptb	Number	External mission port(s) aggregrate count
internal-port-received-packets	iprp	Number	Internal mission port(s) aggregrate count
internal-port-transmitted-packets	iptp	Number	Internal mission port(s) aggregrate count
internal-port-received-bytes	iprb	Number	Internal mission port(s) aggregrate count
internal-port-transmitted-bytes	iptb	Number	Internal mission port(s) aggregrate count

Top Destination IP Addresses Type

Every 60 seconds the CMS creates a message that contains the most frequently seen destination IP addresses over the time period as determined by sflow sampling of the data inbound from the Internet into the site. There will be up to 10 entries in each message and each one contains the IP address and its percent of the samples. The following is an example how this message is sent to configured syslog receivers and a description of the contents.

```
Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-
cms.corero.com cat=network,type=top-dip,
v=1,cl=all,pg=all,samples=6,ip1=208.87.136.155,pcen
t1=67,ip2=98.229.163.222,pcent2=17,
ip3=66.203.81.178,pcent3=17,
```

Example Top Destination IP Address Message

The specific content for the top-dip message is shown below:

Field Name	Short	Specific Value /	Description

	Name	Type	
Category	cat	network	Network event category
Type	type	top-dip	Top destination IP addresses
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
Samples	samples	Number	Number of sampled packets for this time period
IP <rank>	ipx	IP Address	The destination IP address at the rank of <rank> for this time period
Percentage<rank>	pcentx	Number	The percent of the sampled packets at this rank that this IP address represents

Top Source Ports Type

Every 60 seconds the CMS creates a message that contains the most frequently seen source IP ports over the time period as determined by sflow sampling of the data inbound from the Internet into the site. There will be up to 10 entries in each message and each one contains the IP port and its percent of the samples. The following is an example how this message is sent to configured syslog receivers and a description of the contents.

```
Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-
cms.corero.com cat=network,type=top-sprt,
v=1,cl=all,pg=all,samples=6,port1=41520,pcent1=67,p
ort2=64009,pcent2=17,port3=0,pcent3=17,
```

Example Top Source Port Message

The specific content for the top-sip message is show below:

Field Name	Short Name	Specific Value / Type	Description
Category	cat	network	Network event category
Type	type	top-sprt	Top source ports
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
Samples	samples	Number	Number of sampled packets for this time period
Port <rank>	portx	Number	The source port at the rank of <rank> for this time period
Percentage <rank>	pcentx	Number	The percent of the sampled packets at this rank that this port represents

Top Destination Ports Type

Every 60 seconds the CMS creates a message that contains the most frequently seen destination IP ports over the time period as determined by sflow sampling of the data inbound from the Internet into the site. There will be up to 10 entries in each message and each one contains the port and its percent of the samples. The following is an example how this message is sent to configured syslog receivers and a description of the contents.

```
Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-
cms.corero.com cat=network,type=top-dprt,
v=1,cl=all,pg=all,samples=6,port1=41520,pcent1=67,po
rt2=64009,pcent2=17,port3=0,pcent3=17,
```

Mar =6,port1=41520,pcent1=67,port2=64009,pcent2=17,port3=0,pcent3=17,

Example Top Destination Port Message

The specific content for the top-dprt message is shown below:

Field Name	Short Name	Specific Value / Type	Description
Category	cat	network	Network event category
Type	type	top-dprt	Top destination ports
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
Samples	samples	Number	Number of sampled packets for this time period
Port <rank>	portx	Number	The destination port at the rank of <rank> for this time period
Percentage <rank>	pcentx	Number	The percent of the sampled packets at this rank that this port represents

Top Time-to-Live (ttl) Type

Every 60 seconds the CMS creates a message that contains the most frequently seen Time-to-Live (ttl) fields over the time period as determined by sflow sampling of the data inbound from the Internet into the site. There will be up to 10 entries in each message and each one contains the ttl and its percent of the samples. The following is an example how this message is sent to configured syslog receivers and a description of the contents.

```
Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-
cms.corero.com cat=network,type=top-ttl,
v=1,cl=all,pg=all,samples=6,ttl1=255,pcent1=67,ttl2=
64,pcent2=33,
```

Mar =6,port1=41520,pcent2=64009,pcent2=17,port3=0,pcent3=17,

Example Top Time-to-Live Message

The specific content for the top-ttl message is shown below:

Field Name	Short Name	Specific Value / Type	Description
Category	cat	network	Network event category
Type	type	top-ttl	Top Time-to-Live
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
Samples	samples	Number	Number of sampled packets for this time period
TTL <rank>	ttlx	Number	The TTL at the rank of <rank> for this time period
Percentage <rank>	pcentx	Number	The percent of the sampled packets at this rank that this ttl represents

Top Packet Length Type

Every 60 seconds the CMS creates a message that contains the most frequently seen specific packets lengths fields over the time period as determined by sflow sampling of the data inbound from the Internet into the site. There will be up to 10 entries in each message and each one contains the length and its percent of the samples. The following is an example how this message is sent to configured syslog receivers and a description of the contents.

```
Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-
cms.corero.com cat=network,type=top-length,
v=1,cl=all,pg=all,samples=665,length1=78,pcent1=72,l
ength2=64,pcent2=28,
```

Mar =6,port1=41520,pcent2=64009,pcent2=17,port3=0,pcent3=17,

Example Top Packet Length Message

The specific content for the top-length message is shown below:

Field Name	Short Name	Specific Value / Type	Description
Category	cat	network	Network event category
Type	type	top-length	Top packet length
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
Samples	samples	Number	Number of sampled packets for this time period
Length<rank>	lengthx	Number	The length at the rank of <rank> for this time period
Percentage <rank>	pcentx	Number	The percent of the sampled packets at this rank that this ttl represents

Usage Statistics Type

Every 60 seconds the CMS creates a report that summarizes usage statistics on the Defense appliances. The following is an example how this report is sent to configured syslog receivers and a description of the contents.

```
Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-
cms.corero.com cat=network,type=usage,
v=1,cl=all,pg=all,iuf=503343,iutf=352434,iuuf=10556
56,iuif=34342, iuof=14566,tps=34340, tpf=120,
```

Example Usage Statistics Message

The specific content for the usage message:

Field Name	Short Name	Specific Value / Type	Description
Category	cat	network	Network event category
Type	type	usage	Key resource usage statistics
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
in-use-flows	iuf	Number	Total in use flows
in-use-tcp-flows	iutf	Number	In use TCP flows
in-use-udp-flows	iuuf	Number	In use UDP flows
in-use-icmp-flows	iuif	Number	In use ICMP flows
in-use-other-flows	iuof	Number	In use non-TCP, UDP, or ICMP flows
total-proxy-setups	tps	Number	Total flow setups using a proxy mechanism
total-proxy-fails	tpf	Number	Total proxy setup failures

Setup Rate Statistics Type

Every 60 seconds the CMS creates a report that summarizes setup rate statistics for flows on the Defense appliances. The following is an example how this report is sent to configured syslog receivers and a description of the contents.

```
Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-
cms.corero.com cat=network,type=setups,
v=1,cl=all,pg=all,tsr=134343,ntsr=74342,usr=73432,i
sr=32220,oisr=42420,
```

Example Setup Rate Statistics Message

The specific content for the setups message is:

Field Name	Short Name	Specific Value / Type	Description
Cat	cat	network	Network event category
Type	type	setups	Key resource usage statistics
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group of these statistics
tcp-setup-rate	tsr	Number	Current TCP flow setup rate
non-tcp-setup-rate	ntsr	Number	Current non-TCP flow setup rate
udp-setup-rate	usr	Number	Current UDP ‘flow’ setup rate
icmp-setup-rate	isr	Number	Current ICMP ‘flow’ setup rate
other-ip-setup-rate	oisr	Number	Current non-UDP and TCP ‘flow’ setup rate

Sflow Event Type

Each Network Threat Defense appliance samples the data stream and sends the samples to the CMS for threat and operational analysis. A subset of these events are sampled as well and sent as syslog messages for further integration in the SIEM analysis stream. Five messages per second are forwarded as syslog events.

Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-cms.corero.com cat=network,type=sflow, v=1, cl=NYC_East,pg=NYC_East_
1,aip=10.20.27.111,dir=inbound,dip=10.222.1.5,dprt=53,iplen=46,prot=17,
svc=0,sip=10.121.106.165,sprt=7319,ttl=64,flags=0,time=1413469709753000,plen=60,
pdu=00000000010200000000010108004500002e000000004011cc540a7498140ade01058a1e00350
01a7eaa000102030405060708090a0b0c0d0e0f1011,

Example SFlow Sample Message

The specific content for the sflow message is:

Field Name	Short Name	Specific Value / Type	Description
Category	cat	network	Network category
Type	type	sflow	Sflow sample event
Version	v	Number	The Corero message format version
Cluster	cl	Text	The cluster of the protection group
Protection Group	pg	Text	The protection group from which the original sample was taken
Device/Appliance IP Address	aip	IP Address	The defense appliance that originated this event
Direction	dir	Text [inbound	The direction of the packet

		| outbound	
Destination IP	dip	IP Address	The destination IP address in the sampled packet
Destination Port	dprt	Number	The destination port in the sampled packet
IP Packet Length	iplen	Number	The length of the IP packet including the header
Protocol	prot	Number	The IP protocol in the sampled packet. Refer to the full list at http://www.iana.org/assignments/protocol-numbers/protocol-numbers.xhtml
Type of Service	tos	Number	The type of service in sampled packet
Source IP	sip	IP Address	The source IP address in the sampled packet
Source Port	sprt	Number	The source port in the sampled packet
Time to Live	ttl	Number	The time to live in the sampled packet
TCP Flags	flags	Hex Value	The encoded TCP flags when applicable
Time	time	Number	The timestamp at the time of generation
Packet Length	plen	Number	The length of the sampled packet including the Ethernet header but not including the 4-byte CRC
PDU	pdu	Hex values	Up to the first 200 bytes of the pdu starting from the mac addresses

System Category

The system category contains messages that indicate the operational health of the Corero appliances and CMS.

Alarm Type

The CMS generates alarms for significant system events. A single type of syslog message carries the information for each alarm. These alarms have fields that can be broken out and indexed, and they also carry the full text.

Some of the types of alarms currently supported in the CMS are:

- Connection failure to an appliance
- Appliance has rebooted
- Interface has gone down

```
Mar 16 11:00:58 10.20.29.52 Mar 16 11:04:46 corpit-
cms.corero.com cat=system,type=alarm, v=1,
aname=Defense_10_20_27_111,v=1,atype=al:connection-
failure,sev=MAJOR,clrd=true,sp=none, txt=Failed to
```

```
connect to device Defense_10_20_27_111: connection
refused: Device not responding
```

Example Alarm Message

The specific content for the alarm message is:

Field Name	Short Name	Specific Value / Type	Description
Cat	cat	system	System event category
Type	type	alarm	CMS alarm type
Version	v	Number	The Corero message format version
Device name	aname	Text	The name provisioned for this device (appliance), if applicable
Severity	sev	Text [CRITICAL, MAJOR, MINOR]	The severity of the alarm
Cleared	clrd	Boolean (True/False)	Alarms cleared state
Specific Problem	sp	Text	Additional detail about specific problem
Full Text	txt	Text	Full text of alarm

Additional Integration Options

Corero can help partners deliver a leadership DDoS managed service through tighter integration between Corero's DDS and the partners' management systems and analysis tools. Such integration can allow partners access to the complete set of security information available in the DDS. Partners can contact Corero to investigate the available options by completing the partner inquiry form at http://www.corero.com/en/partners/become_a_partner or via e-mail at partners@corero.com.

About Corero

Corero Network Security (CNS:LN) is the global provider of the "First Line of Defense". Enterprises rely on Corero to protect their critical online assets from the risks associated with network-borne cyber-threats. Corero is headquartered in Massachusetts, U.S. with offices worldwide.

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Rule Reference

Rules are an essential building block of server protection. They are constructed to identify traffic that may be suspicious in origin, malicious in intent, and harmful in its results. Some rules define very strict conditions that you may not want to apply to every host, while other rules address definite known issues that should be applied universally. You can create a named set of rules, modify the treatment applied to individual rules within the set, and then use the named rule set as a component when defining server protection instances.

NOTE: Changing rule settings can affect system operation adversely. Consult with Corero Network Security before changing these settings.

Security Event Category

Every rule begins with a five-character security event category prefix that enables you to sort by prefix anywhere a list of rules appears. Security event category prefixes and their descriptions follow. Broadly, each category describes two dimensions: whether the identified threat is application-based or network-based (this is the "APP" or "NET" portion of the prefix), and whether the nature of the threat involves access restrictions (AR), rate limits (RL), protocol validation (PV), or integrity analysis (IA).

- ARAPP - Access Restrictions (application threats; layer 7)
- ARNET - Access Restrictions (network threats; layer 2 through layer 4)
- RLAPP - Rate Limits (application threats; layer 7)
- RLNET - Rate Limits (network threats; layer 2 through layer 4)
- PVAPP - Protocol Validation (application threats; layer 7)
- PVNET - Protocol Validation (network threats; layer 2 through layer 4)
- IAAPP - Integrity Analysis (application threats; layer 7)
- IANET - Integrity Analysis (network threats; layer 2 through layer 4)

Each rule definition includes a short description and a long description.

Rule Set Overview

A rule set is a named collection of rules that is configured to identify and handle traffic for a particular purpose. For example, the appliance includes built-in rule sets with names like “Recommended Server Protection,” “Strict Server Protection,” "All Rules Block," and "All Rules Detect," each of which furnishes a collection of rules that have been enabled and configured to support the purpose for which the rule set is named. Every appliance comes with a number of these pre-defined rule sets built in. Every rule is included in every rule set; however, each rule can be enabled or disabled explicitly for a particular rule set, so that even though each rule is present in every rule set, it isn’t necessarily used. In addition, every other aspect of how an enabled rule is handled can be configured explicitly for a rule set; this includes the action to be taken when the rule is met, and any logging that is performed at that time.

There are two types of rule sets:

- Security rule sets: Security rule sets describe behaviors for identifying and addressing traffic based on the nature of the traffic itself, such as its packet formation or its content.

- Reputation rule sets: Reputation rule sets describe behaviors for identifying and addressing traffic based on the trustworthiness of its originating host.

Although the appliance contains default rules sets, you can modify those rule sets. Or, if you prefer, you can copy a rule set and use the copy to create a custom rule set to use in your server protection instances.

Rule List

cns-000006

Description: PVNET: IPv4 packet contains an invalid checksum

Long Description: This rule detects an IPv4 packet in which the calculated checksum does not match the value in the header checksum field. This could indicate the possibility of intentional or unintentional corruption in the IP headers.

cns-000020

Description: PVNET: IP packet contains an invalid version number

Long Description: This rule detects an IP packet with a version number that is not consistent with the type of traffic expected. Either IPv4 or IPv6 traffic was expected but the version number indicates that the traffic belongs to a different protocol.

cns-000050

Description: PVNET: Invalid IP header length detected

Long Description: This rule triggers when the IP version 4 header length in the packet is determined to be less than 20 bytes as this could indicate an intentional or unintentional corruption of the headers.

cns-000056

Description: SYSTM: Protocol decoder issued unknown rule id

Long Description: This rule indicates that the protocol decoder has received an unknown rule ID. Prolonged numerous triggers of this rule must be reported to the Corero customer support team for further investigation as it could indicate a problem with the configuration.

cns-000060

Description: IANET: UDP packet shorter than specified minimum length

Long Description: This rule detects a UDP packet that is shorter than a user-configured minimum length. Small UDP packets are used in several attacks and triggers of this rule might need to be investigated to determine possibility of attacks and appropriate action taken to perform re-configuration of the rule if required.

cns-000061

Description: SYSTM: System resource issue

Long Description: This rule indicates that there is an internal appliance resource issue with memory management. Prolonged numerous triggers of this rule must be reported to the Corero customer support team for further investigation as it could indicate a problem with the configuration.

cns-000062

Description: "SYSTM: TCP Proxy resource issue"

Long Description: This rule triggers when TCP proxy is unable to establish a connection due to a lack of sufficient resources. Prolonged numerous triggers of this rule must be reported to the Corero customer support team for further investigation as it could indicate a problem with the configuration.`

cns-000064

Description: IANET: IPv4 too few hops for non-ICMP packet

Long Description: This rule detects a packet with an IPv4 Hop Limit that is less than the configured limit for a non-ICMP packet. Attackers manipulate this field to evade attacks from network security appliances. Triggers of this rule should be investigated and appropriate action taken to re-configure the rule if required.

cns-000067

Description: ARNET: Header Blocking - Traffic blocked as per configuration

Long Description: This rule detects a packet that matches traffic that is configured to be blocked in the Header Blocking table. The Header Blocking table can be used to block traffic based on protocol types, or tunneling. For example, IPv4 or IPv6 tunneled over any protocol can be configured to be blocked using this rule.

cns-000068

Description: ARNET: Block by next header for protocol traffic that is not allowed

Long Description: This rule detects a packet that matches traffic selected in the Next Header Table (Protocol Configuration). This rule is useful in blocking traffic that is not allowed in the environment and functions based on the values specified in the protocol field of the IP version 4 protocol header or the next header field in the IP version 6 protocol header. Typical values of these fields are available at the reference link: http://www.iana.org/assignments/protocol-numbers/protocol-numbers.xhtml. Example: This rule blocks ICMPv4 when tunneled over IPv6, IPv6 hop by hop over IPv4, etc.

cns-000071

Description: ARNET: Block packets with MPLS labels

Long Description: This rule detects packets that contain MPLS labels. Enabling this rule will block all MPLS traffic.

cns-000072

Description: PVNET: Misclassified MPLS traffic detected

Long Description: This rule detects packets with MPLS labels whose payloads could not be properly inspected as the traffic carried by MPLS could not be identified by our advanced algorithms.

cns-000073

Description: IANET: Traffic is nested in more than one tunnel

Long Description: The Defense appliance blocks all traffic with more than one nested tunnel. Protocol traffic that is encapsulated more than once is blocked if this rule is enabled. Several kinds of malware utilize this technique to be evasive. However triggers must be investigated and re-configuration of the rule done as required.

cns-000075

Description: PVNET: Bad Ethernet CRC

Long Description: This rule detects a packet with an incorrect Ethernet CRC. This could indicate the possibility of some corruption in the ethernet frame.

cns-000076

Description: PVNET: Wrong length in 802.3 frame

Long Description: This rule detects a packet with an incorrect 802.3 frame length. The calculated size of the frame received is different from what is stated in the frame thereby indicating some corruption of data, either intentional or unintentional.

cns-000080

Description: PVNET: IPv4 source/destination IP is loopback address

Long Description: This rule detects for which the IPv4 source/destination address is a loopback address (example in the range: 127.0.0.0/8). These addresses are not expected outside of a host and triggers of the rule should be investigated.

cns-000081

Description: PVNET: IPv4 source IP is a multicast address

Long Description: This rule detects a packet for which the IPv4 source address is a multicast address (typically between 224.0.0.0 and 239.255.255.255 for IP version 4). This is not expected and triggers of the rule should be investigated.

cns-000083

Description: PVNET: IPv4 source IP is a broadcast address

Long Description: This rule detects a packet for which the IPv4 source address is a broadcast address. This could indicate an attack and hence triggers need to be investigated.

cns-000085

Description: PVNET: IPv4 length is less than header length

Long Description: This rule detects a packet for which the IPv4 length is less than its header length. This is an anomaly as the header length is supposed to be included in the total length thereby indicating an intentional or unintentional corruption.

cns-000089

Description: IANET: IPv6 too few hops for non-ICMP Packets

Long Description: This rule detects a packet for which the IPv6 Hop Limit is less than the configured limit for a non-ICMP packet. Attackers manipulate this field to evade attacks from network security

appliances. Triggers of this rule should be investigated and appropriate action taken to re-configure the rule if required.

cns-000090

Description: PVNET: IPv6 source IP is a multicast address

Long Description: This rule detects a packet for which the IPv6 source address is a multicast address. This is typically not expected and triggers of the rule should be investigated.

cns-000091

Description: PVNET: IPv6 invalid source/destination

Long Description: This rule detects a packet for which the IPv6 source/destination addresses are equal, or one is a loopback, or one is null. Each of these is an anomaly and could indicate an attack.

cns-000092

Description: PVNET: IPv6 destination IP is a multicast local scope or reserved address

Long Description: This rule detects a packet for which the IPv6 destination address is a multicast local scope (ffx1::/16) or a reserved address. Each of these is an anomaly and could indicate an attack.

cns-000093

Description: PVNET: IPv6 routing header, routing type != 2

Long Description: This rule detects a packet for which the Routing Type of the IPv6 header is not Type 2. Use of deprecated routing types should not be allowed as there are known attacks using those values.

cns-000095

Description: PVNET: Packet too short

Long Description: This rule detects a packet for which the packet length is too short; its length is compared with the sum of the Ethernet header length and the IP packet length. This could indicate an intentional or unintentional corruption of the packet.

cns-000096

Description: PVNET: IP packet with excessive Ethernet padding

Long Description: This rule detects a packet with 256 bytes or more of ethernet padding. This can add excessive overhead in processing in some situations.

cns-000097

Description: PVNET: IP packet with unnecessary Ethernet padding

Long Description: This rule detects an IP packet that contains unnecessary ethernet padding. This can add excessive overhead in processing in some situations.

cns-000098

Description: PVNET: Cannot validate L4 checksum

long-desc `This rule detects a packet whose TCP or UDP checksum could not be validated. The checksums may or may not be valid.

cns-000099

Description: PVNET: Bad extension header length

Long Description: This rule detects a packet that has an incorrect extension header length. An extension header length of 0 indicates an anomaly and triggers should be further investigated.

cns-000100

Description: IANET: Packet could not be classified within 256 bytes

Long Description: This rule detects a packet that could not be classified within 256 bytes. Typically our advanced algorithms are able to determine the type of the packet within the first 256 bytes. Please report triggers along with the sample of the traffic causing the triggers of this rule to the Corero customer support team for further investigation.

cns-001001

Description: ARNET: Connection from spoofed IP address

Long Description: This rule detects packets that have an IP source address that is not allowed on the specified input circuit (input port). Spoofed source addresses are used in several attacks including DDoS.

cns-001003

Description: IANET: Connection containing reserved IPv4 address

Long Description: This rule detects packets that have an IP version 4 address that is in the Class E range of 240.0.0.0 through 255.255.255.254; these addresses are reserved. This is an anomaly and such connections must be investigated further.

cns-001005

Description: PVNET: Connection containing UDP port zero

Long Description: This rule detects packets that have a value of 0 for either the 16-bit source or the 16-bit destination port number in the UDP header. Several attacks use port 0 and hence this traffic is suspicious.

cns-001006

Description: PVNET: Connection containing TCP port zero

Long Description: This rule detects packets that have a 0 value for either the 16-bit source or the 16-bit destination port number in the TCP header. Several attacks and reconnaissance scans use port 0 and hence this traffic should be considered suspicious.

cns-001007

Description: RLNET: Connection from malicious source IP address

Long Description: This rule detects packets that arrive from a client IP address to which the Defense appliance has currently assigned a threat level of malicious. A "malicious" client is one that has initiated, but not completed a configured number of TCP connections.

cns-001016

Description: ARNET: Fragmented packet received against policy that disallows fragmentation

Long Description: This rule detects a packet that is an IP fragment from an application that matches a policy prohibiting fragmentation. Fragmentation is not expected for a packet conforming to this protocol as per the specified policy.

cns-001017

Description: ARNET: TCP connection with missed setup

Long Description: This rule detects TCP packets for which the Defense appliance did not see the connection setup handshake packets for the service. The Defense appliance views this data stream as a midflow. Attack traffic can flood a network segment by directing packets that have TCP port numbers that are typically allowed through firewall or other intervening security devices.

cns-001018

Description: RLNET: Connection from client that failed handshake in proxy mode

Long Description: This rule detects TCP connection packets from a client that failed to complete a proxied handshake properly. If a client repeatedly fails to respond to its own requests for a connection, the appliance marks that client as suspicious and stops sending the client's connection requests directly to the server. Instead, it proxies requests for the server. If the client behaves properly and completes the three-way handshake, the unit forwards the connection request to the server. TCP connection initiations that fail the client side proxy handshake typically indicate a spoofed client address and a SYN flood attack.

cns-001019

Description: ARNET: Connection to server that fails proxied handshake - Possible SYN flood

Long Description: This rule detects packets from a TCP connection initiation that is not successfully completed by the target (server) destination IP address. With SYN flood protection enabled, the Defense appliance monitors connection requests that the server does not complete and takes steps to confuse and stop the attacker.

cns-001020

Description: RLNET: SynFlood - Connection From New Client During DDoS Attack

Long Description: The rule indicates that a connection from a previously untracked IP address was detected during a DDoS attack. During a DDoS attack, the Defense appliance takes additional steps to protect your resources. One of these steps is to initially deny a TCP connection request by an untracked (potentially spoofed and dangerous) client during the DDoS attack. Once the appliance verifies that the client is a good client, it processes the client's connection requests normally. When a UDP DDoS occurs, the unit uses a proprietary threshold mechanism to identify the attack packets and discard them.

cns-001021

Description: ARNET: Matched by firewall

Long Description: This rule triggers when the Firewall logs a new flow.

cns-001022

Description: RLNET: Non-TCPRateLimit - Excessive non-TCP connection rate.

Long Description: This rule detects an excessive rate of non-TCP (typically UDP or ICMP) connection attempts. Additional packets with the same source and destination IP address (and UDP ports, if applicable) are not counted in this rate.

cns-001024

Description: RLNET: Bandwidth Limiting - IP frame exceeded specified rate bucket limit

Long Description: This rule detects a packet for a rate-limited policy for which traffic had exceeded its allowed bandwidth.

cns-001025

Description: ARNET: TCP connection with missed setup - RST or FIN packet

Long Description: This rule detects a TCP RST or FIN packet for which there is no current flow. Such packets are occasionally seen after a TCP connection has been closed. Such packets are not expected and should be dropped.

cns-001026

Description: RLNET: IpFragRateLimit - IP fragments exceeded fragment rate threshold

Long Description: This rule indicates that the number of IP fragments have exceeded the Fragment Rate Threshold. Excessive fragmentation is an indicator of evasiveness and can be used in attacks.

cns-001027

Description: ARNET: IP address blocked via the Fast Path

Long Description: The external IP address is in a range configured for blocking. Packets received on an EXTERNAL interface have the source address checked; packets received on an INTERNAL interface have the destination address checked and they are dropped via the Fast Path if they are in the configured blacklist. Note: This rule does not generate events in the security event viewer or the dashboard. However, the Blocked and Detected Report does reflect the true count of triggers for this rule.

cns-001028

Description: ARNET: NTP MONLIST response blocked

Long Description: A UDP packet for the NTP protocol (UDP source port 123) contained a MONLIST response and was blocked. This functionality has been deprecated and its usage could indicate a possible NTP amplification attack.

cns-001029

Description: ARNET: NTP MONLIST request blocked

Long Description: A UDP packet for the NTP protocol (UDP destination port 123) contained a MONLIST request and was blocked. This functionality has been deprecated and its usage could indicate a possible NTP amplification attack.

cns-001030

Description: RLNET: SynFlood - Connection From New Client During DDoS Attack

Long Description: During a DDoS attack, the Defense appliance takes additional steps to protect your resources. One of these steps is to initially deny a TCP connection request by a potentially spoofed and dangerous client during the DDoS attack.

cns-001031

Description: RLNET: SynFlood - Connection From New Client During DDoS Attack

Long Description: During a DDoS attack, the Defense appliance takes additional steps to protect your resources. The unit uses a proprietary threshold mechanism to identify the attack packets and discard them.

cns-001032

Description: ARNET: uPNP reply blocked

Long Description: A UDP packet for the uPNP protocol (UDP port 1900) contained a uPNP reply (source port 1900) and was blocked. This functionality has been deprecated and its usage could indicate a possible uPNP amplification attack.

cns-001033

Description: ARNET: Specific reply blocked

Long Description: This rule is used when a combined destination IP address and source port is matched. It can be used to thwart attacks like DNS reflection.

cns-001034

Description: ARNET: Specific fragmented reply blocked

Long Description: This rule is meant to be used as a complement to the Specific reply blocked rule. It is used when a combined destination IP address and a non-first fragment are matched. It can be used to thwart attacks like DNS reflection.

cns-001036

Description: IPFragRateLimit – Fragmented to non-fragmented packet ratio exceeded

Long Description: The ratio of fragmented to non-fragmented packets exceeded configured limit.

cns-001037

Description: ARNET: Generic packet filter

Long Description: A packet matched one of the generic packet filters.

cns-001038

Description: ARNET: Inbound SNMP communication blocked

Long Description: A UDP packet for the SNMP protocol (UDP port 161) contained SNMP reply (source port 161) and was blocked. This could indicate a possible SNMP amplification attack.

cns-001039

Description: ARNET: Chargen responses blocked

Long Description: A UDP packet for the CHARGEN protocol (UDP port 19) contained a CHARGEN reply (source port 19) and was blocked. This could indicate a possible CHARGEN amplification attack.

cns-001040

Description: IP address bypassed via the Fast Path

Long Description: The external IP address is in a range configured for bypass. Packets received on an EXTERNAL interface have the source address checked; packets received on an INTERNAL interface have the destination address checked and they are dropped via the Fast Path if they are in the configured blacklist. Note: This rule does not generate events in the security event viewer or the dashboard. However, the Detected Report does reflect the true count of triggers for this rule.

cns-001100

Description: ARNET: IP address has bad reputation, defined in later version

Long Description: This rule indicates that the source IP address is reputed to be bad. This is a generic bucket and more granular classification will be defined in a subsequent version of the product.

cns-001101

Description: ARNET: IP address reputed to participate in denial of service attacks

Long Description: This rule indicates that the source IP address is reputed to participate in denial of service attacks.

cns-001102

Description: ARNET: IP address reputed to be infected with malware

Long Description: This rule indicates that the source IP address is reputed to be infected with malware.

cns-001103

Description: ARNET: IP address reputed to participate in botnets

Long Description: This rule indicates that the source IP address is reputed to be a part of a botnet and hence considered malicious.

cns-001104

Description: ARNET: IP address reputed to offer Windows exploits

Long Description: This rule indicates that the source IP address is reputed to transmit Windows exploits, possibly distributing malware, shell code, rootkits, worms, or viruses.

cns-001105

Description: ARNET: IP address reputed to scan networks

Long Description: This rule indicates that the source IP address is reputed to scan networks, possibly including reconnaissance such as probes, host scans, domain scans, brute force password crackers.

cns-001106

Description: ARNET: IP address reputed to provide anonymization services

Long Description: This rule indicates that the source IP address is reputed to provide anonymization services, possibly including TOR anonymizer IP addresses.

cns-001107

Description: ARNET: IP address reputed to participate in web attacks

Long Description: This rule indicates that the source IP address is reputed to participate in web attacks, possibly including site scripting, iFrame injection, SQL injection, cross domain injection, or brute force domain password cracking.

cns-001108

Description: "ARNET: IP address reputed to host phishing sites "

Long Description: This rule indicates that the source IP address is reputed to host phishing sites.`

cns-001109

Description: ARNET: IP address reputed to send spam

Long Description: This rule indicates that the source IP address is reputed to send spam, possibly including tunneling spam messages through proxy, anomalous SMTP activities, and forum spam activities.

cns-002001

Description: RLNET: ConnLimit - TCP active connections from single client exceed specified limit

Long Description: This rule detects connection request packets from a client that has already established the maximum number of connections configured for any single client in that client's group. If you have a specific client that legitimately requires more resources than you generally wish to allow, you can raise the connection limit imposed for members in that group, or you can place that client into a unique group and establish higher limits specific to that group and its clients.

cns-002002

Description: RLNET: ConnLimit - TCP active connections from client group exceed specified limit

Long Description: This rule detects connection request packets from clients in a client group for which the total number of established connections exceeds the maximum number of connections configured for that group. As part of server resource protection, you can establish a limit on the number of TCP connections allowed for a client group and for any single member within the group. If you have a client group that legitimately requires more resources than you generally wish to allow, create a new connection limit rule, with higher limits, and apply it to that group.

cns-002003

Description: RLNET: ConnLimit - TCP active connections to single server exceed specified limit

Long Description: This rule detects connection requests to a server that has already made the maximum number of connections configured for any single server in that server's group. This mechanism protects servers from attempting to make more connections than they can handle.

cns-002004

Description: RLNET: ConnLimit - TCP active connections to server group exceed specified limit

Long Description: This rule detects connection requests to a server group that has already made the maximum number of total connections configured for all the servers in that group. As part of server resource protection, you can establish a limit on the number of TCP connections allowed for all the servers in a server group and for any single member within the group. This mechanism protects the servers from attempting to make more connections than they can handle.

cns-002005

Description: RLNET: ClientPacketRateLimit - rate from client to enabled services exceeds specified limit

Long Description: This rule detects TCP/UDP request packets from a client that has exceeded the rate limit for the TCP/UDP services that have been identified and configured to be rate limited. If you have configured TCP/UDP request rate limiting for specified TCP/UDP services, once the client uses up the allowed requests or allowed packets for a given period of time, the Defense appliance drops additional requests/packets. The appliance also tracks the excess requests and uses the total to update the TCP/UDP packets available to the client for the next measured period. Note that traffic belonging to services that are not client rate limited is allowed until the overdraft limit is reached for that client.

cns-002006

Description: RLNET: ClientPacketRateLimit - packet rate from client exceeded overdraft threshold

Long Description: This rule detects incoming TCP/UDP packets from a client that has exceeded an established overdraft limit. In addition to establishing a rate limit for TCP/UDP services, you can establish an "overdraft limit" which the Defense appliance uses to determine when a client has exceeded its request limit by so much that it is likely that the client is malicious. If the client exceeds the client rate limiting overdraft limit, the appliance treats the client as malicious and blocks any packet from that client (for any service, regardless of protocol type). The client automatically returns to "good client" status by reducing its packet rate to a reasonable amount. If you know that a client is now a "good" client and that its IP address is no longer being used for an attack, you can manually zero out the accumulated excessive requests of the client and return it immediately to favorable status.

cns-002007

Description: RLNET: ICMP packet rate exceeded threshold

Long Description: This rule detects ICMP packets that exceed a configured limit.

cns-002008

Description: RLNET: ICMPv6 packet rate exceeded threshold

Long Description: This rule detects ICMPv6 packets that exceed a configured limit.

cns-002009

Description: RLNET: UDP packets exceeded rate threshold

Long Description: This rule indicates that the number of UDP packets have exceeded the UDP Rate Threshold.

cns-002010

Description: RLNET: ServerByteRateLimit - byte rate to server limit exceeded

Long Description: Indicates that the rate of data being sent to a server exceeds the global limit. This limit must be the same for ALL servers and is intended to block high bandwidth attacks that can't be blocked by some more specific rule.

cns-002011

Description: RLNET: ServerPktRateLimit - packet rate to server limit exceeded

Long Description: Indicates that the number of packets being sent to a server exceeds the global limit. This limit must be the same for ALL servers and is intended to block high packet rate attacks that can't be blocked by some more specific rule.

cns-002012

Description: RLNET: ReflectionByteRateLimit - byte rate to server limit exceeded

Long Description: Indicates that the rate of data being sent from a specified source address to a server exceeds the global limit. This is typical of reflection attacks. The limit must be the same for ALL servers and is intended to block high bandwidth attacks that can't be blocked by some more specific rule.

cns-002013

Description: RLNET: ReflectionPktRateLimit - packet rate to server limit exceeded

Long Description: Indicates that the number of packets being sent from a specified source address to a server exceeds the global limit. This is typical of reflection attacks. The limit must be the same for ALL servers and is intended to block high packet rate attacks that can't be blocked by some more specific rule.

cns-003001

Description: IANET: TCP flow aged out early

Long Description: This rule is triggered when a TCP flow is terminated due to inactivity. It is useful for identifying service definitions whose age-out time is not long enough.

cns-003002

Description: PVNET: IP frame destination address equals source addr-LanD

Long Description: This rule detects packets associated with the LanD attack (typically created using an attack program named land.c). The land.c program launches a denial-of-service (DoS) attack by sending a TCP SYN packet (a connection request) giving the target host's address as both the source and destination address, and using the same port on the target host as both the source and destination

port. This attack makes it appear that the host computer is sending packets to itself. Servers, that do not properly validate the input socket pair, can experience TCP/IP stack overflows.

cns-003003

Description: PVNET: ICMP frame length illegal for type or exceeds specified limit

Long Description: This rule detects packets associated with ICMP messages that have illegal lengths or lengths that exceed the specified maximum configured limit. Various packet types including but not limited to router solicitation, traceroute, information requests and responses, etc. are checked. Servers that do not validate the length of the ICMP message they receive are vulnerable to ICMP-related attacks. A famous example of an ICMP attack is the "Ping of Death" which is an echo request contained in a packet that is greater than 65536 bytes.

cns-003004

Description: PVNET: ICMP frame contains illegal header

Long Description: This rule detects a packet with a header containing a Type field that has a value beyond the types that have been assigned, or has a Code field with a value that is greater than the maximum value allowed for the given Type field associated with that code.

cns-003005

Description: PVNET: UDP frame length mismatch with IP length UDP bomb

Long Description: This rule detects UDP packets that have a UDP length value that does not match the actual length of the data payload and the UDP header. This could indicate unintentional or intentional data corruption.

cns-003006

Description: PVNET: TCP frame header contains illegal flag combinations

Long Description: This rule detects packets that have illegal combinations of TCP flag values. Several combinations of flags are checked and only traffic containing permissible values is allowed.

cns-003007

Description: IANET: TCP frame URG bit set to port 139-WinNuke

Long Description: This rule detects packets that are sent to port 139 with the URG bit set. This could indicate a potential WinNuke DoS attack. These attacks use the "urgent" mechanism to alter and abuse a NetBIOS Session (TCP port 139)

cns-003011

Description: PVNET: IP frame source IP address zero

Long Description: This rule detects an IP packet with a source address of 0.0.0.0 in the IP version 4 header or :: in the IP version 6 header. This rule applies to all protocols except UDP. This is not a typically expected value for the source IP address field.

cns-003012

Description: PVNET: IP frame destination address zero

Long Description: This rule detects an IP packet with a destination address of 0.0.0.0 in the IP version 4 header or :: in the IP version 6 header. This is not a valid destination address.

cns-003015

Description: IANET: UDP echo has multicast destination IP - Fraggle

Long Description: This rule detects a UDP packet directed to port 7 (echo request) that is flooded to all the addresses in the network or subnet. This could indicate a DDoS attack called Fraggle which is a variant of the Smurf attack. By sending out multiple requests, each responded to by multiple hosts, the attacker consumes network bandwidth and creates a denial-of-service condition for the target host.

cns-003016

Description: IANET: Slow TCP Client

Long Description: This rule detects attacks where the client purposely delays sending all of a request so as to tie up server resources. Slow Loris is one example.

cns-003017

Description: IANET: Slow TCP Server

Long Description: This rule detects attacks where TCP traffic from a server is progressing so slowly that there is likely an attack going on. For example, the slowhtptest attack tool manipulates the client's TCP receive window to slow down server-to-client traffic.

cns-003018

Description: IANET: ICMP unsolicited echo response

Long Description: This rule detects an ICMP Echo response packet sent to a host that did not send an Echo request. This could be an indicator of a reconnaissance scan or a potential DoS attack.

cns-003020

Description: IANET: DNS unsolicited response

Long Description: This rule detects a DNS (port 53) response packet sent to a host that did not issue a corresponding DNS request. A large number of triggers could indicate a DNS Reflection attack.

cns-003021

Description: IANET: TCP SYN retry limit exceeded

Long Description: This rule detects a TCP connection that is retransmitting SYN packets faster than allowed by RFC 6298. This indicates an attempted SYN flood over a single connection.

cns-003022

Description: IANET: TCP initial sequence number has changed

Long Description: This rule detects a second SYN packet with a different initial sequence number that is received for a TCP flow that is already set up. This may indicate an attempt to hijack an existing TCP connection.

cns-003023

Description: IANET: TCP frame contains bad sequence number

Long Description: This rule detects a sequence number that is beyond a reasonable range determined by the sequence and ACK numbers from previous packets and the TCP window size. This may indicate an attempt to hijack an existing TCP connection.

cns-003024

Description: IANET: TCP frame contains bad ACK number

Long Description: This rule detects an ACK number that is beyond a reasonable range determined by the sequence and ACK numbers from previous packets and the TCP window size. This may indicate an attempt to hijack an existing TCP connection.

cns-003025

Description: PVNET: ICMPv6 frame contains illegal header

Long Description: This rule detects a packet with a header containing a Type field that has a value beyond the types that have been assigned, or has a Code field with a value that is greater than the maximum value allowed for the given Type field associated with that code.

cns-003026

Description: IANET: ICMPv6 unsolicited echo response

Long Description: This rule detects an ICMPv6 Echo response packet sent to a host that did not send an Echo request. This could be an indicator of a reconnaissance scan as it is a technique used by attackers to gain more knowledge about the hosts behind the firewall.

cns-003027

Description: PVNET: TCP SYN packet fragmented

Long Description: This rule detects TCP SYN packets that are fragmented. Multiple triggers could indicate the possibility of a SYN Fragment attack.

cns-003028

Description: IANET: TCP SYN retry limit exceeded on a proxied connection

Long Description: This rule detects a proxied connection on which many SYN packets have arrived. This indicates an attempted SYN flood over a single connection.

cns-003029

Description: IANET: TCP Proxy client connection reset.

Long Description: This rule detects a proxied connection for which the client closed the connection instead of completing the opening handshake. In other words, the client sent a TCP RST packet in response to the SYN/ACK sent by the server. This could be an indication of a port scanner or spoofed connection attempt.

cns-003030

Description: PVNET: TCP frame header contains illegal flag combinations.

Long Description: This rule detects packets that have illegal combinations of TCP flag values. Several combinations of flags are checked and only traffic containing permissible values is allowed.

cns-033001

Description: ARNET: IP address shunning service detect on label 001

Long Description: This rule indicates that a packet has been shunned by shunning service label 001.

cns-033002

Description: ARNET: IP address shunning service detect on label 002

Long Description: This rule indicates that a packet has been shunned by shunning service label 002.

cns-033003

Description: ARNET: IP address shunning service detect on label 003

Long Description: This rule indicates that a packet has been shunned by shunning service label 003.

cns-033004

Description: ARNET: IP address shunning service detect on label 004

Long Description: This rule indicates that a packet has been shunned by shunning service label 004.

cns-033005

Description: ARNET: IP address shunning service detect on label 005

Long Description: This rule indicates that a packet has been shunned by shunning service label 005.

cns-033006

Description: ARNET: IP address shunning service detect on label 006

Long Description: This rule indicates that a packet has been shunned by shunning service label 006.

cns-033007

Description: ARNET: IP address shunning service detect on label 007

Long Description: This rule indicates that a packet has been shunned by shunning service label 007.

cns-033008

Description: ARNET: IP address shunning service detect on label 008

Long Description: This rule indicates that a packet has been shunned by shunning service label 008.

cns-033009

Description: ARNET: IP address shunning service detect on label 009

Long Description: This rule indicates that a packet has been shunned by shunning service label 009.

cns-033010

Description: ARNET: IP address shunning service detect on label 010

Long Description: This rule indicates that a packet has been shunned by shunning service label 010.

cns-033011

Description: ARNET: IP address shunning service detect on label 011

Long Description: This rule indicates that a packet has been shunned by shunning service label 011.

cns-033012

Description: ARNET: IP address shunning service detect on label 012

Long Description: This rule indicates that a packet has been shunned by shunning service label 012.

cns-033013

Description: ARNET: IP address shunning service detect on label 013

Long Description: This rule indicates that a packet has been shunned by shunning service label 013.

cns-033014

Description: ARNET: IP address shunning service detect on label 014

Long Description: This rule indicates that a packet has been shunned by shunning service label 014.

cns-033015

Description: ARNET: IP address shunning service detect on label 015

Long Description: This rule indicates that a packet has been shunned by shunning service label 015.

cns-033016

Description: ARNET: IP address shunning service detect on label 016

Long Description: `This rule indicates that a packet has been shunned by shunning service label 016.

cns-033017

Description: ARNET: IP address shunning service detect on label 017

Long Description: This rule indicates that a packet has been shunned by shunning service label 017.

cns-033018

Description: ARNET: IP address shunning service detect on label 018

Long Description: This rule indicates that a packet has been shunned by shunning service label 018.

cns-033019

Description: ARNET: IP address shunning service detect on label 019

Long Description: This rule indicates that a packet has been shunned by shunning service label 019.

cns-033020

Description: ARNET: IP address shunning service detect on label 020

Long Description: This rule indicates that a packet has been shunned by shunning service label 020.

cns-033021

Description: ARNET: IP address shunning service detect on label 021

Long Description: This rule indicates that a packet has been shunned by shunning service label 021.

cns-033022

Description: ARNET: IP address shunning service detect on label 022

Long Description: This rule indicates that a packet has been shunned by shunning service label 022.

cns-033023

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Description: ARNET: IP address shunning service detect on label 026

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cns-033030

Description: ARNET: IP address shunning service detect on label 030

Long Description: This rule indicates that a packet has been shunned by shunning service label 030.

cns-033031

Description: ARNET: IP address shunning service detect on label 031

Long Description: This rule indicates that a packet has been shunned by shunning service label 031.

cns-033032

Description: ARNET: IP address shunning service detect on label 032

Long Description: This rule indicates that a packet has been shunned by shunning service label 032.

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Description: ARNET: IP address shunning service detect on label 042

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cns-033047

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cns-033048

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cns-033050

Description: ARNET: IP address shunning service detect on label 050

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cns-033051

Description: ARNET: IP address shunning service detect on label 051

Long Description: This rule indicates that a packet has been shunned by shunning service label 051.

cns-033052

Description: ARNET: IP address shunning service detect on label 052

long-desc `This rule indicates that a packet has been shunned by shunning service label 052.

cns-033053

Description: ARNET: IP address shunning service detect on label 053

Long Description: This rule indicates that a packet has been shunned by shunning service label 053.

cns-033054

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cns-033056

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cns-033057

Description: ARNET: IP address shunning service detect on label 057

long-desc `This rule indicates that a packet has been shunned by shunning service label 057`

cns-033058

Description: ARNET: IP address shunning service detect on label 058

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cns-033059

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cns-033060

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cns-033067

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cns-033068

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cns-033069

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cns-033070

Description: ARNET: IP address shunning service detect on label 070

long-desc `This rule indicates that a packet has been shunned by shunning service label 070.

cns-033071

Description: ARNET: IP address shunning service detect on label 071

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cns-033072

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cns-033073

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cns-033075

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cns-033076

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cns-033077

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cns-033079

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cns-033080

Description: ARNET: IP address shunning service detect on label 080

Long Description: This rule indicates that a packet has been shunned by shunning service label 080.

cns-033081

Description: ARNET: IP address shunning service detect on label 081

Long Description: This rule indicates that a packet has been shunned by shunning service label 081.

cns-033082

Description: ARNET: IP address shunning service detect on label 082

Long Description: This rule indicates that a packet has been shunned by shunning service label 082.

cns-033083

Description: ARNET: IP address shunning service detect on label 083

Long Description: This rule indicates that a packet has been shunned by shunning service label 083.

cns-033084

Description: ARNET: IP address shunning service detect on label 084

Long Description: This rule indicates that a packet has been shunned by shunning service label 084.

cns-033085

Description: ARNET: IP address shunning service detect on label 085

Long Description: This rule indicates that a packet has been shunned by shunning service label 085.

cns-033086

Description: ARNET: IP address shunning service detect on label 086

Long Description: This rule indicates that a packet has been shunned by shunning service label 086.

cns-033087

Description: ARNET: IP address shunning service detect on label 087

Long Description: This rule indicates that a packet has been shunned by shunning service label 087.

cns-033088

Description: ARNET: IP address shunning service detect on label 088

Long Description: This rule indicates that a packet has been shunned by shunning service label 088.

cns-033089

Description: ARNET: IP address shunning service detect on label 089

Long Description: This rule indicates that a packet has been shunned by shunning service label 089`

cns-033090

Description: ARNET: IP address shunning service detect on label 090

Long Description: This rule indicates that a packet has been shunned by shunning service label 090.

cns-033091

Description: "ARNET: IP address shunning service detect on label 091"

Long Description: This rule indicates that a packet has been shunned by shunning service label 091.`

cns-033092

Description: ARNET: IP address shunning service detect on label 092

Long Description: This rule indicates that a packet has been shunned by shunning service label 092.

cns-033093

Description: ARNET: IP address shunning service detect on label 093

Long Description: This rule indicates that a packet has been shunned by shunning service label 093.

cns-033094

Description: ARNET: IP address shunning service detect on label 094

Long Description: This rule indicates that a packet has been shunned by shunning service label 094.

cns-033095

Description: ARNET: IP address shunning service detect on label 095

Long Description: This rule indicates that a packet has been shunned by shunning service label 095.

cns-033096

Description: ARNET: IP address shunning service detect on label 096

Long Description: This rule indicates that a packet has been shunned by shunning service label 096.

cns-033097

Description: ARNET: IP address shunning service detect on label 097

Long Description: This rule indicates that a packet has been shunned by shunning service label 097.

cns-033098

Description: ARNET: IP address shunning service detect on label 098

Long Description: This rule indicates that a packet has been shunned by shunning service label 098.

cns-033099

Description: ARNET: IP address shunning service detect on label 099

Long Description: This rule indicates that a packet has been shunned by shunning service label 099.

cns-033100

Description: ARNET: IP address shunning service detect on label 100

Long Description: `This rule indicates that a packet has been shunned by shunning service label 100.`

cns-033101

Description: ARNET: IP address shunning service detect on label 101

Long Description: This rule indicates that a packet has been shunned by shunning service label 101.

cns-033102

Description: ARNET: IP address shunning service detect on label 102

Long Description: This rule indicates that a packet has been shunned by shunning service label 102.

cns-033103

Description: ARNET: IP address shunning service detect on label 103

Long Description: This rule indicates that a packet has been shunned by shunning service label 103.

cns-033104

Description: ARNET: IP address shunning service detect on label 104

Long Description: This rule indicates that a packet has been shunned by shunning service label 104.

cns-033105

Description: ARNET: IP address shunning service detect on label 105

Long Description: This rule indicates that a packet has been shunned by shunning service label 105.

cns-033106

Description: ARNET: IP address shunning service detect on label 106

Long Description: This rule indicates that a packet has been shunned by shunning service label 106.

cns-033107

Description: ARNET: IP address shunning service detect on label 107

Long Description: This rule indicates that a packet has been shunned by shunning service label 107.

cns-033108

Description: ARNET: IP address shunning service detect on label 108

Long Description: This rule indicates that a packet has been shunned by shunning service label 108.

cns-033109

Description: ARNET: IP address shunning service detect on label 109

Long Description: This rule indicates that a packet has been shunned by shunning service label 109.

cns-033110

Description: ARNET: IP address shunning service detect on label 110

Long Description: This rule indicates that a packet has been shunned by shunning service label 110.

cns-033111

Description: ARNET: IP address shunning service detect on label 111

Long Description: This rule indicates that a packet has been shunned by shunning service label 111.

cns-033112

Description: ARNET: IP address shunning service detect on label 112

Long Description: This rule indicates that a packet has been shunned by shunning service label 112.

cns-033113

Description: ARNET: IP address shunning service detect on label 113

Long Description: This rule indicates that a packet has been shunned by shunning service label 113.

cns-033114

Description: ARNET: IP address shunning service detect on label 114

Long Description: This rule indicates that a packet has been shunned by shunning service label 114.

cns-033115

Description: ARNET: IP address shunning service detect on label 115

Long Description: This rule indicates that a packet has been shunned by shunning service label 115.

cns-033116

Description: ARNET: IP address shunning service detect on label 116

Long Description: This rule indicates that a packet has been shunned by shunning service label 116.

cns-033117

Description: ARNET: IP address shunning service detect on label 117

Long Description: `This rule indicates that a packet has been shunned by shunning service label 117.

cns-033118

Description: ARNET: IP address shunning service detect on label 118

Long Description: This rule indicates that a packet has been shunned by shunning service label 118.

cns-033119

Description: ARNET: IP address shunning service detect on label 119

Long Description: This rule indicates that a packet has been shunned by shunning service label 119.

cns-033120

Description: ARNET: IP address shunning service detect on label 120

Long Description: This rule indicates that a packet has been shunned by shunning service label 120.

cns-033121

Description: ARNET: IP address shunning service detect on label 121

Long Description: This rule indicates that a packet has been shunned by shunning service label 121.

cns-033122

Description: ARNET: IP address shunning service detect on label 122

Long Description: This rule indicates that a packet has been shunned by shunning service label 122.

cns-033123

Description: ARNET: IP address shunning service detect on label 123

Long Description: This rule indicates that a packet has been shunned by shunning service label 123.

cns-033124

Description: ARNET: IP address shunning service detect on label 124

Long Description: This rule indicates that a packet has been shunned by shunning service label 124.

cns-033125

Description: ARNET: IP address shunning service detect on label 125

Long Description: This rule indicates that a packet has been shunned by shunning service label 125.

cns-033126

Description: ARNET: IP address shunning service detect on label 126

Long Description: This rule indicates that a packet has been shunned by shunning service label 126.

cns-033127

Description: ARNET: IP address shunning service detect on label 127

Long Description: This rule indicates that a packet has been shunned by shunning service label 127.

cns-033128

Description: ARNET: IP address shunning service detect on label 128

Long Description: This rule indicates that a packet has been shunned by shunning service label 128.

cns-033129

Description: ARNET: IP address shunning service detect on label 129

Long Description: This rule indicates that a packet has been shunned by shunning service label 129.

cns-033130

Description: ARNET: IP address shunning service detect on label 130

Long Description: This rule indicates that a packet has been shunned by shunning service label 130.

cns-033131

Description: ARNET: IP address shunning service detect on label 131

Long Description: This rule indicates that a packet has been shunned by shunning service label 131.

cns-033132

Description: ARNET: IP address shunning service detect on label 132

Long Description: This rule indicates that a packet has been shunned by shunning service label 132.

cns-033133

Description: ARNET: IP address shunning service detect on label 133

Long Description: This rule indicates that a packet has been shunned by shunning service label 133.

cns-033134

Description: ARNET: IP address shunning service detect on label 134

Long Description: This rule indicates that a packet has been shunned by shunning service label 134.

cns-033135

Description: ARNET: IP address shunning service detect on label 135

Long Description: This rule indicates that a packet has been shunned by shunning service label 135.

cns-033136

Description: ARNET: IP address shunning service detect on label 136

Long Description: This rule indicates that a packet has been shunned by shunning service label 136.

cns-033137

Description: ARNET: IP address shunning service detect on label 137

long-desc `This rule indicates that a packet has been shunned by shunning service label 137.

cns-033138

Description: ARNET: IP address shunning service detect on label 138

Long Description: This rule indicates that a packet has been shunned by shunning service label 138.

cns-033139

Description: ARNET: IP address shunning service detect on label 139

Long Description: This rule indicates that a packet has been shunned by shunning service label 139.

cns-033140

Description: ARNET: IP address shunning service detect on label 140

Long Description: This rule indicates that a packet has been shunned by shunning service label 140.

cns-033141

Description: ARNET: IP address shunning service detect on label 141

Long Description: This rule indicates that a packet has been shunned by shunning service label 141.

cns-033142

Description: ARNET: IP address shunning service detect on label 142

Long Description: This rule indicates that a packet has been shunned by shunning service label 142.

cns-033143

Description: ARNET: IP address shunning service detect on label 143

Long Description: This rule indicates that a packet has been shunned by shunning service label 143.

cns-033144

Description: ARNET: IP address shunning service detect on label 144

Long Description: This rule indicates that a packet has been shunned by shunning service label 144.

cns-033145

Description: ARNET: IP address shunning service detect on label 145

Long Description: This rule indicates that a packet has been shunned by shunning service label 145.

cns-033146

Description: ARNET: IP address shunning service detect on label 146

Long Description: This rule indicates that a packet has been shunned by shunning service label 146.

cns-033147

Description: ARNET: IP address shunning service detect on label 147

Long Description: This rule indicates that a packet has been shunned by shunning service label 147.

cns-033148

Description: ARNET: IP address shunning service detect on label 148

Long Description: This rule indicates that a packet has been shunned by shunning service label 148.

cns-033149

Description: ARNET: IP address shunning service detect on label 149

Long Description: This rule indicates that a packet has been shunned by shunning service label 149.

cns-033150

Description: ARNET: IP address shunning service detect on label 150

Long Description: This rule indicates that a packet has been shunned by shunning service label 150.

cns-033151

Description: ARNET: IP address shunning service detect on label 151

Long Description: This rule indicates that a packet has been shunned by shunning service label 151.

cns-033152

Description: ARNET: IP address shunning service detect on label 152

Long Description: This rule indicates that a packet has been shunned by shunning service label 152.

cns-033153

Description: ARNET: IP address shunning service detect on label 153

Long Description: This rule indicates that a packet has been shunned by shunning service label 153.

cns-033154

Description: ARNET: IP address shunning service detect on label 154

Long Description: This rule indicates that a packet has been shunned by shunning service label 154.

cns-033155

Description:"ARNET: IP address shunning service detect on label 155

Long Description: This rule indicates that a packet has been shunned by shunning service label 155.

cns-033156

Description: ARNET: IP address shunning service detect on label 156

Long Description: This rule indicates that a packet has been shunned by shunning service label 156.

cns-033157

Description: ARNET: IP address shunning service detect on label 157

Long Description: This rule indicates that a packet has been shunned by shunning service label 157.

cns-033158

Description: ARNET: IP address shunning service detect on label 158

Long Description: This rule indicates that a packet has been shunned by shunning service label 158.

cns-033159

Description: ARNET: IP address shunning service detect on label 159

Long Description: This rule indicates that a packet has been shunned by shunning service label 159.

cns-033160

Description: ARNET: IP address shunning service detect on label 160

Long Description: This rule indicates that a packet has been shunned by shunning service label 160.

cns-033161

Description: ARNET: IP address shunning service detect on label 161

Long Description: This rule indicates that a packet has been shunned by shunning service label 161.

cns-033162

Description: ARNET: IP address shunning service detect on label 162

Long Description: This rule indicates that a packet has been shunned by shunning service label 162.

cns-033163

Description: ARNET: IP address shunning service detect on label 163

Long Description: This rule indicates that a packet has been shunned by shunning service label 163.

cns-033164

Description: ARNET: IP address shunning service detect on label 164

Long Description: This rule indicates that a packet has been shunned by shunning service label 164.

cns-033165

Description: ARNET: IP address shunning service detect on label 165

Long Description: This rule indicates that a packet has been shunned by shunning service label 165.

cns-033166

Description: ARNET: IP address shunning service detect on label 166

Long Description: This rule indicates that a packet has been shunned by shunning service label 166.

cns-033167

Description:"ARNET: IP address shunning service detect on label 167

Long Description: This rule indicates that a packet has been shunned by shunning service label 167.

cns-033168

Description: ARNET: IP address shunning service detect on label 168

Long Description: This rule indicates that a packet has been shunned by shunning service label 168.

cns-033169

Description: ARNET: IP address shunning service detect on label 169

Long Description: This rule indicates that a packet has been shunned by shunning service label 169.

cns-033170

Description: ARNET: IP address shunning service detect on label 170

Long Description: This rule indicates that a packet has been shunned by shunning service label 170.

cns-033171

Description: ARNET: IP address shunning service detect on label 171

Long Description: This rule indicates that a packet has been shunned by shunning service label 171.

cns-033172

Description: ARNET: IP address shunning service detect on label 172

Long Description: This rule indicates that a packet has been shunned by shunning service label 172.

cns-033173

Description: ARNET: IP address shunning service detect on label 173

Long Description: This rule indicates that a packet has been shunned by shunning service label 173.

cns-033174

Description: ARNET: IP address shunning service detect on label 174

Long Description: This rule indicates that a packet has been shunned by shunning service label 174.

cns-033175

Description: ARNET: IP address shunning service detect on label 175

Long Description: This rule indicates that a packet has been shunned by shunning service label 175.

cns-033176

Description: ARNET: IP address shunning service detect on label 176

Long Description: This rule indicates that a packet has been shunned by shunning service label 176.

cns-033177

Description: ARNET: IP address shunning service detect on label 177

l**Long Description:** This rule indicates that a packet has been shunned by shunning service label 177.

cns-033178

Description: ARNET: IP address shunning service detect on label 178

Long Description: This rule indicates that a packet has been shunned by shunning service label 178.

cns-033179

Description: ARNET: IP address shunning service detect on label 179

Long Description: This rule indicates that a packet has been shunned by shunning service label 179.

cns-033180

Description: ARNET: IP address shunning service detect on label 180

Long Description: This rule indicates that a packet has been shunned by shunning service label 180.

cns-033181

Description: ARNET: IP address shunning service detect on label 181

Long Description: This rule indicates that a packet has been shunned by shunning service label 181.

cns-033182

Description: ARNET: IP address shunning service detect on label 182

Long Description: This rule indicates that a packet has been shunned by shunning service label 182.

cns-033183

Description: ARNET: IP address shunning service detect on label 183

Long Description: This rule indicates that a packet has been shunned by shunning service label 183.

cns-033184

Description: ARNET: IP address shunning service detect on label 184

Long Description: This rule indicates that a packet has been shunned by shunning service label 184.

cns-033185

Description: ARNET: IP address shunning service detect on label 185

Long Description: This rule indicates that a packet has been shunned by shunning service label 185.

cns-033186

Description: ARNET: IP address shunning service detect on label 186

Long Description: This rule indicates that a packet has been shunned by shunning service label 186.

cns-033187

Description: ARNET: IP address shunning service detect on label 187

Long Description: This rule indicates that a packet has been shunned by shunning service label 187.

cns-033188

Description: ARNET: IP address shunning service detect on label 188

Long Description: This rule indicates that a packet has been shunned by shunning service label 188.

cns-033189

Description: ARNET: IP address shunning service detect on label 189

Long Description: `This rule indicates that a packet has been shunned by shunning service label 189.

cns-033190

Description: ARNET: IP address shunning service detect on label 190

Long Description: This rule indicates that a packet has been shunned by shunning service label 190.

cns-033191

Description: ARNET: IP address shunning service detect on label 191

Long Description: This rule indicates that a packet has been shunned by shunning service label 191.

cns-033192

Description: ARNET: IP address shunning service detect on label 192

Long Description: This rule indicates that a packet has been shunned by shunning service label 192.

cns-033193

Description: ARNET: IP address shunning service detect on label 193

Long Description: This rule indicates that a packet has been shunned by shunning service label 193.

cns-033194

Description: ARNET: IP address shunning service detect on label 194

Long Description: This rule indicates that a packet has been shunned by shunning service label 194.

cns-033195

Description: ARNET: IP address shunning service detect on label 195

Long Description: This rule indicates that a packet has been shunned by shunning service label 195.

cns-033196

Description: ARNET: IP address shunning service detect on label 196

Long Description: This rule indicates that a packet has been shunned by shunning service label 196.

cns-033197

Description: ARNET: IP address shunning service detect on label 197

Long Description: This rule indicates that a packet has been shunned by shunning service label 197.

cns-033198

Description: ARNET: IP address shunning service detect on label 198

Long Description: This rule indicates that a packet has been shunned by shunning service label 198.

cns-033199

Description: ARNET: IP address shunning service detect on label 199

Long Description: This rule indicates that a packet has been shunned by shunning service label 199.

cns-033200

Description: ARNET: IP address shunning service detect on label 200

Long Description: This rule indicates that a packet has been shunned by shunning service label 200.

cns-033201

Description: ARNET: IP address shunning service detect on label 201

Long Description: This rule indicates that a packet has been shunned by shunning service label 201.

cns-033202

Description: ARNET: IP address shunning service detect on label 202

Long Description: This rule indicates that a packet has been shunned by shunning service label 202.

cns-033203

Description: ARNET: IP address shunning service detect on label 203

Long Description: This rule indicates that a packet has been shunned by shunning service label 203.

cns-033204

Description: ARNET: IP address shunning service detect on label 204

Long Description: This rule indicates that a packet has been shunned by shunning service label 204.

cns-033205

Description: ARNET: IP address shunning service detect on label 205

long-desc `This rule indicates that a packet has been shunned by shunning service label 205.

cns-033206

Description: ARNET: IP address shunning service detect on label 206

Long Description: This rule indicates that a packet has been shunned by shunning service label 206.

cns-033207

Description: ARNET: IP address shunning service detect on label 207

Long Description: This rule indicates that a packet has been shunned by shunning service label 207.

cns-033208

Description: ARNET: IP address shunning service detect on label 208

Long Description: This rule indicates that a packet has been shunned by shunning service label 208.

cns-033209

Description: ARNET: IP address shunning service detect on label 209

Long Description: This rule indicates that a packet has been shunned by shunning service label 209.

cns-033210

Description: ARNET: IP address shunning service detect on label 210

Long Description: This rule indicates that a packet has been shunned by shunning service label 210.

cns-033211

Description: ARNET: IP address shunning service detect on label 211

Long Description: This rule indicates that a packet has been shunned by shunning service label 211.

cns-033212

Description: ARNET: IP address shunning service detect on label 212

Long Description: This rule indicates that a packet has been shunned by shunning service label 212.

cns-033213

Description: ARNET: IP address shunning service detect on label 213

l**Long Description:** This rule indicates that a packet has been shunned by shunning service label 213.

cns-033214

Description: ARNET: IP address shunning service detect on label 214

Long Description: This rule indicates that a packet has been shunned by shunning service label 214.

cns-033215

Description: ARNET: IP address shunning service detect on label 215

Long Description: This rule indicates that a packet has been shunned by shunning service label 215.

cns-033216

Description: ARNET: IP address shunning service detect on label 216

Long Description: This rule indicates that a packet has been shunned by shunning service label 216.

cns-033217

Description: ARNET: IP address shunning service detect on label 217

Long Description: This rule indicates that a packet has been shunned by shunning service label 217.

cns-033218

Description: ARNET: IP address shunning service detect on label 218

long-desc `This rule indicates that a packet has been shunned by shunning service label 218.

cns-033219

Description: ARNET: IP address shunning service detect on label 219

Long Description: This rule indicates that a packet has been shunned by shunning service label 219.

cns-033220

Description: ARNET: IP address shunning service detect on label 220

Long Description: This rule indicates that a packet has been shunned by shunning service label 220.

cns-033221

Description: ARNET: IP address shunning service detect on label 221

Long Description: This rule indicates that a packet has been shunned by shunning service label 221.

cns-033222

Description: ARNET: IP address shunning service detect on label 222

Long Description: This rule indicates that a packet has been shunned by shunning service label 222.

cns-033223

Description: ARNET: IP address shunning service detect on label 223

Long Description: This rule indicates that a packet has been shunned by shunning service label 223.

cns-033224

Description: ARNET: IP address shunning service detect on label 224

long-desc `This rule indicates that a packet has been shunned by shunning service label 224.

cns-033225

Description: ARNET: IP address shunning service detect on label 225

Long Description: This rule indicates that a packet has been shunned by shunning service label 225.

cns-033226

Description: ARNET: IP address shunning service detect on label 226

Long Description: This rule indicates that a packet has been shunned by shunning service label 226.

cns-033227

Description: ARNET: IP address shunning service detect on label 227

Long Description: This rule indicates that a packet has been shunned by shunning service label 227.

cns-033228

Description: ARNET: IP address shunning service detect on label 228

Long Description: This rule indicates that a packet has been shunned by shunning service label 228.

cns-033229

Description: ARNET: IP address shunning service detect on label 229

Long Description: This rule indicates that a packet has been shunned by shunning service label 229.

cns-033230

Description: ARNET: IP address shunning service detect on label 230

Long Description: This rule indicates that a packet has been shunned by shunning service label 230.

cns-033231

Description: ARNET: IP address shunning service detect on label 231

Long Description: This rule indicates that a packet has been shunned by shunning service label 231.

cns-033232

Description: ARNET: IP address shunning service detect on label 232

l**Long Description:** This rule indicates that a packet has been shunned by shunning service label 232.

cns-033233

Description: ARNET: IP address shunning service detect on label 233

Long Description: This rule indicates that a packet has been shunned by shunning service label 233.

cns-033234

Description: ARNET: IP address shunning service detect on label 234

Long Description: This rule indicates that a packet has been shunned by shunning service label 234.

cns-033235

Description: ARNET: IP address shunning service detect on label 235

Long Description: This rule indicates that a packet has been shunned by shunning service label 235.

cns-033236

Description: ARNET: IP address shunning service detect on label 236

Long Description: This rule indicates that a packet has been shunned by shunning service label 236.

cns-033237

Description: ARNET: IP address shunning service detect on label 237

Long Description: This rule indicates that a packet has been shunned by shunning service label 237.

cns-033238

Description: ARNET: IP address shunning service detect on label 238

Long Description: This rule indicates that a packet has been shunned by shunning service label 238.

cns-033239

Description: ARNET: IP address shunning service detect on label 239

Long Description: This rule indicates that a packet has been shunned by shunning service label 239.

cns-033240

Description: ARNET: IP address shunning service detect on label 240

Long Description: This rule indicates that a packet has been shunned by shunning service label 240.

cns-033241

Description: ARNET: IP address shunning service detect on label 241

Long Description: This rule indicates that a packet has been shunned by shunning service label 241.

cns-033242

Description: ARNET: IP address shunning service detect on label 242

Long Description: This rule indicates that a packet has been shunned by shunning service label 242.

cns-033243

Description: ARNET: IP address shunning service detect on label 243

Long Description: This rule indicates that a packet has been shunned by shunning service label 243.

cns-033244

Description: ARNET: IP address shunning service detect on label 244

Long Description: This rule indicates that a packet has been shunned by shunning service label 244.

cns-033245

Description: ARNET: IP address shunning service detect on label 245

Long Description: This rule indicates that a packet has been shunned by shunning service label 245.

cns-033246

Description: ARNET: IP address shunning service detect on label 246

Long Description: This rule indicates that a packet has been shunned by shunning service label 246.

cns-033247

Description: ARNET: IP address shunning service detect on label 247

Long Description: This rule indicates that a packet has been shunned by shunning service label 247.

cns-033248

Description: ARNET: IP address shunning service detect on label 248

Long Description: This rule indicates that a packet has been shunned by shunning service label 248.

cns-033249

Description: ARNET: IP address shunning service detect on label 249

Long Description: This rule indicates that a packet has been shunned by shunning service label 249.

cns-033250

Description: ARNET: IP address shunning service detect on label 250

Long Description: This rule indicates that a packet has been shunned by shunning service label 250.

cns-033251

Description: ARNET: IP address shunning service detect on label 251

Long Description: This rule indicates that a packet has been shunned by shunning service label 251.

cns-033252

Description: ARNET: IP address shunning service detect on label 252

Long Description: This rule indicates that a packet has been shunned by shunning service label 252.

cns-033253

Description: ARNET: IP address shunning service detect on label 253

Long Description: This rule indicates that a packet has been shunned by shunning service label 253.

cns-033254

Description: ARNET: IP address shunning service detect on label 254

Long Description: This rule indicates that a packet has been shunned by shunning service label 254.

cns-033255

Description: ARNET: IP address shunning service detect on label 255

Long Description: This rule indicates that a packet has been shunned by shunning service label 255.

cns-033256

Description: ARNET: IP address shunning service detect on label 256

Long Description: This rule indicates that a packet has been shunned by shunning service label 256.`

cns-100003

Description: PVNET: IP fragment offset is beyond last expected offset

Long Description: This rule detects a situation in which the Defense appliance has already seen a fragment for a packet and knows what the offset of the last fragment is. It then receives this IP fragment that contains an offset placing it beyond the fragment, which is not a valid condition, and indicates an attack.

cns-100004

Description: IANET: IP fragment is duplicate

Long Description: This rule detects an attempt by an attacker to inject another copy of a fragment into the stream. As the Defense appliance reassembles the fragments into a complete packet, it detects the duplicate fragment. Duplicate fragments can be used in fragmentation attacks.

cns-100005

Description: IANET: IP fragment is overlapping

Long Description: This rule detects an attempt by an attacker to send a fragment that has an offset that causes it to overlap another fragment. As the Defense appliance reassembles the fragments into a complete packet, it detects the overlapping fragment. Overlapping fragments can be used in fragmentation attacks.`

cns-100008

Description: "PVNET: IP fragment length mismatch - UDP bomb"

Long Description: This rule detects an attempt to send a packet fragment where the length specified in the UDP header is not consistent with the total packet length specified in the IP header. This could be used in denial of service attacks.

cns-100009

Description: PVNET: IP fragment reassembly exceeds max size

Long Description: This rule detects an IP fragment that causes the datagram being reassembled to exceed the maximum size allowed. This could indicate a denial of service attack.

cns-100010

Description: PVNET: IP fragment reserved bit set

Long Description: The rule detects IP traffic that has an abnormal fragment flag combination set. An example of an abnormal fragment flag combination is a packet that has the flag set but the "more fragment" flag is also set. An abnormal flag combination could lead to erroneous server processing in servers that do not perform proper verification.

cns-100011

Description: PVNET: IP fragment missing byte alignment

Long Description: This rule detects any IP fragment which has a length that is not a multiple of 8 bytes, and which is also not the last fragment of the IP datagram. This could lead to erroneous server processing in servers that do not perform proper verification.

cns-100018

Description: IANET: IP fragment number exceeds maximum allowed

Long Description: This rule detects an IP fragment that exceeds the maximum number of fragments allowed by the Defense appliance for a single datagram. Excessive fragmentation is typically an indicator of an attempt to evade the Defense appliance.

cns-100019

Description: IANET: IP fragment reached maximum age

Long Description: This rule detects an IP Fragment that has reached the maximum age allowed for fragments. The Defense appliance collects all the fragments of an IP datagram for reassembly. If it has

not received all the fragments within the maximum age time (default = 15 seconds), it drops all the fragments.

cns-100024

Description: PVNET: TCP frame contains bad checksum

Long Description: This rule detects a TCP frame that contains a bad checksum. This could indicate the possibility of intentional or unintentional corruption of the headers.

cns-100026

Description: IANET: IP fragment size less than minimum length

Long Description: This rule detects an IP fragment with a length that is less than the configured minimum; the default minimum fragment length is 256 octets. Excessive fragmentation is typically an indicator of an attempt to evade the Defense appliance.

cns-100027

Description: PVNET: UDP frame contains bad checksum

Long Description: This rule detects a UDP frame that contains a bad checksum. This could indicate the possibility of intentional or unintentional corruption of the headers.

Books in Adobe PDF

This help set is available as a book in Adobe PDF.

Also, appliance hardware information is available in this book, which is also available in Adobe PDF.

About Corero

This section covers information about Corero Network Security and this Corero Management Server software.

Contacting Corero Customer Support

Corero Network Security offers two options for contacting Customer Services and Support.

- Contact the Customer Services Center by phone at + 1 978-212-1500
  - Support is available for all customers with a Hardware or Software Warranty from 8:00 AM to 5:00 PM (Eastern US Time).
  - If you have purchased the Software Subscription Service, you can obtain service 7x24 by calling the support phone number and pressing Option 2. If the issue is critical, press Option 2 then Option 7.

  NOTE:
  If, for any reason, the primary support phone number does not work, call Corero's answering service at +1.888.324.1246 (US) or +1.603.645.4145 (International) and a support representative will return your call.
- On the web through the Customer Support Portal: https://support.corero.com. The Web Portal is the most effective way to log and track support issues. This Portal provides:
  - Web-based incident management and customer support tracking system
  - Service request communications
  - Access to downloadable files including software and product documentation
  - An extensive knowledge base.

When you contact Customer Services and Support for assistance, have the following information ready:

- The case number, if you are calling about a previous problem
- Your name, and if someone else will be the contact person for the problem, the contact person's name.
- Your company name and location (city, state or province, and country)
- The telephone number (including area code) at which you or the contact person can be reached.
- The email address at which you or the contact person can be reached.
- The product name, model number, and serial number.
- A list of system hardware and software, including revision levels.
- A detailed problem description:
  - Describe the symptom and the activities that preceded it.
  - Include details about any recent configuration changes, if applicable.

- Be as specific as possible.
- Briefly describe your trouble-shooting steps and observations.

Commenting on This Help Set

At Corero Network Security, our goal is to provide the highest quality products and services to our customers. We value customer feedback and encourage users of Corero's systems to send their comments on the product, service, and documentation to our Customer Service Department, so that we can continue to improve our products.

Please send your comments and suggestions to Corero customer service at this Email address: customer.service@corero.com

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